DESINGULARIZATION OF NONDEGENERATE ROTATING VORTEX PATCHES

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Abstract. This paper analyzes the space of steady rotating solutions to the two-dimensional incompressible Euler equations nearby vortex patch solutions satisfying a natural nondegeneracy condition. We address the question of desingularization and prove that such vortex patch states are the limit of rotating Euler solutions that are smooth to infinite order, have compact vorticity support, and respect dihedral symmetry. Our nondegeneracy condition is proved to be satisfied by Kirchhoff ellipses and along the local bifurcation curves emanating from the Rankine vortex.

The construction, which is based on a local stream function formulation in a tubular neighborhood of the patch boundary, is a synthesis of delicate analysis on thin domains, nonlinear a priori estimates, and a custom version of Newton's method. Our techniques are robust enough to additionally allow us to construct exotic families of singular rotating vortex patch-like solutions nearby a given nondegenerate state. To the best of the authors' knowledge, this work constitutes the first desingularization procedure applicable to general families of steady rotating vortex patches.

1. Introduction

1.1. **The Euler equations in the plane and in the disk, steady rotating solutions, and elliptical vortices.** In this paper we study inviscid incompressible fluids whose motion is described by solutions to the two-dimensional Euler system of equations. Two fluid geometries are within the scope of our analysis: the planar case of a fluid defined on all of $\mathbb{R}^2$ and the disk case of a fluid confined to $\overline{B(0,R)}$ for some $R > 0$. To consolidate notation, we shall make the convention that $\overline{B(0,R)} = \mathbb{R}^2$ and $\partial B(0,R) = \varnothing$ when $R = \infty$.

In Eulerian variables, the system of Euler equations relates the velocity vector $u : \mathbb{R}^+ \times \overline{B(0,R)} \to \mathbb{R}^2$ and the scalar pressure $p : \mathbb{R}^+ \times \overline{B(0,R)} \to \mathbb{R}$ via the bulk equations

$$\partial_t u + u \cdot \nabla u + \nabla p = 0 \text{ and } \nabla \cdot u = 0 \tag{1.1.1}$$

and, when $R < \infty$, the no-penetration boundary condition

$$u(t,x) \cdot x/|x| = 0 \text{ for all } x \in \partial B(0,R),\ t \in \mathbb{R}^+. \tag{1.1.2}$$

As is well-known (see, e.g., [49, 6, 56, 50, 4]), the above equations (1.1.1) and (1.1.2) can be equivalently formulated in terms of the vorticity $\widetilde{\omega} : \mathbb{R}^+ \times \overline{B(0,R)} \to \mathbb{R}$ and the stream function $\psi : \mathbb{R}^+ \times \overline{B(0,R)} \to \mathbb{R}$ via the bulk equations

$$\partial_t \widetilde{\omega} + \nabla^\perp \psi \cdot \nabla \widetilde{\omega} = 0 \text{ and } \Delta \psi = \widetilde{\omega}, \tag{1.1.3}$$

where $\nabla^\perp = (-\partial_2, \partial_1)$ denotes the counterclockwise rotated gradient, and the boundary condition

$$\nabla^\perp \psi(t,x) \cdot x/|x| = 0 \text{ for all } x \in \partial B(0,R),\ t \in \mathbb{R}. \tag{1.1.4}$$

One recovers the velocity and pressure of (1.1.1) from the vorticity and stream function of (1.1.3) and (1.1.4) via the relationships

$$u = \nabla^\perp \psi \text{ and } -\Delta p = \nabla \cdot \nabla \cdot (u \otimes u). \tag{1.1.5}$$

We are interested in the class of *steady rotating* solutions to equations (1.1.3) and (1.1.4); these are time-independent when viewed in a coordinate system rotating at a constant angular velocity. Let us fix

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an arbitrary angular velocity $\Omega \in \mathbb{R}$ and make the ansatz that there exists $\omega, \Psi : \overline{B(0,R)} \to \mathbb{R}$ such that for all $t \in \mathbb{R}^+$ and $x \in \overline{B(0,R)}$ we have

$$\widetilde{\omega}(t,x) = \omega(\mathbf{R}(\Omega t)x) \text{ and } \psi(t,x) = \Psi(\mathbf{R}(\Omega t)x) + \Omega|\mathbf{R}(\Omega t)x|^2/2 \tag{1.1.6}$$

where for $s \in \mathbb{R}^+$ we have defined the rotation matrix

$$\mathbf{R}(s) = \begin{pmatrix} \cos(s) & -\sin(s) \\ \sin(s) & \cos(s) \end{pmatrix}. \tag{1.1.7}$$

The corresponding bulk equations for the *time-independent* scalars ω and Ψ, called the vorticity and the relative stream function, respectively, are then computed to be the following:

$$\nabla^\perp \Psi \cdot \nabla \omega = 0 \text{ and } \Delta \Psi = \omega - 2\Omega. \tag{1.1.8}$$

When $R < \infty$, we additionally have the boundary condition

$$\nabla^\perp \Psi(x) \cdot x/|x| = 0 \text{ for all } x \in \partial B(0,R). \tag{1.1.9}$$

On the other hand, when $R = \infty$, we supplement the second equation of (1.1.8) with the condition $\nabla(\Psi + \Omega|\cdot|^2/2)(x) \to 0$ as $|x| \to \infty$.

While the equations (1.1.8) and (1.1.9) were derived from the parent Euler system (1.1.1) under a sufficient amount of assumed smoothness, to properly cast our analysis and results we must interpret the equations more broadly in a sense of weak solutions. This is made precise by the following definition.

Definition 1.1 (Steady rotating weak solutions to the Euler system)**.** *Let $R \in (0,\infty]$. We say that $\Omega \in \mathbb{R}$, $\Psi \in C^1(\overline{B(0,R)})$, and $\omega \in (L^1 \cap L^\infty)(\overline{B(0,R)})$ are a weak solution to equations* (1.1.8) *and* (1.1.9) *if the following hold.*

(1) For almost every $x \in \overline{B(0,R)}$ we have satisfied

$$\nabla \Psi(x) = -\Omega x - \int_{B(0,R)} \mathbf{K}_R(x,y)^\perp \omega(y)\,\mathrm{d}y \tag{1.1.10}$$

with $\mathbf{K}_R$ the Biot-Savart kernel, given for $R < \infty$ by

$$\mathbf{K}_R(x,y) = \frac{1}{2\pi}\frac{(x-y)^\perp}{|x-y|^2} + \frac{1}{2\pi}\frac{(R^2 y - |y|^2 x)^\perp}{R^4 - 2R^2 x\cdot y + |x|^2|y|^2}. \tag{1.1.11}$$

When $R = \infty$, $\mathbf{K}_\infty$ is simply the above right hand side's first term.

(2) For all $f \in C^1_{\mathrm{c}}(\overline{B(0,R)})$ we have satisfied

$$\int_{B(0,R)} \omega(y)\nabla^\perp\Psi(y)\cdot\nabla f(y)\,\mathrm{d}y = 0. \tag{1.1.12}$$

(3) For all $x \in \partial B(0,R)$ we have satisfied $\nabla^\perp\Psi(x)\cdot x/|x| = 0$.

Weak solutions to equation (1.1.8) in the sense of Definition 1.1 give rise to so-called Yudovich weak solutions (see Section 3.1 of [66], Section 8.2 of [6], or Section 2.3 in [50]) to system (1.1.1). Explicitly if $R \in (0,\infty]$, $\Omega \in \mathbb{R}$, $\Psi \in C^1(\overline{B(0,R)})$, and $\omega \in (L^1 \cap L^\infty)(\overline{B(0,R)})$ are a steady rotating weak solution in the sense of Definition 1.1 and we define the dynamic scalars $\widetilde{\omega}$ and ψ as in equation (1.1.6) and let $u = \nabla^\perp\psi$, then the following hold.

- We have the inclusions $\widetilde{\omega} \in L^\infty([0,\infty);(L^1\cap L^\infty)(B(0,R)))$ and $u \in L^\infty([0,\infty);\mathrm{LL}(\overline{B(0,R)}))$ where LL denotes the collection of logarithmic-Lipschitz functions (see Section 1.4).
- We have the identities

$$u = \int_{B(0,R)} \mathbf{K}_R(\cdot,y)\widetilde{\omega}(y)\,\mathrm{d}y,\ (\nabla^\perp\cdot u) = \widetilde{\omega} \text{ and } \nabla\cdot u = 0. \tag{1.1.13}$$

  in the sense of distributions on $B(0,R)$ where $\mathbf{K}_R$ is as in (1.1.11). We also have, in the case $R < \infty$, the satisfaction of the no penetration boundary condition

$$u(x)\cdot x/|x| = 0 \text{ for all } x \in \partial B(0,R). \tag{1.1.14}$$

- For all $T > 0$ and all $f \in C^1([0,T]; C^1_c(B(0,R)))$ it holds that

$$\int_{B(0,R)} f(T,\cdot)\widetilde{\omega}(T,\cdot) - \int_{B(0,R)} f(0,\cdot)\widetilde{\omega}(0,\cdot) = \int_0^T \int_{B(0,R)} (\partial_t f + u \cdot \nabla f)\widetilde{\omega} \tag{1.1.15}$$

An important and well-studied class of solutions to the two-dimensional steady rotating system Euler equations are the *steady rotating vortex patches*, which are also known as *V-states*, see, e.g., [23, 8, 64, 52, 40, 9, 10, 21, 37, 38, 65] and references therein. These are weak solutions to the steady rotating Euler equations in the sense of Definition 1.1 in which the vorticity is singular, being the discontinuous characteristic function of an open, bounded, and simply connected set with boundary belonging to some regularity class. Among this class of steady rotating vortex patches, there is an *explicit* subfamily known as the Kirchhoff elliptical vortices (see Kirchhoff [44] or Article 159 of Chapter VII in Lamb [45]). Modulo the symmetries of the Euler equations, these form a one parameter family, that we enumerate in a particular way. For $\xi \in \mathbb{R}^+$ we define

$$E^\xi = \{x \in \mathbb{R}^2 \ : \ x_1^2 + (x_2/\tanh(\xi))^2 < 1\} \subset \mathbb{R}^2 \text{ and } \Omega^\xi = \tanh(\xi)/(1+\tanh(\xi))^2. \tag{1.1.16}$$

The ξ-Kirchhoff vortex is a steady rotating (with angular velocity Ω^ξ) weak solution (in the sense of Definition 1.1 with $R = \infty$) to (1.1.8) with vorticity $\omega = \mathbb{1}_{E^\xi}$ and, for some constant $c^\xi \in \mathbb{R}$, relative stream function $\Psi = \Psi^\xi$ satisfying

$$\Psi^\xi(x) = (\mathbf{N} * \mathbb{1}_{E^\xi})(x) - \Omega^\xi|x|^2/2 - c^\xi \text{ for } x \in \mathbb{R}^2 \text{ and } \Psi^\xi(\partial E^\xi) = \{0\} \tag{1.1.17}$$

with

$$\mathbf{N}(x) = (1/2\pi)\log|x| \text{ for } x \in \mathbb{R}^2 \setminus \{0\} \tag{1.1.18}$$

the Newtonian potential. In the extreme case $\xi = \infty$ we have that $E^\infty = B(0,1)$ is simply the open unit disk and $\Omega^\infty = 1/4$. Due to radial symmetry, $\omega = \mathbb{1}_{B(0,1)}$ is the vorticity of a weak solution to (1.1.8) not just for $\Omega = 1/4$, but for all $\Omega \in \mathbb{R}$. This family of circular vortex solutions are known as the Rankine vortices [55].

A motivation for this work's forthcoming analysis is the following natural question: Are the class of singular solutions given by the steady rotating vortex patches able to be realized as limits of classical C^∞-smooth solutions having the same qualitative properties, such as steady rotation and compact vorticity support, while respecting discrete symmetries? A positive answer to the previous desingularization question would show that these singular solutions are further from being pathological and, in a sense, are more physically meaningful.

In this paper we provide a partial positive answer to the preceding query. We formulate a notion of nondegeneracy for steady rotating vortex patches that is satisfied when a particular related linear operator (having the form of a compact perturbation of the identity) has a trivial kernel. When this condition is satisfied by a patch, we prove, among several other things, that the patch can be desingularized in the sense that it is the limit of smooth solutions to equations (1.1.8) and (1.1.9) that respect certain desirable qualitative properties.

It is clearly important to verify that there exist steady rotating patches that satisfy our nondegeneracy condition. We provide two uncountable families of examples. The first are the Kirchhoff elliptical vortices; more precisely, we show that there exists a countable set $\mathcal{N} \subset \mathbb{R}^+$ such that for all $\xi \in \mathbb{R}^+ \setminus \mathcal{N}$ the ξ-Kirchhoff vortex of equations (1.1.16) and (1.1.17) is nondegenerate and hence can be desingularized. The second family of examples is given by the m-fold symmetric V-states of Burbea [8] that locally bifurcate from the Rankine vortex (these are discussed in detail in Section 2.2). The authors conjecture – but make no attempt to prove in this work – that our nondegeneracy condition, that is precisely stated in the fourth item of Definition 1.2 below, is a generic property of steady rotating vortex patches.

1.2. **Survey of prior work.** The incompressible Euler equations (1.1.1) are among the first PDEs to be written down and have been at the focal point of intense research for nearly a quarter of a millennium. As such, we make no attempt to give a comprehensive literature review; rather we focus just on the results for the two-dimensional equations that are most closely related to our own.

The local and global well-posedness of the initial value problem for system (1.1.1) in the class of classical solutions is due to Lichtenstein [47], Hölder [41], and Wolibner [63]; modern treatments of these facts are found in Kato and Ponce [43]. The corresponding global well-posedness result in a class of weak solutions for the vorticity formulation was proved by Yudovich [66]; see also Bardos [3]. Extensions to larger classes of (more singular) vortex data have been made by Yudovich [67], Serfati [57], and Elgindi, Murray, and Said [28].

Among the most well-known explicit solutions to (1.1.1) are the Kirchhoff elliptical vortices. Regimes of spectral stability and instability for the Kirchhoff elliptical vortices were first established by Love [48]. Promotion to nonlinear stability came with the work of Tang [60] and Wan [61]. In the complementary regime, nonlinear instability was proved by Guo, Hallstrom, and Spirn [34]. Pulvirenti and Wan [62] established nonlinear stability of the Rankine vortices.

The distinguishing feature of two-dimensional Euler flow governed by equation (1.1.3) is the transportation of the vorticity along the velocity vector field; hence, when the initial value problem is given an initial vorticity that is the characteristic function of a set, this property is propagated in time. These weak solutions are known as *vortex patches*. Global well-posedness of vortex patches in spaces of sufficient boundary regularity was proved by Chemin [12, 13], Serfati [58], and Bertozzi-Constantin [5]; see also Radu [54]. Analysis of more singular vortex patch solutions can be found in the work of Danchin [19, 20], Elgindi-Jeong [26, 25], and Elgindi-Jo [27].

The steady rotating vortex patch solutions discussed in Section 1.1 are evidently a special type of vortex patch solution whose dynamics are described simply by constant rotation. The only known explicit and nonradial rotating vortex patches are the Kirchhoff ellipses. There has long been an industry, starting with the numerical work of Deem and Zabusky [23], devoted to the construction of other steady rotating vortex patch solutions (also known as V-states). The first approach to obtaining such states as local bifurcations from the Rankine vortices was introduced by Burbea [8]. The method was then revisited and placed on more solid technical ground by Hmidi, Mateu, and Verdera [40]. Local bifurcations near Kirchhoff ellipses were constructed by Castro, Córdoba, and Gómez-Serrano [10] and Hmidi and Mateu [37]. Analytic regularity of the bifurcating patch's boundary was proved by Castro, Córdoba, and Gómez-Serrano [10]. Global bifurcations from the Rankine vortices were first obtained by Hassainia, Masmoudi, and Wheeler [36]. We also mention that: Hmidi and Mateu [39] constructed disconnected steady rotating vortex patches; Hoz, Hmidi, and Mateu [22] found doubly connected planar vortex states; and Hoz, Hassainia, Hmidi, and Mateu [21] uncovered nontrivial rotating vortex patches confined to the disk.

The literature related to the specific task of steady vortex patch desingularization for the Euler equations is much sparser, consisting of, to the best of the authors' knowledge, two specific results. Castro, Córdoba, and Gómez-Serrano [11] construct via a delicate local bifurcation argument starting from Rankine vortices uniformly rotating Euler solutions with compact and class C^2 vorticity and without radial symmetry. Elgindi and Huang [24], employing the Schauder fixed point theorem, desingularize the spatially periodic Bahouri-Chemin patch.

Our work in this paper, therefore, stands as the first general desingularization procedure for steady rotating vortex patches; moreover, we desingularize to the class of C^∞_{c}-smooth states. We address *any* nondegenerate V-state and prove that two well-known families of V-states – Kirchhoff's ellipses and Burbea's Rankine perturbations – are generically nondegenerate.

Finally, let us remark that our results belong to the general structural study of the space of stationary or rigidly evolving solutions to the two-dimensional system of Euler equations. We list a few representative results of this subfield: Choffrut and Šverák [15]; Choffrut and Székelyhidi [14]; Constantin, Drivas, and Ginsberg [16]; and Coti Zelati, Elgindi, and Widmayer [17].

1.3. **Main results and discussion.** As we alluded to in Section 1.1, our main theorems apply for a class of *admissible* steady rotating vortex patches obeying a certain nondegeneracy condition. We now make these precise in the next definition. Recall that $\mathbf{N}$ from (1.1.18) denotes the Newtonian potential.

Definition 1.2 (Admissible steady rotating vortex patch solutions)**.** *We say that* $\Psi \in C^{1,1}(\mathbb{R}^2)$*,* $\Omega, c \in \mathbb{R}$*,* $\varnothing \neq \widetilde{U} \subset \mathbb{R}^2$*,* $m \in \mathbb{N}^+$ *form an admissible tuple if the following are satisfied.*

(1) The set $\widetilde{U}$ is open, bounded, simply connected, and has m-fold dihedral symmetry in the sense that

$$x = (x_1, x_2) \in \widetilde{U} \text{ if and only if } (x_1, -x_2) \in \widetilde{U} \text{ if and only if } \mathbf{R}(2\pi/m)x \in \widetilde{U}, \tag{1.3.1}$$

where $\mathbf{R}$ is defined in (1.1.7). The boundary $\Sigma = \partial\widetilde{U}$ is class $C^{1,1}$.

(2) For all $x \in \Sigma$ we have $\Psi(x) = 0$ and if $\nu : \Sigma \to \mathbb{R}^2$ denotes the outward pointing unit normal, then $\min_\Sigma(\nu \cdot \nabla\Psi) > 0$.

(3) For all $x \in \mathbb{R}^2$ it holds that

$$\Psi(x) = (\mathbf{N} * \mathbb{1}_{\widetilde{U}})(x) - \Omega|x|^2/2 - c. \tag{1.3.2}$$

(4) If $\phi \in C^0(\Sigma)$ has m-fold dihedral symmetry in the sense that

$$\phi(x) = \phi(x_1, -x_2) = \phi(\mathbf{R}(2\pi/m)x) \tag{1.3.3}$$

for all $x = (x_1, x_2) \in \Sigma$, and satisfies for all $x \in \Sigma$ the identity

$$\phi(x) + \int_\Sigma \mathbf{N}(x-y)\frac{\phi(y)}{\nu(y)\cdot\nabla\Psi(y)}\,\mathrm{d}\mathcal{H}^1(y) = 0, \tag{1.3.4}$$

then $\phi = 0$. Here $\mathrm{d}\mathcal{H}^1$ denotes integration with respect to the one-dimensional Hausdorff measure.

The first three items of Definition 1.2 guarantee in particular that the triple Ω, Ψ, and $\omega = \mathbb{1}_{\widetilde{U}}$ are a steady rotating weak solution in the sense of Definition 1.1 with $R = \infty$. It is worth noting that the positive normal derivative assertion of the second item of Definition 1.2 holds automatically in the case that $\Omega < 1/2$, as a simple consequence of the Hopf boundary point lemma.

The fourth item is the nondegeneracy condition. Roughly speaking, this condition is saying that the linearization at the particular patch solution of (1.3.2) has no kernel. As we shall prove in Theorem IV below, the collection of admissible states in the sense of Definition 1.2 is far from being empty.

At last, we are ready to state our main theorems, the proofs of which are found in Section 3.4. Our first result gives a positive answer to the question of classical realization for rotating patches.

Theorem I (Desingularization of admissible states)**.** *Let Ψ, Ω, c, $\widetilde{U}$, and m be an admissible steady rotating vortex patch in the sense of Definition 1.2 and let $\gamma \in C^\infty(\mathbb{R})$ be any function that satisfies $\gamma(t) = 0$ for $t \leqslant -1$ and $\gamma(t) = 1$ for $t \geqslant 1$. There exists $\varepsilon_\star \in (0,1]$, $0 < C < \infty$, and an open and bounded neighborhood $U \supset \Sigma$ with the property that for all $0 < \varepsilon \leqslant \varepsilon_\star$ there exists $\omega_\varepsilon \in C^\infty_{\mathrm{c}}(\mathbb{R}^2)$ and $\Psi_\varepsilon \in C^\infty(\mathbb{R}^2)$ such that*

$$\forall\, \alpha \in (0,1),\ \|\nabla(\Psi_\varepsilon - \Psi)\|_{C^\alpha_{\mathrm{b}}(\mathbb{R}^2)} \to 0 \text{ as } \varepsilon \to 0 \tag{1.3.5}$$

and the following hold.

(1) For all $x \in \mathbb{R}^2$

$$\Psi_\varepsilon = (\mathbf{N} * \omega_\varepsilon)(x) - (\Omega/2)|x|^2 - c \text{ and } \nabla^\perp\Psi_\varepsilon(x)\cdot\nabla\omega_\varepsilon(x) = 0. \tag{1.3.6}$$

In particular, the vorticity-relative stream function formulation of the steady rotating planar Euler system (1.1.8) is classically satisfied with angular velocity Ω, vorticity ω_ε, and relative stream function Ψ_ε.

(2) We have $C\|\omega_\varepsilon\|_{C^1(\mathbb{R}^2)} \geqslant 1/\varepsilon$ and the deviation estimates

$$\|\omega_\varepsilon - \mathbb{1}_{\widetilde{U}}\|_{L^1(\mathbb{R}^2)} + |\|\nabla\omega_\varepsilon\|_{\mathrm{TV}(\mathbb{R}^2)} - \|\nabla\mathbb{1}_{\widetilde{U}}\|_{\mathrm{TV}(\mathbb{R}^2)}| + \|\nabla(\Psi_\varepsilon - \Psi)\|_{L^\infty(\mathbb{R}^2)} \leqslant C\varepsilon\log(1/\varepsilon), \tag{1.3.7}$$

where $\|\cdot\|_{\mathrm{TV}(\mathbb{R}^2)}$ denotes the total variation norm on the space of vector valued measures.

(3) We have

$$\forall\, x \in U,\ \omega_\varepsilon(x) = \gamma\Big(-\frac{\Psi_\varepsilon(x)}{\varepsilon}\Big) \text{ and } \forall\, x \in \mathbb{R}^2 \setminus U,\ \omega(x) \in \{0,1\}. \tag{1.3.8}$$

(4) The functions Ψ_ε and ω_ε have m-fold dihedral symmetry.

The machinery we develop to prove Theorem I, which is the content of Section 3, is quite robust and finds applications beyond just the task of admissible patch desingularization. The two subsequent main theorems highlight some of what is possible in this direction.

Our next result, also addressing admissible states, shows that any neighborhood of the rotating vortex patch contains a whole zoo of singular solutions that look like linear combinations of nested patches.

Thus one can think of the given nondegenerate patch solution as able to be split into sums of multiple patches in many different ways.

Theorem II (Splitting of admissible states). *Let Ψ, Ω, c, $\widetilde{U}$, and m be an admissible steady rotating vortex patch in the sense of Definition 1.2, $M \in \mathbb{N}^+$, and $\{\sigma_k\}_{k=-M}^{M} \subset \mathbb{R}$ satisfy $\sum_{k=-M}^{M} \sigma_k = 1$. There exists $\rho_\star \in (0,1]$, $0 < C < \infty$, and an open and bounded neighborhood $U \supset \Sigma$ with the property that for all $0 < \rho \leqslant \rho_\star$ there exists $\omega_\rho \in (L^\infty \cap \mathrm{BV})_{\mathrm{c}}(\mathbb{R}^2)$ (with the subscript 'c' indicating compact support) and $\Psi_\rho \in \bigcap_{0<\alpha<1} C^{1,\alpha}(\mathbb{R}^2)$ such that*

$$\forall\, \alpha \in (0,1),\ \|\nabla(\Psi_\rho - \Psi)\|_{C^\alpha_{\mathrm{b}}(\mathbb{R}^2)} \to 0 \text{ as } \rho \to 0 \tag{1.3.9}$$

and the following hold.

(1) The steady rotating planar Euler system is satisfied, in the weak sense of Definition 1.1 with $R = \infty$, by the angular velocity Ω, vorticity ω_ρ, and relative stream function Ψ_ρ.

(2) There exist open, bounded domains of class $\bigcap_{0<\alpha<1} C^{1,\alpha}$, denoted $\{W_\rho^k\}_{k=-M}^{M}$, satisfying the following.

(a) The monotone inclusion relations

$$W_\rho^M \subset \cdots \subset W_\rho^0 \subset \cdots \subset W_\rho^{-M} \tag{1.3.10}$$

hold.

(b) The vorticity is given by the linear combination

$$\omega_\rho = \sum_{k=-M}^{M} \sigma_k \mathbb{1}_{W_\rho^k} \tag{1.3.11}$$

almost everywhere in $\mathbb{R}^2$.

(c) The distance estimates

$$C \cdot \mathrm{dist}(\partial W_\rho^k, \partial W_\rho^{k+1}) \geqslant \rho,\ k \in \{-M, \dots, M-1\} \tag{1.3.12}$$

hold.

(3) We have the deviation estimates

$$\|\omega_\rho - \mathbb{1}_{\widetilde{U}}\|_{L^1(\mathbb{R}^2)} + \big|\|\nabla\omega_\rho\|_{\mathrm{TV}(\mathbb{R}^2)} - \|\nabla \mathbb{1}_{\widetilde{U}}\|_{\mathrm{TV}(\mathbb{R}^2)}\big| + \|\nabla(\Psi_\rho - \Psi)\|_{L^\infty(\mathbb{R}^2)} \leqslant C\rho\log(1/\rho), \tag{1.3.13}$$

where, again, $\|\cdot\|_{\mathrm{TV}(\mathbb{R}^2)}$ denotes the total variation norm.

(4) We have

$$\omega_\rho = \sum_{k=-M}^{M} \sigma_k \mathbb{1}_{(0,\infty)}(-\Psi_\rho - k\rho) \text{ a.e. in } U \tag{1.3.14}$$

and, in particular, for all $k \in \{-M, \dots, M\}$ it holds that $\Psi_\rho + k\rho = 0$ on ∂W_ρ^k.

(5) The functions Ψ_ρ and ω_ρ have m-fold dihedral symmetry.

Our final construction around admissible states addresses what we call trapping. The vortex patches in the scope of Definition 1.2 are solutions to the Euler equations over the entire plane $\mathbb{R}^2$. One may wonder if the solutions can be confined to a circular disk region of a large but finite radius at the cost of a small perturbation in the patch and in the stream function. The following result shows that this is indeed true.

Theorem III (Trapping of admissible states). *Let Ψ, Ω, c, $\widetilde{U}$, and m be an admissible steady rotating vortex patch in the sense of Definition 1.2. There exists $1 \leqslant R_\star < \infty$, $0 < C, C_1 < \infty$ with $C_1 \leqslant R_\star$, and an open and bounded neighborhood $U \supset \Sigma$ with the property that for all $R_\star \leqslant R < \infty$ there exists $\omega_R \in (L^\infty \cap \mathrm{BV})_{\mathrm{c}}(B(0,R/2))$ (again with the subscript 'c' indicating compact support) and $\Psi_R \in \bigcap_{0<\alpha<1} C^{1,\alpha}(\overline{B(0,R)})$ such that*

$$\forall \widetilde{R} \in \mathbb{R}^+,\ \forall\, \alpha \in (0,1),\ \|\nabla(\Psi_R - \Psi)\|_{C^\alpha_{\mathrm{b}}(\overline{B(0,\min\{R,\widetilde{R}\})})} \to 0 \text{ as } R \to \infty \tag{1.3.15}$$

and the following hold.

(1) The steady rotating planar Euler system is satisfied, in the weak sense of Definition 1.1, in the domain $B(0,R)$ *with angular velocity* Ω*, vorticity* ω_R*, and relative stream function* Ψ_R*.*

(2) There exists an open, bounded, and class $\bigcap_{0<\alpha<1} C^{1,\alpha}$ *domain* $W_R \subset B(0,C_1)$ *such that* $\omega_R = \mathbb{1}_{W_R}$ *and*

$$\partial W_R = \{x \in U \ : \ \Psi_R(x) = 0\}. \tag{1.3.16}$$

(3) We have the deviation estimates

$$\|\omega_R - \mathbb{1}_{\widetilde{U}}\|_{L^1(B(0,R))} + |\|\nabla\omega_R\|_{\mathrm{TV}(B(0,R))} - \|\nabla\mathbb{1}_{\widetilde{U}}\|_{\mathrm{TV}(B(0,R))}| + \|\nabla(\Psi_R - \Psi)\|_{L^\infty(B(0,R))} \leqslant C\log(R)/R^2, \tag{1.3.17}$$

where $\|\cdot\|_{\mathrm{TV}(B(0,R))}$ *denotes the total variation norm.*

(4) The functions Ψ_R *and* ω_R *have* m*-fold dihedral symmetry.*

The last main theorem of the paper, whose proof is the content of Section 2, gives two examples of families of rotating vortex patches satisfying the conditions of Definition 1.2.

Theorem IV (Existence of admissible states)**.** *Definition 1.2 is nonvacuous. In fact:*

(1) There exists a countable set $\mathcal{N} \subset \mathbb{R}^+$ *such that for all* $\xi \in \mathbb{R}^+ \setminus \mathcal{N}$ *the* ξ*-Kirchhoff elliptical vortex, defined in* (1.1.16) *and* (1.1.17)*, is admissible in the sense of Definition 1.2 with* $\Psi = \Psi^\xi$*,* $\Omega = \Omega^\xi$*,* $c = c^\xi$*,* $\widetilde{U} = E^\xi$*, and* $m = 2$*.*

(2) See Theorem 2.9 for a precise statement; informally: for every $\mathbb{N} \ni m \geqslant 2$*, the* m*-fold symmetric states locally bifurcating from the circular Rankine vortex with angular velocity* $\Omega_m = (m-1)/2m$ *(in other words Burbea's* m*-fold symmetric V-states) are admissible in the sense of Definition 1.2.*

The remainder of this subsection is devoted to a brief and high-level discussion of the proof of our main results. Let Ψ, Ω, c, $\widetilde{U}$, and m be an admissible steady rotating vortex patch solution in the sense of Definition 1.2. The starting point is the equation (1.3.2) that relates the stream function Ψ to the interior vorticity support $\widetilde{U}$. For the purposes of this discussion, let us suppose that $\Psi < 0$ in $\widetilde{U}$. This condition is not built into Definition 1.2, but is satisfied thanks to a maximum principle argument, if we know additionally that $\Omega < 1/2$. Let us also pretend that $\Psi > 0$ in all of $\mathbb{R}^2 \setminus \overline{\widetilde{U}}$ – this is not really true, but it is convenient for the ensuing imprecise discussion. These assumptions allow us to imagine that

$$\mathbb{1}_{\widetilde{U}} = \mathbb{1}_{(0,\infty)}(-\Psi). \tag{1.3.18}$$

Then, we obtain a closed equation in terms of the stream function with a Heaviside *vorticity function*; we write this as an abstract operator equation $\mathsf{F}[\Psi] = 0$ with

$$(\mathsf{F}[\Psi])(x) = \Psi(x) - \frac{1}{2\pi}\int_{\mathbb{R}^2} \log|x-y|\mathbb{1}_{(0,\infty)}(-\Psi(y))\,\mathrm{d}y + \Omega|x|^2/2 + c = 0 \text{ for all } x \in \mathbb{R}^2. \tag{1.3.19}$$

The goal then is to construct nearby Euler solutions $\widetilde{\Psi}$ that solve a perturbed operator equation of the form $\mathsf{F}_{\mathsf{v}}[\widetilde{\Psi}] = 0$ with

$$(\mathsf{F}_{\mathsf{v}}[\widetilde{\Psi}])(x) = \widetilde{\Psi}(x) - \frac{1}{2\pi}\int_{\mathbb{R}^2} \log|x-y|\mathsf{v}(-\widetilde{\Psi}(y))\,\mathrm{d}y + \Omega|x|^2/2 + c \text{ for all } x \in \mathbb{R}^2. \tag{1.3.20}$$

where $\mathsf{v} : \mathbb{R} \to \mathbb{R}$ is a vorticity function that is in an appropriate sense close to the Heaviside function $\mathbb{1}_{(0,\infty)}$. For example, in Theorem I we take (with $\varepsilon > 0$ small)

$$\mathsf{v}(t) = \gamma(t/\varepsilon) \text{ with } \gamma \in C^\infty(\mathbb{R}),\ \gamma(t) = 0 \text{ for } t \leqslant -1 \text{ and } \gamma(t) = 1 \text{ for } t \geqslant 1; \tag{1.3.21}$$

on the other hand, for Theorem II, we take (with $\rho > 0$ small)

$$\mathsf{v}(t) = \sum_{k=-M}^{M} \sigma_k \mathbb{1}_{(0,\infty)}(t - k\rho) \text{ with } \sum_{k=-M}^{M} \sigma_k = 1. \tag{1.3.22}$$

Note that taking $\sigma_0 = 1$ and $\sigma_k = 0$ for $k \neq 0$ in (1.3.22) recovers the Heaviside vorticity function.

One might hope to address these perturbed equations (1.3.20) through some sort of implicit function theorem (IFT) or Banach fixed point argument. Indeed, it is this hope that is the inspiration for the instantiation of our nondegeneracy condition given in the fourth item of Definition 1.2. By formally taking the Fréchet derivative of the expression for F given in equation (1.3.19) at the admissible stream function Ψ in an arbitrary direction ϕ and restricting the result to Σ, we get exactly the left hand side of (1.3.4). Thus we see precisely that the nondegeneracy condition is equivalent to the invertibility of this formal functional derivative.

However, substantial technical obstructions arise that preclude a direct invocation of the IFT or contraction mapping principle. More specifically, the operator F (1.3.19) appears not to be C^1 in the Fréchet sense or even locally Lipschitz for any choice of suitable function spaces. The reason for this failure is precisely the discontinuous vorticity function $\mathsf{v} = \mathbb{1}_{(0,\infty)}$.

As a first step in solving the perturbed equations $\mathsf{F}_{\mathsf{v}}[\widetilde{\Psi}] = 0$, we can replace the Heaviside vorticity function of (1.3.19) with a nearby one that is qualitatively smooth. Executing this swap restores Fréchet smoothness and a local Lipschitz condition, at the expense of introducing a source term. We define the following regularized versions of (1.3.22):

$$\mathsf{v}_{\varepsilon,\rho}(t) = \sum_{k=-M}^{M} \sigma_k \gamma\Big(\frac{t-k\rho}{\varepsilon}\Big). \tag{1.3.23}$$

The singular case of the vorticity function is ultimately recovered by taking the limit as $\varepsilon \to 0$ in the above – of course, this requires carefully tracking various quantities.

We let $X = C^1_{\mathrm{b}}(\mathbb{R}^2)$ be an ambient Banach space and set up the operator

$$\mathsf{G}_{\varepsilon,\rho} : B_X(0,r) \to X \text{ with action } \mathsf{G}_{\varepsilon,\rho}[\phi] = \mathsf{F}_{\mathsf{v}_{\varepsilon,\rho}}[\Psi+\phi] - \mathsf{F}_{\mathsf{v}_{\varepsilon,\rho}}[\Psi] \text{ for } \phi \in B_X(0,r). \tag{1.3.24}$$

We endeavor to solve, at least for sufficiently small ε and ρ, the equation, that is equivalent to $\mathsf{F}_{\mathsf{v}}[\widetilde{\Psi}] = 0$,

$$\mathsf{G}_{\varepsilon,\rho}[\psi] = \mathsf{g}_{\varepsilon,\rho}, \text{ for the source term } \mathsf{g}_{\varepsilon,\rho} = \mathsf{F}[\Psi] - \mathsf{F}_{\mathsf{v}_{\varepsilon,\rho}}[\Psi] \in X \tag{1.3.25}$$

with $\psi = \widetilde{\Psi} - \Psi$ denoting the relative stream function perturbation. Note crucially that in the formulation above, the operator $\mathsf{G}_{\varepsilon,\rho}$ is smooth; however, the norms of its derivatives are blowing up in the limit $\varepsilon \to 0$.

The following two facts are sufficient to guarantee the existence of solutions to the perturbed operator equation (1.3.25).

(1) We have the limit: $\lim_{\varepsilon,\rho\to 0}\|\mathsf{g}_{\varepsilon,\rho}\|_X = 0$. This essentially follows from the coarea formula.
(2) Uniform local surjectivity. More precisely: there exists $r_1, \varepsilon_1, \rho_1 > 0$ such that for all $0 < \varepsilon \leqslant \varepsilon_1$ and $0 \leqslant \rho < \rho_1$ we have $B_X(0,r_1) \subset \mathsf{G}_{\varepsilon,\rho}[B_X(0,r)]$.

The proof of the latter fact is a delicate matter due to the development of singularities in the limit as $\varepsilon \to 0$ and its competition with the requirement that the radius $r_1 > 0$ be taken *independent* of small $\varepsilon, \rho > 0$. To wit: the nondegeneracy condition of Definition 1.2 does ensure that $D\mathsf{G}_{\varepsilon,\rho}[0] : X \to X$ is invertible and so the IFT indeed gives local invertibility of $\mathsf{G}_{\varepsilon,\rho}$ near zero; the problem, however, is that the operator norms satisfy $\min\{\|DG_{\varepsilon,\rho}[0]\|_{\mathcal{L}(X)}, \|DG_{\varepsilon,\rho}[0]^{-1}\|_{\mathcal{L}(X)}\} \to \infty$ as $\varepsilon \to 0$. Thus, the domains granted by the IFT are shrinking to nothing and we do not yet satisfy the second fact above.

Our construction of the local inverse that does satisfy the uniform local surjectivity condition instead proceeds via a version of Newton's method capable of handling this singular limit. We formulate this tool abstractly in Theorem 3.13 below, but the main hypotheses that guarantee a local inverse are:

(1) A nonlinear a priori estimate. There exists c_1, c_2 with $c_2 < r$ (r being the domain radius from (1.3.24)) such that (uniformly for small ε and ρ) $\|\mathsf{G}_{\varepsilon,\rho}[\phi]\|_X \leqslant c_1$ implies that $\|\phi\|_X \leqslant c_2$.
(2) Qualitative invertibility of the derivative in a uniform neighborhood. There exists $c_2 < c_3 \leqslant r$ (again uniform for small ε and ρ) such that for all $\|\phi\|_X \leqslant c_3$ we have that $D\mathsf{G}_{\varepsilon,\rho}[\phi] : X \to X$ is an invertible map.

The key feature is that the second item allows for the norm of the derivative to become unboundedly large in the singular limit of small parameters. What is important is that the size of the neighborhood in which the derivative's inverse exists is not collapsing. The role of the nonlinear a priori estimate is to ensure that the iterates in Newton's method do not leave a bounded region, thus facilitating the convergence. It is a pleasant fact that the two hypotheses of our uniform local surjectivity theorem when applied to the maps $\mathsf{G}_{\varepsilon,\rho}$ (1.3.24) are satisfied as consequences of the nondegeneracy condition (1.3.4) from Definition 1.2.

After we have constructed the uniform local inverses of the maps $\mathsf{G}_{\varepsilon,\rho}$, we, roughly speaking, obtain Theorem I by setting $\rho = 0$. On the other hand, Theorem II is obtained by taking the limit $\varepsilon \to 0$ for each fixed $\rho > 0$. Theorem III is ultimately proved by a strategy similar to the one described above, but we introduce another type of perturbation to the base operator equation (1.3.19) by varying the Green's function.

The next subsection is a notation interlude. The plan for the remainder of the paper is as follows. Section 2 is devoted to the proof of Theorem IV, with Sections 2.1 and 2.2 handling the first and second items of this main theorem, respectively. Next, in Section 3, we essentially prove Theorems I, II, and III simultaneously. Section 3.1 witnesses the derivation of a flurry of preliminary tools and estimates. Next, in Section 3.2, we establish the most salient consequences of the nondegeneracy condition of the fourth item of Definition 1.2. After that, in Section 3.3 we construct our abstract quantitative local surjectivity theorem based on Newton's method and then apply it to problem at hand. We conclude Section 3 with the proof of our main theorems in Section 3.4. Finally, Appendix A records some useful analysis supplements.

1.4. **Conventions of notation.** The set of natural numbers is denoted by $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$ and we also denote $\mathbb{N}^+ = \mathbb{N} \setminus \{0\}$. The set of positive real numbers is denoted by $\mathbb{R}^+ = (0, \infty)$. The characteristic function of a set $E_0 \subset E_1$ is $\mathbb{1}_{E_0} : E_1 \to \{0, 1\}$. If E is a subset of any topological space, the closure and interior are denoted $\overline{E}$ and $\mathrm{int}E$, respectively. If A and B are subsets of a topological space then $A \Subset B$ means that $\overline{A} \subset \mathrm{int}B$. Let $m \in \mathbb{N}^+$; we say that $E \subset \mathbb{R}^2$ has m-fold dihedral symmetry if

$$x = (x_1, x_2) \in E \text{ if and only if } (x_1, -x_2) \in E \text{ if and only if } \mathbf{R}(2\pi/m)x \in E \tag{1.4.1}$$

where $\mathbf{R}$ is from (1.1.7).

If $\varnothing \neq U \subset \mathbb{R}^2$ is open, $k \in \mathbb{N}$, and $\alpha \in [0, 1]$ we let $C^{k,\alpha}(U)$ denote the vector space of k-times differentiable functions that, together with their first k derivatives, are locally α-Hölder continuous in U. If U is bounded, we shall also consider the Banach subspace of functions which are $C^{k,\alpha}$ up to the boundary:

$$C^{k,\alpha}(\overline{U}) = \{f \in C^{k,\alpha}(U) \ : \ \|f\|_{C^{k,\alpha}(\overline{U})} < \infty\} \tag{1.4.2}$$

where

$$\|f\|_{C^{k,\alpha}(\overline{U})} = \sum_{j=0}^{k} \Big(\sup_{x \in U} |\nabla^j f(x)| + \sup_{x \neq \widetilde{x}} \frac{|\nabla^j f(x) - \nabla^j f(\widetilde{x})|}{|x - \widetilde{x}|^\alpha} \Big) \tag{1.4.3}$$

When $\alpha = 0$, we write $C^k(U)$ and $C^k(\overline{U})$ in place of $C^{k,0}(U)$ and $C^{k,0}(\overline{U})$, respectively.

For a bounded open set $U \subset \mathbb{R}^2$ the collection of logarithmic-Lipschitz functions is the Banach space

$$\mathrm{LL}(\overline{U}) = \{f \in C^0(\overline{U}) \ : \ \|f\|_{\mathrm{LL}(\overline{U})} < \infty\}, \ \|f\|_{\mathrm{LL}(\overline{U})} = \|f\|_{C^0(\overline{U})} + \sup_{\substack{x,y \in U \\ x \neq y}} \frac{|f(x) - f(y)|}{|x - y|(1 + |\log|x - y||)}. \tag{1.4.4}$$

We shall also define the related space

$$\mathrm{LL}^1(\overline{U}) = \{f \in C^1(\overline{U}) \ : \ \nabla f \in \mathrm{LL}(\overline{U}; \mathbb{R}^2)\} \text{ with the norm } \|f\|_{\mathrm{LL}^1(\overline{U})} = \|f\|_{C^1(\overline{U})} + \|\nabla f\|_{\mathrm{LL}(\overline{U})}. \tag{1.4.5}$$

If $U \subset \mathbb{R}^2$ has m-fold dihedral symmetry and $X(U)$ is a Banach space of functions defined on U, we let $X_{(m)}(U)$ denote the closed subspace of functions having m-fold dihedral symmetry, precisely:

$$X_{(m)}(U) = \{f \in X(U) \ : \ \forall\, x = (x_1, x_2) \in U, \ f(x) = f(x_1, -x_2) = f(\mathbf{R}(2\pi/m)x)\}. \tag{1.4.6}$$

2. Existence of admissible states

In this subsection we establish the existence of admissible steady rotating vortex patches that satisfy Definition 1.1. The lion's share of the work is, perhaps unsurprisingly, devoted to checking that the fourth item of the aforementioned definition – the nondegeneracy condition – holds. In Section 2.1 we consider the Kirchhoff elliptical vortices and most of the verification is explicit computation. In contrast, the content of Section 2.2, which addresses m-fold symmetric perturbations of the circular vortex, is a more involved indirect argument exploiting subtle connections between various equivalent formulations of the steady rotating patch equations.

2.1. **Kirchhoff elliptical vortices.** Recall the one parameter families of ellipses $\{E^\xi\}_{\xi\in\mathbb{R}^+}$ and angular velocities $\{\Omega^\xi\}_{\xi\in\mathbb{R}^+}$ defined in (1.1.16). The purpose of the following result is to be more precise about how these continua form the family of Kirchhoff vortex patch solutions and to set up notation for subsequent results.

Lemma 2.1 (Stream functions for Kirchhoff vortices)**.** *For each $\xi\in\mathbb{R}^+$ there exists $c^\xi\in\mathbb{R}$ such that the function $\Psi^\xi\in C^{1,1}_{(2)}(\mathbb{R}^2)$ defined via*

$$\Psi^\xi(x)=(\mathbf{N}*\mathbb{1}_{E^\xi})(x)-\Omega^\xi|x|^2/2-c^\xi \text{ for } x\in\mathbb{R}^2 \tag{2.1.1}$$

obeys the succeeding items.

(1) For all $x\in\partial E^\xi$ we have $\Psi^\xi(x)=0$.

(2) If $\widetilde{\Psi}^\xi:\mathbb{R}^+\times\mathbb{R}\to\mathbb{R}$, defined via

$$\widetilde{\Psi}^\xi(\zeta,\eta)=\Psi^\xi(\operatorname{sech}(\xi)\cosh(\zeta)\cos(\eta),\operatorname{sech}(\xi)\sinh(\zeta)\sin(\eta)) \text{ for all } (\zeta,\eta)\in\mathbb{R}^+\times\mathbb{R}, \tag{2.1.2}$$

denotes Ψ^ξ in elliptical coordinates, then for all $\zeta\in[\xi,\infty)$ and $\eta\in\mathbb{R}$ we have the formula

$$\widetilde{\Psi}^\xi(\zeta,\eta)=\frac{\tanh(\xi)}{4}\Big(2(\zeta-\xi)-\frac{\cosh(2\zeta)-\cosh(2\xi)}{e^{2\xi}}\Big)+\frac{\tanh(\xi)}{4}\Big(\frac{1}{e^{2\zeta}}-\frac{1}{e^{2\xi}}\Big)\cos(2\eta). \tag{2.1.3}$$

(3) For all $\eta\in\mathbb{R}$ we have

$$\partial_1\widetilde{\Psi}^\xi(\xi,\eta)=\frac{\tanh(\xi)}{2e^{2\xi}}(\cosh(2\xi)-\cos(2\eta)). \tag{2.1.4}$$

and hence $\min_{\partial E^\xi}|\nabla\Psi^\xi|>0$.

Proof. First, let $\widetilde{\Psi}_{\mathrm{rot}}:\mathbb{R}^+\times\mathbb{R}\to\mathbb{R}$ denote the elliptical coordinate description of the stream function associated with uniform rotation by the angular velocity Ω^ξ; more precisely

$$\widetilde{\Psi}^\xi_{\mathrm{rot}}(\zeta,\eta)=-\frac{\Omega^\xi}{2}((\operatorname{sech}(\xi)\cosh(\zeta)\cos(\eta))^2+(\operatorname{sech}(\xi)\sinh(\zeta)\sin(\eta))^2)=-\frac{\tanh(\xi)}{4e^{2\xi}}(\cosh(2\zeta)+\cos(2\eta)) \tag{2.1.5}$$

for all $\zeta\in\mathbb{R}^+$ and $\eta\in\mathbb{R}$.

Next, by repeating the computations in Article 159 of Chapter VII of Lamb [45] we obtain a particular $\widetilde{\Psi}^\xi_{\mathrm{ell}}:\mathbb{R}^+\times\mathbb{R}\to\mathbb{R}$ that satisfies for $\xi\leqslant\zeta<\infty$ and $\eta\in\mathbb{R}$ the identity

$$\widetilde{\Psi}^\xi_{\mathrm{ell}}(\zeta,\eta)=\frac{\tanh(\xi)}{4}\Big(2\zeta+\frac{\cos(2\eta)}{e^{2\zeta}}\Big). \tag{2.1.6}$$

Now we define $\widetilde{\Psi}^\xi:\mathbb{R}^+\times\mathbb{R}\to\mathbb{R}$ via

$$\widetilde{\Psi}^\xi(\zeta,\eta)=(\widetilde{\Psi}^\xi_{\mathrm{rot}}+\widetilde{\Psi}^\xi_{\mathrm{ell}})(\zeta,\eta)-(\widetilde{\Psi}^\xi_{\mathrm{rot}}+\widetilde{\Psi}^\xi_{\mathrm{ell}})(\xi,\eta). \tag{2.1.7}$$

After routine computation, it is clear that (2.1.7) agrees with the right hand side of (2.1.3) on the set $\zeta\geqslant\xi$, $\eta\in\mathbb{R}$. The aforementioned construction in Lamb [45] ensures that there exists $\Psi^\xi\in C^{1,1}_{(2)}(\mathbb{R}^2)$ uniquely determined by the property that

$$\Psi^\xi(\operatorname{sech}(\xi)\cosh(\zeta)\cos(\eta),\operatorname{sech}(\xi)\sinh(\zeta)\sin(\eta))=\widetilde{\Psi}^\xi(\zeta,\eta) \text{ for all } \zeta\in\mathbb{R}^+,\ \eta\in[0,2\pi). \tag{2.1.8}$$

Let us now see that Ψ^ξ is the desired relative stream function that satisfies (2.1.1). The previous construction following Lamb [45] guarantees that

$$\Delta\Psi^\xi=\mathbb{1}_{E^\xi}-2\Omega^\xi \text{ in } \mathbb{R}^2,\ \Psi^\xi=0 \text{ on } \partial E^\xi, \text{ and } \lim_{|x|\to\infty}(\nabla\Psi^\xi(x)+\Omega^\xi x)=0. \tag{2.1.9}$$

Therefore, Liouville's theorem for harmonic functions (see, e.g., Theorem 2.16 in Folland [31]) assures us that

$$\nabla(\Psi^\xi+\Omega^\xi|\cdot|^2/2-\mathbf{N}*\mathbb{1}_{E^\xi})=0 \text{ in } \mathbb{R}^2. \tag{2.1.10}$$

The existence of the number c^ξ seen in (2.1.1) is now immediate. $\square$

Armed with sufficiently explicit formulae for the Kirchhoff vortices' relative stream functions, we are in a position to unpack the nondegeneracy condition of the fourth item of Definition 1.2.

Proposition 2.2 (Nondegeneracy on Kirchhoff stream functions, I)**.** *Fix $\xi \in \mathbb{R}^+$ and let $\Sigma^\xi = \partial E^\xi$. The following are equivalent for $\phi \in C^0_{(2)}(\Sigma^\xi)$.*

(1) For all $x \in \Sigma^\xi$ we have
$$\phi(x) + \int_{\Sigma^\xi} \mathbf{N}(x-y)\frac{\phi(y)}{|\nabla\Psi^\xi(y)|}\,\mathrm{d}\mathcal{H}^1(y) = 0 \tag{2.1.11}$$
where $\mathrm{d}\mathcal{H}^1$ denotes integration with respect to the one-dimensional Hausdorff measure.

(2) If we let $\rho \in C^0(2\pi\mathbb{T})$ be defined via $\rho(\eta) = \phi(\cos(\eta), \tanh(\xi)\sin(\eta))$ for $\eta \in 2\pi\mathbb{T}$ then
$$(\sinh(2\xi) - 2e^{2\xi}\log(1+e^{-2\xi}))\mathscr{F}[\rho](0) = 0 \tag{2.1.12}$$
and for $k \in \mathbb{Z}\setminus\{0\}$
$$(|k|\sinh(2\xi) - e^{2\xi}(1+e^{-2\xi|k|}))\mathscr{F}[\rho](k) = 0. \tag{2.1.13}$$
Here $\mathscr{F}[\rho]$ denotes the Fourier transform of ρ.

Proof. The first stage of the proof is to write the right hand integral operator of ϕ in (2.1.11) in terms of ρ. Let $\phi \in C^0_{(2)}(\Sigma^\xi)$ and $\boldsymbol{\phi} \in C^0(\mathbb{R}^2)$ be such that $\phi = \boldsymbol{\phi}$ on Σ^ξ. Such a $\boldsymbol{\phi}$ exists according to the Tietze extension theorem; see, e.g. Theorem 4.16 in Folland [32]. By using the coarea formula (see, e.g., Theorem 3.2.11 in [30] or Theorem 3.13 in [29]), or arguing as in the proof of Proposition 3.8 below, one sees that for every $x \in \Sigma^\xi$
$$\int_{\Sigma^\xi} \mathbf{N}(x-y)\frac{\phi(y)}{|\nabla\Psi^\xi(y)|}\,\mathrm{d}\mathcal{H}^1(y) = \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\int_{U\cap\{|\Psi^\xi|<\varepsilon\}} \mathbf{N}(x-y)\boldsymbol{\phi}(y)\,\mathrm{d}y \tag{2.1.14}$$
where $U \supset \Sigma^\xi$ is a bounded open set with the property that $\{x \in U \;:\; \Psi^\xi(x) = 0\} = \Sigma^\xi$ and $\min_{\overline{U}}|\nabla\Psi^\xi| > 0$ (see the third item of Lemma 2.1).

Now for $\varepsilon \in \mathbb{R}^+$ sufficiently small, we change variables in the right hand side integral (2.1.14) according to the elliptical coordinates map
$$\mathbf{e} : \mathbb{R}^+ \times 2\pi\mathbb{T} \to \mathbb{R}^2 \text{ defined via } \mathbf{e}(\zeta,\eta) = (\operatorname{sech}(\xi)\cosh(\zeta)\cos(\eta), \operatorname{sech}(\xi)\sinh(\zeta)\sin(\eta)) \tag{2.1.15}$$
which has Jacobian
$$\mathbf{J} : \mathbb{R}^+ \times 2\pi\mathbb{T} \to \mathbb{R}^+ \text{ defined via } \mathbf{J}(\zeta,\eta) = \operatorname{sech}^2(\xi)(\cosh(2\zeta) - \cos(2\eta))/2. \tag{2.1.16}$$
Thus, upon writing $x = \mathbf{e}(\xi,\widetilde{\eta})$ for some $\widetilde{\eta} \in 2\pi\mathbb{T}$ we have
$$\begin{aligned}\int_{U\cap\{|\Psi^\xi|<\varepsilon\}} &\mathbf{N}(x-y)\boldsymbol{\phi}(y)\,\mathrm{d}y \\ &= \frac{1}{\varepsilon}\int_0^\infty\int_0^{2\pi} \mathbf{N}(\mathbf{e}(\xi,\widetilde{\eta}) - \mathbf{e}(\zeta,\eta))(\boldsymbol{\phi}\circ\mathbf{e})(\zeta,\eta)(\mathbb{1}_{\mathbf{e}\in U\cap\{|\widetilde{\Psi}^\xi|<\varepsilon\}}\mathbf{J})(\zeta,\eta)\,\mathrm{d}\eta\,\mathrm{d}\zeta,\end{aligned} \tag{2.1.17}$$
with $\widetilde{\Psi}^\xi$ as defined in (2.1.2). Now we pass to the limit as $\varepsilon \to 0$ on the right hand side of (2.1.17) while using an argument similar to the justification of (2.1.14) to obtain
$$\int_{\Sigma^\xi} \mathbf{N}(x-y)\frac{\phi(y)}{|\nabla\Psi^\xi(y)|}\,\mathrm{d}\mathcal{H}^1(y) = \int_0^{2\pi} \mathbf{N}(\mathbf{e}(\xi,\widetilde{\eta}) - \mathbf{e}(\xi,\eta))\rho(\eta)\frac{\mathbf{J}(\xi,\eta)}{\partial_1\widetilde{\Psi}^\xi(\xi,\eta)}\,\mathrm{d}\eta \tag{2.1.18}$$
with $\rho = \phi\circ\mathbf{e}(\xi,\cdot)$. Note that in deriving (2.1.18), we have tacitly used the identity $|\nabla\widetilde{\Psi}^\xi(\xi,\eta)| = \partial_1\widetilde{\Psi}^\xi(\xi,\eta)$ on $\{\widetilde{\Psi}^\xi = 0\} \cap U$, which is a consequence of Lemma 2.1. Now we have the following key equality, that follows from (2.1.4) and (2.1.16):
$$\frac{\mathbf{J}(\xi,\eta)}{\partial_1\widetilde{\Psi}^\xi(\xi,\eta)} = \frac{e^{2\xi}\operatorname{sech}^2(\xi)}{\tanh(\xi)} = \frac{2e^{2\xi}}{\sinh(2\xi)}. \tag{2.1.19}$$

Notice that crucially the dependence on η drops out of the right hand side.

Before substituting (2.1.19) into (2.1.18), we turn our attention to simplification of the kernel expression involving $\mathbf{N}$. One computes that

$$|\mathbf{e}(\xi,\widetilde{\eta})-\mathbf{e}(\xi,\eta)|^2 = 4\operatorname{sech}^2(\xi)\sin^2((\eta-\widetilde{\eta})/2)(\sinh^2(\xi)+\sin^2((\eta+\widetilde{\eta})/2)) \tag{2.1.20}$$

and hence for the kernels

$$K_1(\eta) = \log\sin^2(\eta/2)/4\pi \text{ and } K_2(\eta) = \log(\sinh^2(\xi)+\sin^2(\eta/2))/4\pi \tag{2.1.21}$$

we have

$$\mathbf{N}(\mathbf{e}(\xi,\widetilde{\eta})-\mathbf{e}(\xi,\eta)) = \log(2\operatorname{sech}(\xi))/2\pi + K_1(\eta-\widetilde{\eta}) + K_2(\eta+\widetilde{\eta}). \tag{2.1.22}$$

Synthesizing (2.1.18), (2.1.19), and (2.1.22) gives us our sought after relationship:

$$\int_{\Sigma^\xi}\mathbf{N}(x-y)\frac{\phi(y)}{|\nabla\Psi^\xi(y)|}\,\mathrm{d}\mathcal{H}^1(y) = \frac{2e^{2\xi}}{\sinh(2\xi)}\int_0^{2\pi}\Big(\frac{1}{2\pi}\log(2\operatorname{sech}(\xi)) + K_1(\eta-\widetilde{\eta}) + K_2(\eta+\widetilde{\eta})\Big)\rho(\eta)\,\mathrm{d}\eta. \tag{2.1.23}$$

The second stage of the proof is to analyze the action in Fourier space of the following integral operators:

$$(O_1\rho)(\widetilde{\eta}) = \int_0^{2\pi}K_1(\eta-\widetilde{\eta})\rho(\eta)\,\mathrm{d}\eta \text{ and } (O_2\rho)(\widetilde{\eta}) = \int_0^{2\pi}K_2(\eta+\widetilde{\eta})\rho(\eta)\,\mathrm{d}\eta, \tag{2.1.24}$$

defined for $\rho\in C^0(2\pi\mathbb{T})$ and $\widetilde{\eta}\in 2\pi\mathbb{T}$. The following identities for $k\in\mathbb{Z}$ are consequences of Lemmas 3.3 and 3.4 in Castro, Córdoba, and Gómez-Serrano [10]:

$$\int_0^{2\pi}K_1(\eta)e^{-\mathrm{i}k\eta}\,\mathrm{d}\eta = -\frac{1}{2}\begin{cases}\log 4 & \text{if } k=0,\\ |k|^{-1} & \text{if } k\neq 0,\end{cases} \text{ and } \int_0^{2\pi}K_2(\eta)e^{-\mathrm{i}k\eta}\,\mathrm{d}\eta = -\frac{1}{2}\begin{cases}\log 4-2\xi & \text{if } k=0,\\ e^{-2|k|\xi}|k|^{-1} & \text{if } k\neq 0.\end{cases} \tag{2.1.25}$$

Therefore, for any $\rho\in C^0(2\pi\mathbb{T})$ and $k\in\mathbb{Z}$ we can compute

$$\mathscr{F}[O_1\rho](k) = -\frac{\mathscr{F}[\rho](k)}{2}\begin{cases}\log 4 & \text{if } k=0,\\ |k|^{-1} & \text{if } k\neq 0,\end{cases} \text{ and } \mathscr{F}[O_2\rho](k) = -\frac{\overline{\mathscr{F}[\rho](k)}}{2}\begin{cases}\log 4-2\xi & \text{if } k=0,\\ e^{-2|k|\xi}|k|^{-1} & \text{if } k\neq 0.\end{cases} \tag{2.1.26}$$

We now have all of the tools we need to prove the stated equivalence. Identity (2.1.11) holds for a $\phi\in C^0(\Sigma^\xi)$ if and only if for $\rho=\phi\circ\mathbf{e}(\xi,\cdot)$ we have

$$\rho + 2e^{2\xi}(\log(2\operatorname{sech}(\xi))\mathscr{F}[\rho](0) + O_1\rho + O_2\rho)/\sinh(2\xi) = 0. \tag{2.1.27}$$

Then equality (2.1.27) holds if and only if each Fourier coefficient of the left hand side vanishes. Since ϕ is e_2-symmetric, we deduce $\rho(\eta)=\rho(-\eta)$ for all $\eta\in 2\pi\mathbb{T}$ and $\mathscr{F}[\rho](k)=\mathrm{Re}\mathscr{F}[\rho](|k|)$ for all $k\in\mathbb{Z}$. We then may use the previous computations to see that this latter condition is equivalent to the desired identities (2.1.12) and (2.1.13). □

Corollary 2.3 (Nondegeneracy on Kirchhoff stream functions, II)**.** *There exists an at most countable set $\mathcal{N}\subset\mathbb{R}^+$ such that for all $\xi\in\mathbb{R}^+\setminus\mathcal{N}$ the only solution $\phi\in C^0_{(2)}(\Sigma_\xi)$ to equation (2.1.11) is $\phi=0$.*

Proof. As suggested by the second item of Proposition 2.2, we consider the sequence of functions $\{\lambda_k\}_{k\in\mathbb{N}}$ defined for $\xi\in\mathbb{R}^+$ via

$$\lambda_0(\xi) = \sinh(2\xi) - 2e^{2\xi}\log(1+e^{-2\xi}) \text{ and for } k>0,\ \lambda_k(\xi) = k\sinh(2\xi) - e^{2\xi}(1+e^{-2k\xi}). \tag{2.1.28}$$

We deduce from Proposition 2.2 that for a fixed $\xi\in\mathbb{R}^+$ there exists $\phi\in C^0_{(2)}(\Sigma^\xi)\setminus\{0\}$ satisfying (2.1.11) if and only if there exists $k\in\mathbb{N}$ with $\lambda_k(\xi)=0$. Thus the result will follow as soon as we show that for each $k\in\mathbb{N}$ there exists at most one $\xi\in\mathbb{R}^+$ with $\lambda_k(\xi)=0$.

For $k=0$ we show that λ_0 has a unique positive zero. This follows from the observations that 1) $\lim_{\xi\to 0}\lambda_0(\xi)=-2\log 2$, 2) $\lim_{\xi\to\infty}\lambda_0(\xi)=\infty$, and 3) $\lambda_0'(\xi)>0$.

For $k\in\{1,2\}$, we show that λ_k has no zeros. For $k=1$, we have $\lim_{\xi\to 0}\lambda_1(\xi)=-2$ and $\lambda_1'(\xi)<0$ for all $\xi\in\mathbb{R}^+$ and hence $\lambda_1(\xi)<-2$ for all $\xi\in\mathbb{R}^+$. For $k=2$, we calculate that $\lambda_2(\xi)<0$ for all $\xi\in\mathbb{R}^+$.

For $\mathbb{N} \ni k \geqslant 3$, we show that λ_k has a unique positive zero. This follows from the observations: 1) $\lim_{\xi\to 0} \lambda_k(\xi) = -2$, 2) $\lim_{\xi\to\infty} \lambda_k(\xi) = \infty$, and 3) $\lambda_k'(\xi) > 0$ for all $\xi \in \mathbb{R}^+$. □

Synthesizing the previous calculations gives us the main result of this subsection.

Theorem 2.4 (Admissibility of Kirchhoff elliptical vortices)**.** *Let $\mathcal{N} \subset \mathbb{R}^+$ be the countable set from Corollary 2.3. For all $\xi \in \mathbb{R}^+ \setminus \mathcal{N}$ the Kirchhoff elliptical vortex defined in* (1.1.16) *and* (1.1.17) *is an admissible steady rotating vortex patch in the sense of Definition 1.2.*

Proof. To verify Definition 1.2 we take $\xi \in \mathbb{R}^+ \setminus \mathcal{N}$, $\Psi = \Psi^\xi$, $\Omega = \Omega^\xi$, $c = c^\xi$, and $\widetilde{U} = E^\xi$. The satisfaction of the first item of the definition is immediate. The second and third items follow from Lemma 2.1. The final item is a consequence of Corollary 2.3. □

2.2. **Perturbations of the Rankine vortex.** We now turn our focus to the verification of the admissibility of the steady rotating patches with discrete rotation symmetry, locally bifurcating from the Rankine vortex. We start by recalling the set-up of their construction, following Burbea [8], Hmidi and Mateu [40], and Hassainia, Masmoudi, and Wheeler [36].

Let D be a bounded, simply connected domain corresponding to a rotating vortex patch solution, and denote by $\mathbb{D} = B(0,1)$ the open unit-disk with boundary $\mathbb{S}^1 = \partial\mathbb{D}$. Assuming that the boundary ∂D is of $C^{k,\alpha}$ regularity ($k \in \mathbb{N}^+, \alpha \in (0,1)$), we infer that a conformal mapping $\Phi : \mathbb{C} \setminus \overline{\mathbb{D}} \to \mathbb{C} \setminus \overline{D}$ extends to a $C^{k,\alpha}$ mapping $\Phi : \mathbb{C} \setminus \mathbb{D} \to \mathbb{C} \setminus D$, such that $\Phi : \mathbb{S}^1 \to \partial D$ is a $C^{k,\alpha}$ parametrization of the boundary of the patch (see, e.g., Theorems 3.5 and 3.6 in Pommerenke [53]). Interpreting the gradient of the relative stream function as a complex scalar under the usual identification $\nabla\Psi = \partial_1\Psi + \mathrm{i}\partial_2\Psi$, equation (1.1.10) on the boundary ∂D of the patch reads

$$-\overline{\nabla\Psi}(\Phi(w)) = \Omega\overline{\Phi(w)} + \frac{1}{4\pi\mathrm{i}}\int_{\mathbb{S}^1} \frac{\overline{\Phi(\tau)} - \overline{\Phi(w)}}{\Phi(\tau) - \Phi(w)}\Phi'(\tau)\,\mathrm{d}\tau, \tag{2.2.1}$$

for all $w \in \mathbb{S}^1$. On the other hand, since $\Psi(\Phi(w)) = 0$ for all $w \in \mathbb{S}^1$ and $\mathrm{i}w\Phi'(w)$ is the tangent vector at $\Phi(w)$, we have that

$$\mathrm{Re}\{\overline{\nabla\Psi}(\Phi(w))\mathrm{i}w\Phi'(w)\} = 0, \tag{2.2.2}$$

which when combined with (2.2.1) implies

$$\mathrm{Im}\Big\{\Big(\Omega\overline{\Phi(w)} + \frac{1}{4\pi\mathrm{i}}\int_{\mathbb{S}^1} \frac{\overline{\Phi(\tau)} - \overline{\Phi(w)}}{\Phi(\tau) - \Phi(w)}\Phi'(\tau)\,\mathrm{d}\tau\Big)w\Phi'(w)\Big\} = 0. \tag{2.2.3}$$

This is a version of the Burbea map, first introduced in [8]. Following Hassainia, Masmoudi, and Wheeler [36], we condense the notation in (2.2.3) in terms of the Cauchy integral operator

$$\mathcal{C}(\Phi) : f \mapsto \frac{1}{2\pi\mathrm{i}}\int_{\mathbb{S}^1} \frac{f(\tau) - f(w)}{\Phi(\tau) - \Phi(w)}\Phi'(\tau)\,\mathrm{d}\tau. \tag{2.2.4}$$

Equation (2.2.3), then, becomes

$$\mathrm{Im}\{(\Omega\overline{\Phi} + \mathcal{C}(\Phi)\overline{\Phi}/2)w\Phi'\} = 0. \tag{2.2.5}$$

It is not difficult to see that, for all $\Omega \in \mathbb{R}$, $\Phi(z) = z$ solves the equation above on $\mathbb{S}^1$.

In the following definition, we instantiate the Burbea map on perturbations around the Rankine vortex that satisfy the reflection symmetry $\overline{\phi(w)} = \phi(\overline{w})$.

Definition 2.5 (Burbea map around the Rankine vortex)**.** *Let $k \geqslant 1$ and $\alpha \in (0,1)$. Consider the Banach spaces*

$$X^{k,\alpha} = \Big\{\phi \in C^{k,\alpha}(\mathbb{S}^1;\mathbb{C}) \;:\; \phi(w) = \sum_{n=0}^{\infty} a_n\overline{w}^n; a_n \in \mathbb{R}\Big\} \tag{2.2.6}$$

and

$$Y^{k-1,\alpha} = \{f \in C^{k-1,\alpha}(\mathbb{S}^1;\mathbb{R}) \;:\; f(\overline{w}) = -f(w)\}, \tag{2.2.7}$$

and the open subset $\varnothing \neq U^{k,\alpha} \subset X^{k,\alpha}$ *defined by*

$$U^{k,\alpha} = \Big\{\phi \in X^{k,\alpha} \ : \ \inf_{\tau\neq w}\Big|1 + \frac{\phi(\tau)-\phi(w)}{\tau - w}\Big| > 0\Big\}. \tag{2.2.8}$$

We define the Burbea map around the Rankine vortex $\mathcal{F} : U^{k,\alpha} \times \mathbb{R} \to Y^{k-1,\alpha}$ *by*

$$\mathcal{F}(\phi, \Omega) = \mathrm{Im}\{(\Omega(\overline{w} + \overline{\phi}) + \mathcal{C}(w+\phi)(\overline{w}+\overline{\phi})/2)w(1+\phi')\}. \tag{2.2.9}$$

In the previous definition, the space $X^{k,\alpha}$ does not encode any discrete rotation symmetry for the boundary of the patch. We remark, however, that the spaces can be modified in a way that keeps track of this information.

Remark 2.6. *The Burbea map considered in Definition 2.5 is well-defined and analytic thanks to the analyticity of the Cauchy integral operator on* $w + U^{k,\alpha}$ *(see* [46]*; also Lemma 2.6 of* [36] *for a very similar setting to the one we consider here). Moreover, given* $m \in \mathbb{N}^+$*, the spaces* $X^{k,\alpha}$ *and* $Y^{k-1,\alpha}$*, and the open set* $U^{k,\alpha}$ *can be replaced by*

$$X^{k,\alpha}_m = \Big\{\phi \in X^{k,\alpha} \ : \ \phi(w) = \sum_{n=1}^{\infty} a_n \overline{w}^{nm-1}\Big\}, \ Y^{k-1,\alpha}_m = \{f \in Y^{k-1,\alpha} \ : \ f(e^{2\pi \mathrm{i}/m}w) = f(w)\} \tag{2.2.10}$$

and $U^{k,\alpha}_m = U^{k,\alpha} \cap X^{k,\alpha}_m$*, respectively (see Section 2.2 in* [36] *or Section 3.7 of* [40]*).*

We now state the main existence result for nontrivial steady rotating vortex patches with dihedral symmetry. Most of the content of the following theorem is already proven in the literature in various forms, the closest to the formulation below being that of [36]. The novelty consists solely in the triviality of the kernel of $D_1\mathcal{F}$ on the bifurcating curve, locally near the bifurcation point.

Theorem 2.7 (Bifurcations from the Rankine vortex)**.** *For* $\mathbb{N} \ni m \geqslant 2$*, let* $\Omega_m = (m-1)/2m$*. There exists* $\epsilon, \widetilde{\epsilon} > 0$ *with* $\widetilde{\epsilon} \leqslant \epsilon$*, an open set* $(0, \Omega_m) \in V \subset U^{k,\alpha} \times \mathbb{R}$*, and analytic functions* $(\phi, \lambda) : (-\epsilon, \epsilon) \to V$ *such that the following hold.*

(1) For the map $\mathcal{F}$ *the curve* (ϕ, λ) *is the unique bifurcating branch of solutions in* V*:*

$$(V \cap \mathcal{F}^{-1}\{0\}) \setminus (\{0\} \times \mathbb{R}) = \{(\phi(s), \lambda(s)) \ : \ 0 < |s| < \epsilon\}. \tag{2.2.11}$$

(2) The branch bifurcates from the Rankine vortex (i.e. $\phi(0) = 0$*) through the* m*-fold symmetric class (i.e.* $\phi(s) \in X^{k,\alpha}_m$ *for all* $s \in (-\epsilon, \epsilon)$*) and satisfies* $\phi'(0) = \overline{w}^{m-1}$*.*

(3) The bifurcation is of subcritical pitchfork type; in other words $\lambda(0) = \Omega_m$*,* $\lambda'(0) = 0$*,* $\lambda''(0) = -(m-1)/2 < 0$*. Consequently, for all* $0 < |s| \leqslant \widetilde{\epsilon}$*,* $D_1\mathcal{F}(\phi(s), \lambda(s)) : X^{k,\alpha} \to Y^{k-1,\alpha}$ *has trivial kernel and* $\lambda(s) < \Omega_m < 1/2$*.*

Proof. The proof is based on an application of the Crandall-Rabinowitz Theorem [18]. With the aim of verifying its assumptions, we begin by computing the Fréchet derivative of $\mathcal{F}$:

$$D_1\mathcal{F}(\phi, \Omega)[\xi] = \mathrm{Im}\{(\Omega\overline{\xi} + D\mathcal{C}(\Phi)[\xi]\overline{\Phi}/2 + \mathcal{C}(\Phi)\overline{\xi}/2)w\Phi' + (\Omega\overline{\Phi} + \mathcal{C}(\Phi)\overline{\Phi}/2)w\xi'\}, \tag{2.2.12}$$

where we have used the notation $\Phi(w) = w + \phi(w)$ and

$$D\mathcal{C}(\Phi)[\xi]f(w) = \frac{1}{2\pi \mathrm{i}}\int_{\mathbb{S}^1} \frac{f(\tau)-f(w)}{\Phi(\tau)-\Phi(w)}\xi'(\tau)\,\mathrm{d}\tau - \frac{1}{2\pi \mathrm{i}}\int_{\mathbb{S}^1}\frac{(f(\tau)-f(w))(\xi(\tau)-\xi(w))}{(\Phi(\tau)-\Phi(w))^2}\Phi'(\tau)\,\mathrm{d}\tau. \tag{2.2.13}$$

When we consider $\phi = 0$, simple computations (see, for instance, the proof of Lemma 3.2[1] of [36]) show that $D\mathcal{C}(w)[\xi]\overline{w} = \mathcal{C}(w)\overline{\xi} = 0$, and $\mathcal{C}(w)\overline{w} = -\overline{w}$. It follows, then, that

$$D_1\mathcal{F}(0, \Omega)[\xi] = \mathrm{Im}\{(\Omega - 1/2)\xi' + \Omega w\overline{\xi}\}. \tag{2.2.14}$$

By using the series expansion of $\xi \in X^{k,\alpha}$, $\xi(w) = \sum_{n=0}^{\infty} a_n \overline{w}^n$, we find

$$D_1\mathcal{F}(0, \Omega)[\xi] = \frac{1}{2\mathrm{i}}\sum_{n=0}^{\infty} a_n\Big(\Omega + n\Big(\Omega - \frac{1}{2}\Big)\Big)(w^{n+1} - \overline{w}^{n+1}). \tag{2.2.15}$$

[1]Note that the statement of Lemma 3.2. in [36] does not precisely match the form we claim in (2.2.14). This is due to a typo in equation (3.1) in the statement of the cited lemma. Following their proof, however, yields the claimed result.

When $\Omega = \Omega_m$, it is clear that

$$\ker D_1\mathcal{F}(0,\Omega_m) = \mathrm{span}\{\overline{w}^{m-1}\} \in X_m^{k,\alpha}. \tag{2.2.16}$$

Moreover, the range of $D_1\mathcal{F}(0,\Omega_m)$ is of codimension one both when viewed as a mapping from $X^{k,\alpha} \to Y^{k-1,\alpha}$, and when viewed as a mapping from $X_m^{k,\alpha} \to Y_m^{k-1,\alpha}$. Finally, the following transversality condition holds:

$$D_1D_2\mathcal{F}(0,\Omega_m)[\overline{w}^{m-1}] = m(w^m - \overline{w}^m)/2\mathrm{i} \notin \mathrm{ran} D_1\mathcal{F}(0,\Omega_m). \tag{2.2.17}$$

Thus the assumptions of Theorem in 1.7 in Crandall and Rabinowitz [18] are satisfied; however, here we employ the analytic version given by Theorem 8.3.1 in Buffoni and Toland [7]. The latter result implies the existence of $\epsilon > 0$ and the analytic functions (ϕ,λ) claimed in the statement of the theorem such that the first item above holds, $\phi(0) = 0$, $\phi'(0) = \overline{w}^{m-1}$, and $\lambda(0) = \Omega_m$. Moreover, the fact that $\phi(s) \in X_m^{k,\alpha}$ follows from the application of the Crandall-Rabinowitz Theorem both to $\mathcal{F} : U^{k,\alpha} \times \mathbb{R} \to Y^{k-1,\alpha}$ and to its restriction $\mathcal{F} : U_m^{k,\alpha} \times \mathbb{R} \to Y_m^{k-1,\alpha}$, together with the uniqueness of the bifurcating branch. It remains, then, to verify the third item.

The fact that $\lambda'(0) = 0$ (i.e. that the bifurcation is pitchfork) can be readily obtained from symmetry considerations (see the proof of Theorem 3.3. in [36]). Nevertheless, we obtain this result here as well, through direct computation, in the process of proving the subcriticality condition $\lambda''(0) < 0$. We claim that

$$0 = D_1^2\mathcal{F}(0,\Omega_m)[\phi'(0),\phi'(0)] + 2D_1D_2\mathcal{F}(0,\Omega_m)[\phi'(0)]\lambda'(0) + D_1\mathcal{F}(0,\Omega_m)[\phi''(0)], \tag{2.2.18}$$

and, assuming that $\lambda'(0) = 0$, it holds that

$$\begin{aligned} 0 = D_1^3\mathcal{F}(0,\Omega_m)&[\phi'(0),\phi'(0),\phi'(0)] + 3D_1^2\mathcal{F}(0,\Omega_m)[\phi'(0),\phi''(0)] \\ &+ 3D_1D_2\mathcal{F}(0,\Omega_m)[\phi'(0)]\lambda''(0) + D_1\mathcal{F}(0,\Omega_m)[\phi'''(0)]. \end{aligned} \tag{2.2.19}$$

By differentiating with respect to s in $\mathcal{F}(\phi(s),\lambda(s)) = 0$, we obtain

$$D_1\mathcal{F}(\phi(s),\lambda(s))[\phi'(s)] + D_2\mathcal{F}(\phi(s),\lambda(s))\lambda'(s) = 0. \tag{2.2.20}$$

Differentiating once more in s we obtain

$$\begin{aligned} 0 = D_1^2\mathcal{F}(\phi(s),\lambda(s))&[\phi'(s),\phi'(s)] + 2D_1D_2\mathcal{F}(\phi(s),\lambda(s))[\phi'(s)]\lambda'(s) + D_1\mathcal{F}(\phi(s),\lambda(s))[\phi''(s)] \\ &+ D_2^2\mathcal{F}(\phi(s),\lambda(s))(\lambda'(s))^2 + D_2\mathcal{F}(\phi(s),\lambda(s))\lambda''(s). \end{aligned} \tag{2.2.21}$$

On the other hand, $D_2^k\mathcal{F}(0,\Omega) = 0$ for all $\Omega \in \mathbb{R}$ and $k \in \mathbb{N}$. Consequently, evaluating the previous expression at $s = 0$ yields equation (2.2.18). To obtain (2.2.19), we differentiate the above once more in s and then evaluate at $s = 0$, while using the assumption $\lambda'(0) = 0$.

We now claim that

$$D_1^2\mathcal{F}(0,\Omega_m)[\overline{w}^{m-1},\overline{w}^{m-1}] = 0 \tag{2.2.22}$$

and that

$$D_1^3\mathcal{F}(0,\Omega_m)[\overline{w}^{m-1},\overline{w}^{m-1},\overline{w}^{m-1}] = 3(m-1)m(w^m - \overline{w}^m)/4\mathrm{i} \notin \mathrm{ran} D_1\mathcal{F}(0,\Omega_m) \tag{2.2.23}$$

Assuming equations (2.2.22) and (2.2.23) hold for the moment, note that (2.2.18) becomes

$$2D_1D_2\mathcal{F}(0,\Omega_m)[\overline{w}^{m-1}]\lambda'(0) + D_1\mathcal{F}(0,\Omega_m)[\phi''(0)] = 0. \tag{2.2.24}$$

By virtue of the transversality condition (2.2.17), we obtain $\lambda'(0) = 0$ and, consequently, that $\phi''(0) \in \ker D_1\mathcal{F}(0,\Omega_m)$, which implies $\phi''(0) = a\overline{w}^{m-1}$ for some $a \in \mathbb{R}$. Knowing now that $\lambda'(0) = 0$, equation (2.2.19) holds unconditionally, and, under the assumptions (2.2.22) and (2.2.23), this equation becomes

$$D_1^3\mathcal{F}(0,\Omega_m)[\overline{w}^{m-1},\overline{w}^{m-1},\overline{w}^{m-1}] + 3D_1D_2\mathcal{F}(0,\Omega_m)\lambda''(0) + D_1\mathcal{F}(0,\Omega_m)[\phi'''(0)] = 0. \tag{2.2.25}$$

Since the first two terms above are not in $\mathrm{ran} D_1\mathcal{F}(0,\Omega_m)$, we obtain $\lambda''(0) = -(m-1)/2$, as desired.

With the aim of proving relations (2.2.22) and (2.2.23), we take the second Fréchet derivative in (2.2.12):

$$D_1^2\mathcal{F}(\phi,\Omega)[\xi,\xi] = \mathrm{Im}\{(D^2\mathcal{C}(\Phi)[\xi,\xi]\overline{\Phi}/2 + D\mathcal{C}(\Phi)[\xi]\overline{\xi})w\Phi' + (2\Omega\overline{\xi} + D\mathcal{C}(\Phi)[\xi]\overline{\Phi} + \mathcal{C}(\Phi)\overline{\xi})w\xi'\} \quad (2.2.26)$$

where

$$D^2\mathcal{C}(\Phi)[\xi,\xi]f(w) = \frac{2}{2\pi\mathsf{i}}\int_{\mathbb{S}^1}\frac{(f(\tau)-f(w))(\xi(\tau)-\xi(w))^2}{(\Phi(\tau)-\Phi(w))^3}\Phi'(\tau)\,\mathrm{d}\tau - \frac{2}{2\pi\mathsf{i}}\int_{\mathbb{S}^1}\frac{(f(\tau)-f(w))(\xi(\tau)-\xi(w))}{(\Phi(\tau)-\Phi(w))^2}\xi'(\tau)\,\mathrm{d}\tau. \quad (2.2.27)$$

Restricting now to the case $\Phi(w) = w$ and $\xi(w) = \overline{w}^{m-1}$, we find that

$$\begin{aligned} D^2\mathcal{C}(w)[\overline{w}^{m-1},\overline{w}^{m-1}]\overline{w} &= \frac{2}{2\pi\mathsf{i}}\int_{\mathbb{S}^1}\frac{(\overline{\tau}-\overline{w})(\overline{\tau}^{m-1}-\overline{w}^{m-1})^2}{(\tau-w)^3}\,\mathrm{d}\tau \\ &\quad + \frac{2(m-1)}{2\pi\mathsf{i}}\int_{\mathbb{S}^1}\frac{(\overline{\tau}-\overline{w})(\overline{\tau}^{m-1}-\overline{w}^{m-1})}{(\tau-w)^2}\frac{1}{\tau^m}\,\mathrm{d}\tau \\ &= -\frac{2}{2\pi\mathsf{i}}\int_{\mathbb{S}^1}\frac{(\tau^{m-2}+...+w^{m-2})^2}{w^{2m-1}\tau^{2m-1}}\,\mathrm{d}\tau + \frac{2(m-1)}{2\pi\mathsf{i}}\int_{\mathbb{S}^1}\frac{\tau^{m-2}+...+w^{m-2}}{w^m\tau^{2m}}\,\mathrm{d}\tau = 0. \end{aligned} \quad (2.2.28)$$

On the other hand,

$$\begin{aligned} D\mathcal{C}(w)[\overline{w}^{m-1}]w^{m-1} &= -\frac{m-1}{2\pi\mathsf{i}}\int_{\mathbb{S}^1}\frac{\tau^{m-1}-w^{m-1}}{\tau-w}\frac{1}{\tau^m}\,\mathrm{d}\tau - \frac{1}{2\pi\mathsf{i}}\int_{\mathbb{S}^1}\frac{(\tau^{m-1}-w^{m-1})(\overline{\tau}^{m-1}-\overline{w}^{m-1})}{(\tau-w)^2}\,\mathrm{d}\tau \\ &= -\frac{m-1}{2\pi\mathsf{i}}\int_{\mathbb{S}^1}\frac{\tau^{m-2}+...+w^{m-2}}{\tau^m}\,\mathrm{d}\tau + \frac{1}{2\pi\mathsf{i}}\int_{\mathbb{S}^1}\frac{(\tau^{m-2}+...+w^{m-2})^2}{\tau^{m-1}w^{m-1}}\,\mathrm{d}\tau = \frac{m-1}{w}. \end{aligned} \quad (2.2.29)$$

We have calculated, then, that

$$D_1^2\mathcal{F}(0,\Omega_m)[\overline{w}^{m-1},\overline{w}^{m-1}] = \mathrm{Im}\{(m-1)(1-2\Omega_m)\} = 0, \quad (2.2.30)$$

and (2.2.22) is proven. To show the validity of (2.2.23), we take the third Fréchet derivative in (2.2.26):

$$\begin{aligned} D_1^3\mathcal{F}(\phi,\Omega)[\xi,\xi,\xi] &= \mathrm{Im}\{(D^3\mathcal{C}(\Phi)[\xi,\xi,\xi]\overline{\Phi}/2 + 3D^2\mathcal{C}(\Phi)[\xi,\xi]\overline{\xi}/2)w\Phi' \\ &\quad + (3D^2\mathcal{C}(\Phi)[\xi,\xi]\overline{\Phi}/2 + 3D\mathcal{C}(\Phi)[\xi]\overline{\xi})w\xi'\}, \end{aligned} \quad (2.2.31)$$

where

$$\begin{aligned} &D^3\mathcal{C}(\Phi)[\xi,\xi,\xi]f(w) = \\ &\frac{6}{2\pi\mathsf{i}}\int_{\mathbb{S}^1}\frac{(f(\tau)-f(w))(\xi(\tau)-\xi(w))^2}{(\Phi(\tau)-\Phi(w))^3}\xi'(\tau)\,\mathrm{d}\tau - \frac{6}{2\pi\mathsf{i}}\int_{\mathbb{S}^1}\frac{(f(\tau)-f(w))(\xi(\tau)-\xi(w))^3}{(\Phi(\tau)-\Phi(w))^4}\Phi'(\tau)\,\mathrm{d}\tau. \end{aligned} \quad (2.2.32)$$

For $\Phi(w) = w$ and $\xi(w) = \overline{w}^{m-1}$, we have

$$\begin{aligned} &D^3\mathcal{C}(w)[\overline{w}^{m-1},\overline{w}^{m-1},\overline{w}^{m-1}]\overline{w} = \\ &\quad -\frac{6(m-1)}{2\pi\mathsf{i}}\int_{\mathbb{S}^1}\frac{(\overline{\tau}-\overline{w})(\overline{\tau}^{m-1}-\overline{w}^{m-1})^2}{(\tau-w)^3}\frac{1}{\tau^m}\,\mathrm{d}\tau - \frac{6}{2\pi\mathsf{i}}\int_{\mathbb{S}^1}\frac{(\overline{\tau}-\overline{w})(\overline{\tau}^{m-1}-\overline{w}^{m-1})^3}{(\tau-w)^4}\,\mathrm{d}\tau \\ &\quad = \frac{6(m-1)}{2\pi\mathsf{i}}\int_{\mathbb{T}}\frac{(\tau^{m-2}+...+w^{m-2})^2}{\tau^{3m-1}w^{2m-1}}\,\mathrm{d}\tau - \frac{6}{2\pi\mathsf{i}}\int_{\mathbb{S}^1}\frac{(\tau^{m-2}+...+w^{m-2})^3}{\tau^{3m-2}w^{3m-2}}\,\mathrm{d}\tau = 0, \end{aligned} \quad (2.2.33)$$

and

$$
\begin{aligned}
&D^2\mathcal{C}(w)[\overline{w}^{m-1},\overline{w}^{m-1}]w^{m-1} = \\
&\frac{2}{2\pi\mathrm{i}}\int_{\mathbb{S}^1}\frac{(\tau^{m-1}-w^{m-1})(\overline{\tau}^{m-1}-\overline{w}^{m-1})^2}{(\tau-w)^3}\,\mathrm{d}\tau+\frac{2(m-1)}{2\pi\mathrm{i}}\int_{\mathbb{S}^1}\frac{(\tau^{m-1}-w^{m-1})(\overline{\tau}^{m-1}-\overline{w}^{m-1})}{(\tau-w)^2}\frac{1}{\tau^m}\,\mathrm{d}\tau \\
&\qquad=\frac{2}{2\pi\mathrm{i}}\int_{\mathbb{S}^1}\frac{(\tau^{m-2}+\dots+w^{m-2})^3}{\tau^{2m-2}w^{2m-2}}\,\mathrm{d}\tau-\frac{2(m-1)}{2\pi\mathrm{i}}\int_{\mathbb{S}^1}\frac{(\tau^{m-2}+\dots+w^{m-2})^2}{\tau^{2m-1}w^{m-1}}\,\mathrm{d}\tau \\
&\qquad=2\sum_{\substack{k_1,k_2,k_3=0\\k_1+k_2+k_3=2m-3}}^{m-2}\frac{w^{3(m-2)-(k_1+k_2+k_3)}}{w^{2m-2}}=\frac{(m-1)(m-2)}{w^{m+1}}. \qquad (2.2.34)
\end{aligned}
$$

By substituting what we have found in (2.2.31), we recover (2.2.23).

It remains to argue that the kernel of $D_1\mathcal{F}(\phi(s),\lambda(s))$ is trivial for $|s|>0$ sufficiently small. Since $\lambda'(0)=0$ and $\lambda''(0)\neq 0$, it is the case that $\lambda'(s)\neq 0$ for all $s\neq 0$ sufficiently close to zero. The conclusion follows by Theorem 1.17 in Crandall and Rabinowitz [18]. ☐

Let $\Phi_m:\mathbb{S}^1\to\mathbb{C}$ now be defined by $\Phi_m(w)=w+\phi(s)(w)$, with $\phi(s)\in X_m^{k,\alpha}$, for some $0<|s|<\widetilde{\epsilon}$ such that $\ker D_1\mathcal{F}(\phi(s),\lambda(s))=\{0\}$ and $\lambda(s)<\Omega_m<1/2$. Since $\phi(s)\in U^{k,\alpha}$, we have that

$$\inf_{\tau\neq w}\left|\frac{\Phi_m(\tau)-\Phi_m(w)}{\tau-w}\right|>0, \qquad (2.2.35)$$

which implies the injectivity of Φ_m. Consequently, $\Phi_m(\mathbb{S}^1)$ is a $C^{k,\alpha}$ simple closed curve bounding a domain D_m. Moreover, since $\phi(s)\in X_m^{k,\alpha}$, we have that

$$\Phi_m(w)=w\Big(1+\sum_{n=1}^{\infty}a_n\overline{w}^{nm}\Big),$$

for some sequence $\{a_n\}_{n=1}^{\infty}\subset\mathbb{R}$. Therefore, Φ_m satisfies the reflection symmetry $\overline{\Phi_m(w)}=\Phi_m(\overline{w})$ and the discrete rotation symmetry

$$\Phi_m(e^{2\pi\mathrm{i}/m}w)=e^{2\pi\mathrm{i}/m}\Phi_m(w). \qquad (2.2.36)$$

These symmetries translate to e_2-reflection and m-fold symmetry, respectively, for the domain D_m. Let

$$\widetilde{\Psi}_m(x)=\mathbf{N}*\mathbb{1}_{D_m}-\lambda_m|x|^2/2 \text{ for } x\in\mathbb{R}^2, \qquad (2.2.37)$$

where we denote $\lambda(s)=\lambda_m$. Then, (2.2.1) holds with $\Psi=\widetilde{\Psi}_m$, $\Phi=\Phi_m$, and $\Omega=\lambda_m$, while (Φ_m,λ_m) solves (2.2.3). Consequently $\widetilde{\Psi}_m(\Phi_m(w))=c_m\in\mathbb{R}$. We then let $\Psi_m=\widetilde{\Psi}_m-c_m$. The tuple $(\Psi_m,\lambda_m,c_m,D_m,m)$ satisfies the first, second, and third items of Definition 1.2. Indeed, the regularity of Ψ_m follows from Proposition 1 in Bertozzi-Constantin [5], the regularity of ∂D_m is satisfied as long as we choose $k\geqslant 2$, the fact that Ψ_m vanishes on ∂D_m holds by construction as the third item, and the condition $\min_{\partial D_m}(\nu\cdot\nabla\Psi_m)>0$ follows from the Hopf boundary point lemma (see, e.g., Lemma 3.4 in Gilbarg and Trudinger [33]), since $\Delta\Psi_m=1-2\lambda_m>0$ in D_m.

The remainder of this subsection is devoted to showing that the fourth item in Definition 1.2 also holds and, thus, that these m-fold symmetric patches are admissible. We first prove a lemma which will aid us in mapping a non-trivial solution to the nondegeneracy condition to a non-trivial kernel of the linearization of the Burbea map.

Lemma 2.8. *Let $k\geqslant 2$ and $\alpha\in(0,1)$, $(\Psi_m,\lambda_m,c_m,D_m,m)$ be one of the tuples described in the preceding paragraphs corresponding to a solution $(\phi(s),\lambda(s))\in U_m^{k,\alpha}\times\mathbb{R}$ given by Theorem 2.7, and $\varphi\in C^0_{(m)}(\partial D_m)$ satisfying*

$$\varphi(x)+\frac{1}{2\pi}\int_{\partial D_m}\log|x-y|\frac{\varphi(y)}{|\nabla\Psi_m(y)|}\,\mathrm{d}\mathcal{H}^1(y)=0, \qquad (2.2.38)$$

for all $x \in \partial D_m$. Then, there exists a solution $\delta \in C^{k-1,\alpha}(\mathbb{S}^1;\mathbb{C})$ to the Riemann-Hilbert problem

$$\operatorname{Re}\{\delta(w)\overline{\nabla\Psi_m}(\Phi_m(w))\} = \varphi(\Phi_m(w)) \text{ for all } w \in \mathbb{S}^1, \tag{2.2.39}$$

such that

$$\delta(w) = \sum_{\mathbb{Z}\ni n\leqslant 1} a_n w^n, \tag{2.2.40}$$

with $\{a_n\}_{\mathbb{Z}\ni n\leqslant 1} \subset \mathbb{R}$.

Proof. We claim first that $\varphi(\Phi_m) \in C^{k,\alpha}(\mathbb{S}^1;\mathbb{R})$. Since (Ψ_m, Φ_m) satisfy (2.2.1), the boundedness of the Cauchy integral operator $\mathcal{C}(\Phi_m) : C^{k,\alpha}(\mathbb{S}^1;\mathbb{C}) \to C^{k,\alpha}(\mathbb{S}^1;\mathbb{C})$ implies that $\nabla\Psi_m(\Phi_m) \in C^{k,\alpha}(\mathbb{S}^1;\mathbb{C})$. Moreover, we can express equation (2.2.38) as

$$\varphi(\Phi_m(e^{\mathrm{i}\theta})) + \frac{1}{2\pi}\int_{2\pi\mathbb{T}} \log|\Phi_m(e^{\mathrm{i}\theta}) - \Phi_m(e^{\mathrm{i}\vartheta})| \frac{\varphi(\Phi_m(e^{\mathrm{i}\vartheta}))}{|\nabla\Psi_m(\Phi_m(e^{\mathrm{i}\vartheta}))|}|\Phi_m'(e^{\mathrm{i}\vartheta})|\,\mathrm{d}\vartheta = 0, \quad \theta \in 2\pi\mathbb{T} \tag{2.2.41}$$

Taking a derivative along the unit circle, we obtain

$$\partial_\theta(\varphi\circ\Phi_m)(e^{\mathrm{i}\theta}) + \operatorname{Re}\Big\{\mathrm{i}e^{\mathrm{i}\theta}\Phi_m'(e^{\mathrm{i}\theta})\frac{1}{2\pi}\mathrm{p.v.}\int_{2\pi\mathbb{T}} \frac{1}{\Phi_m(e^{\mathrm{i}\theta}) - \Phi_m(e^{\mathrm{i}\vartheta})}\frac{\varphi(\Phi_m(e^{\mathrm{i}\vartheta}))}{|\nabla\Psi_m(\Phi_m(e^{\mathrm{i}\vartheta}))|}|\Phi_m'(e^{\mathrm{i}\vartheta})|\,\mathrm{d}\vartheta\Big\} = 0, \tag{2.2.42}$$

and, thus, with $w = e^{\mathrm{i}\theta}$, we have

$$\partial_\theta(\varphi\circ\Phi_m)(w) + \operatorname{Re}\Big\{w\Phi_m'(w)\frac{1}{2\pi}\mathrm{p.v.}\int_{\mathbb{S}^1}\frac{g(\tau)}{\Phi_m(w)-\Phi_m(\tau)}\Phi_m'(\tau)\,\mathrm{d}\tau\Big\} = 0, \tag{2.2.43}$$

where

$$g(\tau) = \frac{\varphi(\Phi_m(\tau))|\Phi_m'(\tau)|}{|\nabla\Psi_m(\Phi_m)|\tau\Phi_m'(\tau)}. \tag{2.2.44}$$

Consequently, we can express the derivative of $\varphi\circ\Phi_m$ in terms of the Cauchy integral operator:

$$\partial_\theta(\varphi\circ\Phi_m)(w) + \operatorname{Im}\{w\Phi_m'(w)\mathcal{C}(\Phi_m)g\} = 0. \tag{2.2.45}$$

By the proof of Proposition 3.7, $\varphi \in C^0(\partial D_m)$ satisfying (2.2.38) implies $\varphi \in \mathrm{LL}(\partial D_m) \subset C^\alpha(\partial D_m)$. On the other hand, if $\varphi\circ\Phi_m \in C^{j,\alpha}(\mathbb{S}^1;\mathbb{R})$, then $g \in C^{\min\{j,k-1\},\alpha}(\mathbb{S}^1;\mathbb{C})$, so (2.2.45) implies that $\varphi\circ\Phi_m \in C^{\min\{j,k-1\}+1,\alpha}(\mathbb{S}^1;\mathbb{R})$. A finite induction argument proves our claim that $\varphi\circ\Phi_m \in C^{k,\alpha}(\mathbb{S}^1;\mathbb{R})$.

Since $\mathcal{F}(\Phi_m(w) - w, \lambda_m) = 0$, we have that $\operatorname{Im}\{\overline{\nabla\Psi_m}(\Phi_m(w))w\Phi_m'(w)\} = 0$. Consequently, there exists a nowhere-vanishing function $\widetilde{F} \in C^{k-1,\alpha}(\mathbb{S}^1;\mathbb{R})$ such that $\overline{\nabla\Psi_m}(\Phi_m(w))w\Phi_m'(w) = \widetilde{F}(w)$. Note that since Ψ_m is reflection invariant, we have that $\overline{\nabla\Psi_m}(w) = \nabla\Psi_m(\overline{w})$, and, therefore, $\widetilde{F}(\overline{w}) = \widetilde{F}(w)$. We write

$$\overline{\nabla\Psi_m}(\Phi_m(w)) = F(w)\overline{\Phi_m'}(w)\overline{w}, \tag{2.2.46}$$

with

$$F(w) = \widetilde{F}(w)/|\Phi_m'(w)|^2, \tag{2.2.47}$$

and note that F is, likewise, non-vanishing and reflection invariant. By the classical theory for the Riemann-Hilbert problem (see, e.g., Muskhelishvili [51] or Theorem A.1. in [36] for a compact statement), since Φ_m' has vanishing winding number, there exists a holomorphic $f : \mathbb{D} \to \mathbb{C}$, which extends as a $C^{k-1,\alpha}$ mapping to the closure of the disk, is reflection invariant $\overline{f(w)} = f(\overline{w})$, and solves

$$\operatorname{Re}\{\Phi_m'(w)f(w)\} = \varphi(\Phi_m(w))/F(w) \text{ for all } w \in \mathbb{S}^1. \tag{2.2.48}$$

We define, then, $\delta \in C^{k-1,\alpha}(\mathbb{S}^1;\mathbb{C})$ by $\delta(w) = w\overline{f(w)}$. We have that

$$\operatorname{Re}\{\delta(w)\overline{\nabla\Psi_m}(\Phi_m(w))\} = F(w)\operatorname{Re}\{\Phi_m'(w)f(w)\} = \varphi(\Phi_m(w)), \tag{2.2.49}$$

as desired. Moreover, since f is holomorphic in the disk and reflection invariant, there exists a sequence $\{b_n\}_{n\in\mathbb{N}} \subset \mathbb{R}$ such that $f(w) = \sum_{n=0}^\infty b_n w^n$. Identity (2.2.40) is then satisfied provided that $a_n = b_{1-n}$. □

At last, we are now ready to prove the main result of this subsection.

Theorem 2.9 (Admissibility of Rankine vortex perturbations)**.** *Let $\mathbb{N} \ni k \geqslant 2$ and $\alpha \in (0,1)$. There exists $0 < \epsilon' \leqslant \widetilde{\epsilon}$, with $\widetilde{\epsilon} > 0$ given by Theorem 2.7, such that the tuple $(\Psi_m, \lambda_m, c_m, D_m, m)$ corresponding, via Theorem 2.7 and the subsequent discussion, to a solution $(\phi(s), \lambda(s)) \in U_m^{k,\alpha} \times \mathbb{R}$, $0 < |s| \leqslant \epsilon'$ is admissible in the sense of Definition 1.2.*

Proof. As already remarked in the discussion following the proof of Theorem 2.7, the first, second, and third items in Definition 1.2 are easily seen to hold for the tuple $(\Psi_m, \lambda_m, c_m, D_m, m)$. For the purpose of obtaining a contradiction, assume that item four fails to hold; namely, that there exists a nontrivial solution $\varphi \in C^0_{(m)}(\partial D_m) \setminus \{0\}$ satisfying equation (2.2.38). Then, Lemma 2.8 applies and yields a function $\delta \in C^{k-1,\alpha}(\mathbb{S}^1; \mathbb{C})$ which satisfies (2.2.39) and, by the assumed non-triviality of φ, does not vanish identically.

Note that, while we have defined $\mathcal{F}$ on the domains $U^{k,\alpha}$ for the purpose of applying the Crandall-Rabinowitz Theorem, the expression (2.2.9) is also well-defined on the Banach space

$$X^{k-1,\alpha}_{\text{ext}} = \Big\{\phi \in C^{k-1,\alpha}(\mathbb{S}^1, \mathbb{C}) \ : \ \phi(w) = \sum_{n=-1}^{\infty} a_n \overline{w}^n; a_n \in \mathbb{R}\Big\}, \tag{2.2.50}$$

of which δ is an element. The Fréchet derivative $D_1\mathcal{F}(\Phi_m - w, \lambda_m)$ exists, then, on $X^{k,\alpha}_{\text{ext}}$ and its expression coincides with the one in (2.2.12). We claim that $D_1\mathcal{F}(\Phi_m - w, \lambda_m)[\delta] = 0$. To this end, we compute

$$\begin{aligned}\partial_\theta \mathrm{Re}\{\delta(e^{\mathrm{i}\theta})\overline{\nabla\Psi_m}(\Phi_m(e^{\mathrm{i}\theta}))\} = \mathrm{Im}\{&e^{\mathrm{i}\theta}\delta'(e^{\mathrm{i}\theta})(\lambda_m\overline{\Phi_m}(e^{\mathrm{i}\theta}) + \mathcal{C}(\Phi_m)\overline{\Phi}_m(e^{\mathrm{i}\theta})/2) \\ &- \mathrm{i}\delta(e^{\mathrm{i}\theta})(\lambda_m\overline{\partial_\theta\Phi_m}(e^{\mathrm{i}\theta}) + \partial_\theta(\mathcal{C}(\Phi_m)\overline{\Phi}_m)(e^{\mathrm{i}\theta})/2)\}.\end{aligned} \tag{2.2.51}$$

For the tangential derivative of the Cauchy integral operator, we have

$$\begin{aligned}\partial_\theta\mathcal{C}(\Phi_m)\overline{\Phi}_m(e^{\mathrm{i}\theta}) &= \frac{1}{2\pi\mathrm{i}}\mathrm{p.v.}\int_{2\pi\mathbb{T}} \frac{\overline{\partial_\theta\Phi_m}(e^{\mathrm{i}\theta})}{\Phi_m(e^{\mathrm{i}\theta}) - \Phi_m(e^{\mathrm{i}\vartheta})}\mathrm{i}e^{\mathrm{i}\vartheta}\Phi_m'(e^{\mathrm{i}\vartheta})\,\mathrm{d}\vartheta \\ &\quad + \frac{\partial_\theta\Phi_m(e^{\mathrm{i}\theta})}{2\pi\mathrm{i}}\mathrm{p.v.}\int_{2\pi\mathbb{T}}(\overline{\Phi}_m(e^{\mathrm{i}\theta}) - \overline{\Phi}_m(e^{\mathrm{i}\vartheta}))\partial_\vartheta\frac{1}{\Phi_m(e^{\mathrm{i}\vartheta}) - \Phi_m(e^{\mathrm{i}\theta})}\,\mathrm{d}\theta \\ &= -\frac{1}{2}\overline{\partial_\theta\Phi_m}(e^{\mathrm{i}\theta}) + \frac{\partial_\theta\Phi_m(e^{\mathrm{i}\theta})}{2\pi\mathrm{i}}\mathrm{p.v.}\int_{2\pi\mathbb{T}}\frac{\overline{\partial_\vartheta\Phi_m}(e^{\mathrm{i}\vartheta})}{\Phi_m(e^{\mathrm{i}\vartheta}) - \Phi_m(e^{\mathrm{i}\theta})}\,\mathrm{d}\vartheta = \partial_\theta\Phi_m(e^{\mathrm{i}\theta})\mathcal{C}(\Phi_m)\Big[\frac{\overline{\partial_\theta\Phi_m}}{\partial_\theta\Phi_m}\Big](e^{\mathrm{i}\theta}).\end{aligned} \tag{2.2.52}$$

Using the notation $w = e^{\mathrm{i}\theta}$, we can express (2.2.51) as

$$\begin{aligned}\partial_\theta\mathrm{Re}\{\delta\overline{\nabla\Psi_m}(\Phi_m)\}(w) = \mathrm{Im}\{&(\lambda_m\overline{\Phi_m}(w) + \mathcal{C}(\Phi_m)\overline{\Phi_m}(w)/2)w\delta'(w) \\ &+ (\lambda_m\overline{\delta}(w) + \delta(w)\mathcal{C}(\Phi_m)[\overline{\partial_\theta\Phi_m}/\partial_\theta\Phi_m]/2)w\Phi_m'(w)\}.\end{aligned} \tag{2.2.53}$$

On the other hand, with $\varphi \circ \Phi_m = \mathrm{Re}\{\delta\overline{\nabla\Psi_m}(\Phi_m)\}$, equation (2.2.44) becomes

$$\begin{aligned}g(w) &= \frac{1}{w\Phi_m'(w)}\mathrm{Re}\Big\{\delta(w)\frac{\overline{\nabla\Psi_m}(\Phi_m(w))}{|\nabla\Psi_m(\Phi_m(w))|}|\Phi_m'(w)|\Big\} \\ &= \frac{1}{w\Phi_m'(w)}\mathrm{sgn}(F)\mathrm{Re}\{\delta(w)\overline{\Phi_m'(w)}\overline{w}\} = \frac{1}{2}\Big(\overline{\delta}(w) - \delta(w)\frac{\overline{\partial_\theta\Phi_m}}{\partial_\theta\Phi_m}(w)\Big),\end{aligned} \tag{2.2.54}$$

where, for the second equality, we have recalled that F is defined in (2.2.47) and used (2.2.46). For the third equality, the fact that $\mathrm{sgn}(F) = 1$ was invoked. The latter can be deduced as follows: from (2.2.46), it follows that, as long as $\|\Phi_m - w\|_{C^1(\mathbb{S}^1)}$ is sufficiently small,

$$\mathrm{sgn}(F) = \mathrm{sgn}(\mathrm{Re}\{\overline{\nabla\Psi_m}w\Phi_m'\}) = \mathrm{sgn}(\mathrm{Re}\{\overline{\nabla\Psi_m}w\}) = \mathrm{sgn}(\partial_{\mathrm{r}}\Psi_m) = 1, \tag{2.2.55}$$

with ∂_{r} denoting the derivative in the radial direction. Consequently, by plugging (2.2.53) and (2.2.54) into (2.2.45), we obtain

$$\begin{aligned}\mathrm{Im}\{&(\lambda_m\overline{\Phi_m}(w) + \mathcal{C}(\Phi_m)\overline{\Phi_m}(w)/2)w\delta'(w) + (\lambda_m\overline{\delta}(w) + \mathcal{C}(\Phi_m)\overline{\delta}(w)/2 - \mathcal{C}(\Phi_m)[\delta\overline{\partial_\theta\Phi_m}/\partial_\theta\Phi_m]/2 \\ &+ \delta(w)\mathcal{C}(\Phi_m)[\overline{\partial_\theta\Phi_m}/\partial_\theta\Phi_m]/2)w\Phi_m'(w)\} = 0.\end{aligned} \tag{2.2.56}$$

Finally, we note that, by integrating the second term by parts in (2.2.13) with $\xi = \delta$ and $f = \overline{\Phi_m}$, we obtain

$$
\begin{aligned}
&D\mathcal{C}(\Phi_m)[\delta]\overline{\Phi_m}(w) = \\
&\quad -\frac{1}{2\pi\mathrm{i}}\int_{2\pi\mathbb{T}} \frac{\overline{\partial_\vartheta\Phi_m}(e^{\mathrm{i}\vartheta})(\delta(e^{\mathrm{i}\vartheta})-\delta(e^{\mathrm{i}\theta}))}{\Phi_m(e^{\mathrm{i}\vartheta})-\Phi_m(e^{\mathrm{i}\theta})}\,\mathrm{d}\vartheta = -\frac{1}{2\pi\mathrm{i}}\int_{\mathbb{S}^1}\frac{\overline{\partial_\theta\Phi_m}}{\partial_\theta\Phi_m}(\tau)\frac{(\delta(\tau)-\delta(w))}{\Phi_m(\tau)-\Phi_m(w)}\Phi_m'(\tau)\,\mathrm{d}\tau \\
&\qquad = \delta(w)\mathcal{C}(\Phi_m)[\overline{\partial_\theta\Phi_m}/\partial_\theta\Phi_m](w) - \mathcal{C}(\Phi_m)[\delta\overline{\partial_\theta\Phi_m}/\partial_\theta\Phi_m](w). \quad (2.2.57)
\end{aligned}
$$

In view of (2.2.12), equations (2.2.56) and (2.2.57) prove our claim that $D_1\mathcal{F}(\Phi_m - w, \lambda_m)[\delta] = 0$.

If the coefficient $a_1 \in \mathbb{R}$ in the series expansion (2.2.40) vanishes, then δ belongs to the smaller space $X^{k-1,\alpha}$ and hence is a non-trivial element in the kernel of $D_1\mathcal{F}(\Phi_m - w, \lambda_m)$, contradicting item three of Theorem 2.7. Assume, then, $a_1 \neq 0$. First, we have that, for all $t \in \mathbb{R}$, it holds that

$$\mathcal{F}(\Phi_m - w + t\Phi_m, \lambda_m) = (1+t)^2\mathcal{F}(\Phi_m - w, \lambda_m) = 0, \quad (2.2.58)$$

and, thus, $\Phi_m \in \ker D_1\mathcal{F}(\Phi_m - w, \lambda_m)$. Consequently, $\delta - a_1\Phi_m \in X^{k-1,\alpha} \cap \ker D_1\mathcal{F}(\Phi_m - w, \lambda_m)$. We will obtain the desired contradiction once we show that $\delta \neq a_1\Phi_m$.

Under the assumption that it is, in fact, the case that $\delta = a_1\Phi_m$, it follows that $\varphi(\Phi_m) = \mathrm{Re}\{\Phi_m\overline{\nabla\Psi_m}\}$ yields a solution to (2.2.38). Switching to a real-variables formulation, this implies that

$$\varphi(x) = x\cdot\nabla\Psi_m(x) \text{ for all } x \in \partial D_m. \quad (2.2.59)$$

Consequently,

$$
\begin{aligned}
\varphi(x) &= -\frac{1}{2\pi}\int_{\partial D_m}\log|x-y|y\cdot\frac{\nabla\Psi_m(y)}{|\nabla\Psi_m(y)|}\,\mathrm{d}\mathcal{H}^1(y) \\
&= -\frac{1}{2\pi}\int_{D_m}\nabla_y\cdot(\log|x-y|y)\,\mathrm{d}y = -\frac{1}{2\pi}\int_{D_m}\frac{y-x}{|y-x|^2}\cdot(y-x+x)\,\mathrm{d}y - \frac{1}{\pi}\int_{D_m}\log|x-y|\,\mathrm{d}y \\
&= -\frac{1}{2\pi}\mathcal{H}^2(D_m) - \frac{1}{2\pi}x\cdot\int_{D_m}\nabla_y\log|x-y|\,\mathrm{d}y - \frac{1}{\pi}\int_{D_m}\log|x-y|\,\mathrm{d}y,
\end{aligned}
$$

where we have applied the divergence theorem, keeping in mind that $\nabla\Psi_m/|\nabla\Psi_m|$ is the outward-pointing unit normal on ∂D_m, and used the notation $\mathcal{H}^2$ to denote the 2-dimensional Hausdorff measure on $\mathbb{R}^2$. By the definition (2.2.37) of Ψ_m,

$$\nabla\Psi_m(x) + \lambda_m x = \frac{1}{2\pi}\int_{D_m}\nabla_x\log|x-y|\,\mathrm{d}y = -\frac{1}{2\pi}\int_{D_m}\nabla_y\log|x-y|\,\mathrm{d}y, \quad (2.2.60)$$

which implies

$$\frac{1}{2\pi}x\cdot\int_{D_m}\nabla_y\log|x-y|\,\mathrm{d}y = -x\cdot\nabla\Psi_m(x) - \lambda_m|x|^2. \quad (2.2.61)$$

It follows that

$$\varphi(x) = -\frac{1}{2\pi}\mathcal{H}^2(D_m) + x\cdot\nabla\Psi_m(x) + 2\Big(\lambda_m\frac{|x|^2}{2} - \frac{1}{2\pi}\int_{D_m}\log|x-y|\,\mathrm{d}y\Big). \quad (2.2.62)$$

On the other hand, since

$$\Psi_m(x) + \lambda_m\frac{|x|^2}{2} - \frac{1}{2\pi}\int_{D_m}\log|x-y|\,\mathrm{d}y + c_m = 0, \quad (2.2.63)$$

we conclude that

$$\varphi(x) = -\mathcal{H}^2(D_m)/2\pi + x\cdot\nabla\Psi_m(x) - 2\Psi_m(x) - 2c_m. \quad (2.2.64)$$

Restricting to $x \in \partial D_m$, the latter reads

$$\varphi(x) = x\cdot\nabla\Psi_m(x) - (\mathcal{H}^2(D_m) + 4\pi c_m)/2\pi, \quad (2.2.65)$$

which, by equality (2.2.59), is absurd whenever

$$\mathcal{H}^2(D_m) + 4\pi c_m \neq 0. \quad (2.2.66)$$

We show now by a continuity argument that (2.2.66) holds for all m-fold symmetric patches near the Rankine vortex. The stream function of the latter circular vortex has the following explicit form

$$\Psi_\Omega = \frac{1}{4}\begin{cases}(1-2\Omega)(|x|^2-1) & \text{for } x \in \mathbb{D},\\ 2\log|x| - 2\Omega(|x|^2-1) & \text{for } x \in \mathbb{R}^2 \setminus \mathbb{D},\end{cases} \tag{2.2.67}$$

in terms of the angular velocity $\Omega \in \mathbb{R}$ and, therefore, $c = \mathfrak{c}_\Omega = -\Omega/2$ as in (1.3.2). Since $\mathcal{H}^2(\mathbb{D}) = \pi$, we obtain that (2.2.66) holds for all $\Omega \neq 1/2$; in particular, it holds for all bifurcation angular velocities Ω_m given in Theorem 2.7. On the other hand, we have that

$$B(0, 1 - \|\Phi_m - w\|_{C^0(\mathbb{S}^1)}) \subset D_m \subset B(0, 1 + \|\Phi_m - w\|_{C^0(\mathbb{S}^1)}), \tag{2.2.68}$$

Then,

$$\mathcal{H}^2(D_m) \geqslant \mathcal{H}^2(B(0, 1 - \|\Phi_m - w\|_{C^0(\mathbb{S}^1)})) = \mathcal{H}^2(\mathbb{D})(1 - \|\Phi_m - w\|_{C^0(\mathbb{S}^1)})^2 \geqslant \mathcal{H}^2(\mathbb{D})/2, \tag{2.2.69}$$

as long as $\|\Phi_m - w\|_{C^0(\mathbb{S}^1)}$ is sufficiently small. Moreover, for any $x \in \partial D_m$,

$$c_m = -\lambda_m |x|^2/2 + \frac{1}{2\pi}\int_{D_m} \log|x-y| \, \mathrm{d}y. \tag{2.2.70}$$

The latter is clearly a continuous mapping of $(\Phi_m, \lambda_m) \in C^0(\mathbb{S}^1) \times \mathbb{R}$ and, thus, $c_m \geqslant \mathfrak{c}_{\Omega_m}/2$ provided $\|\Phi_m - w\|_{C^0(\mathbb{S}^1)}$ and $|\lambda_m - \Omega_m|$ are chosen sufficiently small. We obtain, then, that

$$\mathcal{H}^2(D_m) + 4\pi c_m \geqslant (\mathcal{H}^2(\mathbb{D}) + 4\pi \mathfrak{c}_{\Omega_m})/2 > 0, \tag{2.2.71}$$

and, therefore (2.2.66) holds, thus showing the non-triviality of $\delta - a_1\Phi_m \in X^{k,\alpha} \cap \ker D_1\mathcal{F}(\Phi_m - w, \lambda_m)$. This was the desired contradiction in the case $a_1 \neq 0$ and we conclude that there is no non-trivial solution $\varphi \in C^0_{(m)}(\partial D_m)$ to (2.2.38). The admissibility of the tuple $(\Psi_m, \lambda_m, c_m, D_m, m)$ is, thus, verified. ☐

3. Analysis near admissible states

Let us instantiate, for the entirety of Section 3, a fixed but arbitrary $m \in \mathbb{N}^+$ and tuple $\Psi \in C^{1,1}_{(m)}(\mathbb{R}^2)$, $\Omega, c \in \mathbb{R}$, and $\varnothing \neq \widetilde{U} \subset \mathbb{R}^2$ satisfying the admissibility conditions of Definition 1.2. We know that such a tuple exists thanks to the results of Section 2. Recall that $\Sigma = \partial\widetilde{U}$ denotes the patch boundary and also has m-fold dihedral symmetry.

We shall also fix the following other objects. Let $\gamma \in C^\infty(\mathbb{R})$ be any smooth function that satisfies $\gamma(t) = 0$ for $t \leqslant -1$ and $\gamma(t) = 1$ for $t \geqslant 1$. Also let $M \in \mathbb{N}$ and $\{\sigma_k\}_{k=-M}^M \subset \mathbb{R}$ satisfy

$$\sum_{k=-M}^{M} \sigma_k = 1. \tag{3.0.1}$$

For $\text{\textgreek{\coppa}} \in [0,1]$, the Newtonian potential or the Poisson kernel is a function

$$\mathbf{N}_{\varrho} : \{(x,y) \in (\mathbb{R}^2)^2 \ : \ x \neq y \text{ and } x \neq y/\varrho|y|^2\} \to \mathbb{R} \tag{3.0.2}$$

defined via

$$\mathbf{N}_{\varrho}(x,y) = \frac{1}{4\pi}\log|x-y|^2 - \frac{1}{4\pi}\log(1 - 2\varrho x\cdot y + \varrho^2|x|^2|y|^2). \tag{3.0.3}$$

Perhaps unsurprisingly, the kernel $\mathbf{N}_{\varrho}$ (3.0.3) is related to the Biot-Savart kernels of equation (1.1.11) via

$$\nabla_1 \mathbf{N}_{\varrho}(x,y) = \mathbf{K}_{\varrho^{-1/2}}(x,y). \tag{3.0.4}$$

The operator given by integration against the kernels (3.0.3) is denoted, by an abuse of notation, $\mathbf{N}_{\varrho} : C^0_{\mathrm{c}}(B(0,\varrho^{-1/2})) \to C^0(B(0,\varrho^{-1/2}))$ and has the action

$$\mathbf{N}_{\varrho}[f](x) = \int_{\mathbb{R}^2} \mathbf{N}_{\varrho}(x,y) f(y) \, \mathrm{d}y, \text{ for } f \in C^0_{\mathrm{c}}(B(0,\varrho^{-1/2})) \text{ and } x \in \mathbb{R}^2. \tag{3.0.5}$$

This operator has the property that, for all $f \in C^0_{\mathrm{c}}(B(0, \qoppa^{-1/2}))$, in the sense of weak solutions the following equations are satisfied

$$\Delta(\mathbf{N}_\qoppa[f]) = f \text{ in } B(0, \qoppa^{-1/2}) \text{ and } \mathbf{N}_\qoppa[f] = -\frac{\log \qoppa}{4\pi} \int_{\mathbb{R}^2} f(y)\, \mathrm{d}y \text{ on } \partial B(0, \qoppa^{-1/2}). \tag{3.0.6}$$

Moreover, the operators $\mathbf{N}_\qoppa$ preserve m-fold dihedral symmetry.

3.1. **Preliminary estimates and identities.** Before we can get to the heart of our desingularization, splitting, and trapping constructions, we need to first set the stage. This, admittedly lengthy and technical, subsection paves the way by setting up the functional framework and operators along with recording and justifying a whirlwind of estimates.

Our first result of the subsection unpacks some important immediate consequences of Definition 1.2.

Lemma 3.1 (Simple consequences of admissibility)**.** *There exists an open, bounded, and connected set $U \subset \mathbb{R}^2$ that has m-fold dihedral symmetry and a class $C^{1,1}$ boundary such that the following hold.*

(1) We have the inclusion $\Sigma \subset U$.
(2) We have the equality $\Sigma = \{x \in U \ : \ \Psi(x) = 0\}$.
(3) We have the estimate $2 \min_{\overline{U}} |\nabla \Psi| \geqslant \min_\Sigma(\nu \cdot \nabla \Psi)$.
(4) The complement $\mathbb{R}^2 \setminus \overline{U}$ has two connected components $U^{\mathrm{in}}, U^{\mathrm{out}} \subset \mathbb{R}^2$; these satisfy

$$U^{\mathrm{in}} \subset \widetilde{U},\ \mathrm{dist}(U^{\mathrm{in}}, U^{\mathrm{out}}) > 0,\ \textit{and } U^{\mathrm{out}} \cap \widetilde{U} = \varnothing. \tag{3.1.1}$$

See Figure 1 for a depiction of U, U^{in}, and U^{out}.

Proof. The U shall be taken as an appropriate topological tubular neighborhood of Σ. By a simple compactness argument using the properties from the second item of Definition 1.2, there exists $0 < \rho \leqslant 1$ such that

$$\hat{U} = \{x \in \mathbb{R}^2 \ : \ \mathrm{dist}(x, \Sigma) < \rho\} \text{ satisfies } \inf\{|\nabla \Psi(x)| \ : \ x \in \hat{U}\} \geqslant \min_\Sigma(\nu \cdot \nabla \Psi)/2. \tag{3.1.2}$$

Now, there exists $\sampi_1, \sampi_2 \in (0, 1]$ sufficiently small so that

$$\{x \in \hat{U} \ : \ |\Psi(x)| \leqslant \sampi_1\} \subset \{x \in \hat{U} \ : \ \mathrm{dist}(x, \partial \hat{U}) > \sampi_2\}. \tag{3.1.3}$$

We therefore may take $U = \{x \in \hat{U} \ : \ |\Psi(x)| < \sampi_1\}$. The items of the lemma are now simple consequences of this definition. □

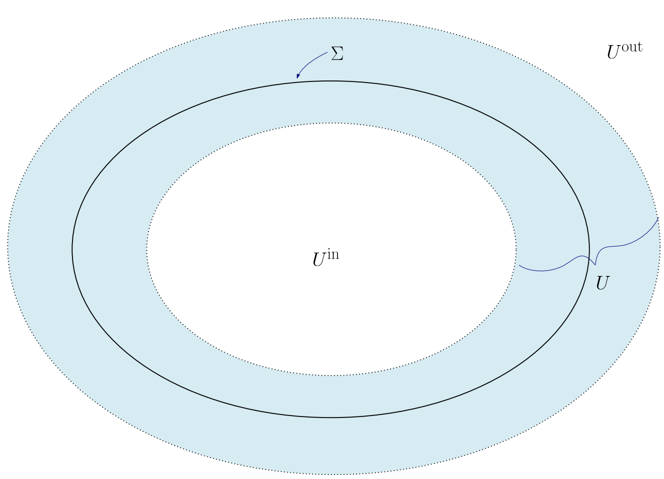

Figure 1. Shown here are the sets U, U^{in}, and U^{out} produced by Lemma 3.1 and their relation to Σ (the curve in solid black). The region in blue is the topological tubular neighborhood U. The set U^{in} is the bounded connected component of the white region. The set U^{out} is the complement of $\overline{U \cup U^{\mathrm{in}}}$. The support of the vorticity $\widetilde{U}$ is not emphasized in the diagram, but it is the bounded simply connected region enclosed by Σ.

In the next result, we introduce a family of spatially localized regularizations to the Heaviside function and identify a few basic properties.

Lemma 3.2 (Regularized Heaviside operators)**.** *There exists* $\varepsilon_0, r_0, \varrho_0, \text{♀}_0 \in (0,1]$ *such that the following hold for all* $0 < \varepsilon \leqslant \varepsilon_0$*.*

(1) For all $\psi \in C^1_{(m)}(\overline{U})$ *satisfying* $\|\psi\|_{C^1(\overline{U})} \leqslant (M+1)r_0$ *we have*

$$\operatorname{dist}(\partial U, \{x \in U \ : \ |\Psi(x) + \psi(x)| \leqslant \varepsilon\}) \geqslant \varrho_0 \text{ and } 4\min_{\overline{U}} |\nabla(\Psi + \psi)| \geqslant \min_{\Sigma}(\nu \cdot \nabla\Psi). \tag{3.1.4}$$

(2) The regularized Heaviside operator $\Gamma^\Psi_\varepsilon : \overline{B(0,(M+1)r_0)} \subset C^1_{(m)}(\overline{U}) \to C^1_{(m),\mathrm{c}}(\mathbb{R}^2)$ *with action given via*

$$[\Gamma^\Psi_\varepsilon(\psi)](x) = \begin{cases} \gamma\big(-\frac{\Psi(x)+\psi(x)}{\varepsilon}\big) & \text{if } x \in \overline{U}, \\ 1 & \text{if } x \in U^{\mathrm{in}}, \\ 0 & \text{if } x \in U^{\mathrm{out}}, \end{cases} \tag{3.1.5}$$

is well defined; moreover its restrictions satisfy $C^\ell_{(m)}(\overline{U}) \to C^\ell_{(m),\mathrm{c}}(\mathbb{R}^2)$ *for every* $\ell \in \mathbb{N}^+$*.*

(3) The derivative regularized Heaviside operator $(\Gamma^\Psi_\varepsilon)' : \overline{B(0,(M+1)r_0)} \subset C^1_{(m)}(\overline{U}) \to C^1_{(m),\mathrm{c}}(\mathbb{R}^2)$ *with action given by*

$$[(\Gamma^\Psi_\varepsilon)'(\psi)](x) = \begin{cases} -\frac{1}{\varepsilon}\gamma'\big(-\frac{\Psi(x)+\psi(x)}{\varepsilon}\big) & \text{if } x \in \overline{U}, \\ 0 & \text{if } x \in \mathbb{R}^2 \setminus \overline{U}, \end{cases} \tag{3.1.6}$$

is well defined; moreover its restrictions satisfy $C^\ell_{(m)}(\overline{U}) \to C^\ell_{(m),\mathrm{c}}(\mathbb{R}^2)$ *for every* $\ell \in \mathbb{N}^+$*.*

(4) For all $\psi \in C^1_{(m)}(\overline{U})$ *we have that* $\operatorname{supp}(\Gamma^\Psi_\varepsilon(\psi)) \subset B(0, \text{♀}_0^{-1/2}/2)$*.*

(5) For all $\psi \in C^1_{(m)}(\overline{U})$ *we have that* $\operatorname{supp}((\Gamma^\Psi_\varepsilon)'(\psi)) \subset U$*.*

Proof. The first item follows simply by selecting sufficiently small parameters. The second item follows from the first in that: For $x \in \overline{U}$ in an open neighborhood of the boundary component $\partial U \cap \partial U^{\mathrm{in}}$ (resp. $\partial U \cap \partial U^{\mathrm{out}}$) we have that the composition $\gamma(-(\Psi(x) + \psi(x))/\varepsilon)$ is identically 1 (resp. 0). Thus the pointwise definition (3.1.5) is smoothly matching. The third item follows by similar observations. The fourth and fifth items are immediate. □

We are now in a position to introduce the main objects of study throughout Section 3. In particular, the third item of the following result introduces the *perturbed* patch equation; the overarching goal of Section 3 is to construct solutions to the operator equation posed in (3.1.11).

Lemma 3.3 (Principal actors and their basic properties)**.** *Let* ε_0*,* r_0*, and* ♀_0 *be as in Lemma 3.2. For* $0 < \varepsilon \leqslant \varepsilon_0$*,* $0 \leqslant \rho \leqslant r_0$*, and* $0 \leqslant \text{♀} < \text{♀}_0$ *the following hold.*

(1) The operators $\boldsymbol{K}_{\varepsilon,\rho,\text{♀}}, \boldsymbol{F}_{\varepsilon,\rho,\text{♀}} : \overline{B(0,r_0)} \subset C^1_{(m)}(\overline{U}) \to C^1_{(m)}(\overline{U})$ *with actions given via*

$$\boldsymbol{K}_{\varepsilon,\rho,\text{♀}}(\psi) = -\sum_{k=-M}^{M} \sigma_k \mathbf{N}_{\text{♀}}[\Gamma^\Psi_\varepsilon(\psi + k\rho) - \Gamma^\Psi_\varepsilon(k\rho)] \text{ and } \boldsymbol{F}_{\varepsilon,\rho,\text{♀}}(\psi) = \psi + \boldsymbol{K}_{\varepsilon,\rho,\text{♀}}(\psi) \tag{3.1.7}$$

are well-defined and, on the interior of their domain, smooth in the Fréchet sense. Here $\mathbf{N}_{\text{♀}}$ *and* Γ^Ψ_ε *are as defined in equations* (3.0.3) *and* (3.1.5)*, respectively.*

(2) For any $\psi \in B(0,r_0) \subset C^1_{(m)}(\overline{U})$ *and* $\phi \in C^1_{(m)}(\overline{U})$ *the derivative of* $\boldsymbol{F}_{\varepsilon,\rho,\text{♀}}$ *obeys the formula*

$$D\boldsymbol{F}_{\varepsilon,\rho,\text{♀}}(\psi)\phi = \phi + \boldsymbol{\kappa}_{\varepsilon,\rho,\text{♀}}(\psi)\phi \tag{3.1.8}$$

with

$$\boldsymbol{\kappa}_{\varepsilon,\rho,\text{♀}}(\psi)\phi = -\sum_{k=-M}^{M} \sigma_k \mathbf{N}_{\text{♀}}[(\Gamma^\Psi_\varepsilon)'(\psi + k\rho)\phi] \tag{3.1.9}$$

(3) With the functions $f_{\varepsilon,\rho,\text{♀}} \in C^1_{(m)}(\overline{U})$ defined via

$$f_{\varepsilon,\rho,\text{♀}} = \sum_{k=-M}^{M} \sigma_k(\mathbf{N}_{\text{♀}}[\Gamma^{\Psi}_{\varepsilon}(k\rho)] - \mathbf{N} * \mathbb{1}_{\widetilde{U}}) \tag{3.1.10}$$

the following two items are equivalent for $\psi \in \overline{B(0,r_0)} \subset C^1_{(m)}(\overline{U})$:

(a) We have satisfied

$$\boldsymbol{F}_{\varepsilon,\rho,\text{♀}}(\psi) = f_{\varepsilon,\rho,\text{♀}}. \tag{3.1.11}$$

(b) We have satisfied

$$\Psi + \psi = \sum_{k=-M}^{M} \sigma_k \mathbf{N}_{\text{♀}}[\Gamma^{\Psi}_{\varepsilon}(\psi + k\rho)] - \Omega|\cdot|^2/2 - c. \tag{3.1.12}$$

Proof. The second and third items are direct computations, as such we omit their simple proofs.

Let us turn our attention to the first item. Since the difference between $\boldsymbol{F}$ and $\boldsymbol{K}$ is merely the identity map, it suffices to observe that $\boldsymbol{K}$ is smooth between the stated spaces. To see this, we note that for $\psi \in B(0,r_0) \subset C^1_{(m)}(\overline{U})$, $\phi \in C^1_{(m)}(\overline{U})$, and $x \in U$ the following formula holds:

$$[\boldsymbol{K}_{\varepsilon,\rho,\text{♀}}(\psi)](x) = -\sum_{k=-M}^{M} \sigma_k \int_U \mathbf{N}_{\text{♀}}(x,y)\Big(\gamma\Big(-\frac{\Psi(y)+\psi(y)+k\rho}{\varepsilon}\Big) - \gamma\Big(-\frac{\Psi(y)+k\rho}{\varepsilon}\Big)\Big)\,\mathrm{d}y. \tag{3.1.13}$$

It is now apparent that $\boldsymbol{K}_{\varepsilon,\rho,\text{♀}}$ is a sum of terms having the form of a bounded linear operator on $C^1_{(m)}(\overline{U})$ acting on outer composition with the smooth function γ. Fréchet smoothness is now a routine consequence of the converse to Taylor's theorem (see, e.g., Section 2.4B in Abraham, Marsden, and Ratiu [1]). □

We now further examine the linear map from equation (3.1.9) by proving a series of simple bounds. Note that in what follows the dependence on ε in the inequalities is likely far from optimal. Indeed, the forthcoming proof invokes truly the most brutal estimates available. It is only the qualitative assertions of this result that are important for the overarching strategy.

Lemma 3.4 (Preliminary mapping estimates, I)**.** *Let ε_0, r_0, and ♀_0 be the positive values granted by Lemma 3.2. There exists $C \in \mathbb{R}^+$ with the property that for all $0 < \varepsilon \leqslant \varepsilon_0$, $0 \leqslant \rho \leqslant r_0$, $0 \leqslant \text{♀} \leqslant \text{♀}_0$, $\psi \in \overline{B(0,r_0)} \subset C^1_{(m)}(\overline{U})$, and $\phi \in C^0_{(m)}(\overline{U})$ we have that $\boldsymbol{\kappa}_{\varepsilon,\rho,\text{♀}}(\psi)\phi \in C^1_{(m)}(\overline{U})$ with the estimate*

$$\|\boldsymbol{\kappa}_{\varepsilon,\rho,\text{♀}}(\psi)\phi\|_{C^1(\overline{U})} \leqslant (C/\varepsilon)\|\phi\|_{C^0(\overline{U})}. \tag{3.1.14}$$

If we suppose additionally that $\phi \in C^1_{(m)}(\overline{U})$, then $\boldsymbol{\kappa}_{\varepsilon,\rho,\text{♀}}(\psi)\phi \in C^2_{(m)}(\overline{U})$ with the estimate

$$\|\boldsymbol{\kappa}_{\varepsilon,\rho,\text{♀}}(\psi)\phi\|_{C^2(\overline{U})} \leqslant (C/\varepsilon^2)\|\phi\|_{C^1(\overline{U})}. \tag{3.1.15}$$

If we suppose further that $\widetilde{\psi} \in \overline{B(0,r_0)} \subset C^1_{(m)}(\overline{U})$, then we have the estimate

$$\|\boldsymbol{\kappa}_{\varepsilon,\rho,\text{♀}}(\psi)\phi - \boldsymbol{\kappa}_{\varepsilon,\rho,\text{♀}}(\widetilde{\psi})\phi\|_{C^1(\overline{U})} \leqslant (C/\varepsilon^2)\|\psi - \widetilde{\psi}\|_{C^0(\overline{U})}\|\phi\|_{C^0(\overline{U})}. \tag{3.1.16}$$

Proof. We take C to be a finite and positive constant that is allowed to change line by line and depend on ε_0, r_0, ♀_0, U, U^{in}, U^{out}, γ, M, $\{\sigma_k\}_{k=-M}^{M}$, and Ψ.

From simple bounds on $\mathbf{N}_{\text{♀}}$ and is derivative, we derive the pointwise estimates for $x \in U$:

$$|[\boldsymbol{\kappa}_{\varepsilon,\rho,\text{♀}}(\psi)\phi](x)| \leqslant \frac{C}{\varepsilon}\|\phi\|_{C^0(\overline{U})}\int_U (|\log|x-y|| + 1)\,\mathrm{d}y$$
$$\text{and } |\nabla[\boldsymbol{\kappa}_{\varepsilon,\rho,\text{♀}}(\psi)\phi](x)| \leqslant \frac{C}{\varepsilon}\|\phi\|_{C^0(\overline{U})}\int_U \Big(\frac{1}{|x-y|} + 1\Big)\,\mathrm{d}y. \tag{3.1.17}$$

Estimate (3.1.14) now follows from (3.1.17).

To prove (3.1.15), we now integrate by parts to derive the pointwise estimate:

$$|\nabla^2[\boldsymbol{\kappa}_{\varepsilon,\rho,\varrho}(\psi)\phi](x)| \leqslant \frac{C}{\varepsilon^2}\|\phi\|_{C^1(\overline{U})}\int_U \left(\frac{1}{|x-y|}+1\right)\mathrm{d}y. \tag{3.1.18}$$

Similarly, inequality (3.1.16) follows from the fundamental theorem of calculus and the pointwise bounds:

$$|[\boldsymbol{\kappa}_{\varepsilon,\rho,\varrho}(\psi)\phi-\boldsymbol{\kappa}_{\varepsilon,\rho,\varrho}(\widetilde{\psi})\phi](x)| \leqslant \frac{C}{\varepsilon^2}\|\phi\|_{C^0(\overline{U})}\|\psi-\widetilde{\psi}\|_{C^0(\overline{U})}\int_U(|\log|x-y||+1)\,\mathrm{d}y$$
$$\text{and } |\nabla[\boldsymbol{\kappa}_{\varepsilon,\rho,\varrho}(\psi)\phi-\boldsymbol{\kappa}_{\varepsilon,\rho,\varrho}(\widetilde{\psi})\phi](x)| \leqslant \frac{C}{\varepsilon^2}\|\phi\|_{C^0(\overline{U})}\|\psi-\widetilde{\psi}\|_{C^0(\overline{U})}\int_U\left(\frac{1}{|x-y|}+1\right)\mathrm{d}y. \tag{3.1.19}$$

□

To establish more subtle estimates than those of Lemma 3.4, we require the indispensable tool of uniform local level set coordinates. The construction of these coordinates is digressive from the main narrative line of this section. As such, the proof of the following result is postponed to Appendix A.1. In what follows we write for $\ell\in\mathbb{R}^+$ the set $\overline{\ell Q}=[-\ell,\ell]^2\subset\mathbb{R}^2$.

Lemma 3.5 (Uniform local level set coordinates). *There exists $0<\widetilde{\varepsilon}_0\leqslant\varepsilon_0$, $0<\widetilde{r}_0\leqslant r_0$, $\delta_-,\delta_+,\ell,R\in\mathbb{R}^+$, $N\in\mathbb{N}^+$, $\{x_i\}_{i=1}^N\subset\Sigma$ such that the following hold for all $i\in\{1,\dots,N\}$, $\psi\in C^1(\overline{U})$ with $\|\psi\|_{C^1(\overline{U})}\leqslant(M+1)\widetilde{r}_0$.*

(1) We have $\delta_-<\delta_+$ and, for all $0<\varepsilon\leqslant\widetilde{\varepsilon}_0$, the inclusions

$$B(x_i,2\delta_+)\subset U \text{ and } \{x\in U\ :\ |\Psi(x)+\psi(x)|\leqslant\varepsilon\}\subset\bigcup_{i=1}^N B(x_i,\delta_-). \tag{3.1.20}$$

(2) There exists a map $G_i^\psi\in C^1(\overline{\ell Q};\mathbb{R}^2)$ which is a diffeomorphism onto its image, satisfies the inclusions

$$B(x_i,2\delta_-)\subset G_i^\psi(\overline{\ell Q})\subset B(x_i,\delta_+), \tag{3.1.21}$$

and for all $y\in\overline{\ell Q}$ we have the identities

$$(\Psi+\psi)\circ G_i^\psi(y)=y_2 \text{ and } |\partial_1 G_i^\psi(y)|=|\nabla(\Psi+\psi)\circ G_i^\psi(y)|\det\nabla G_i^\psi(y) \tag{3.1.22}$$

along with the estimate

$$|\nabla G_i^\psi(y)|+|\nabla G_i^\psi(y)^{-1}|\leqslant R. \tag{3.1.23}$$

(3) The mapping

$$C^1(\overline{U})\supset\overline{B(0,\widetilde{r}_0)}\ni\psi\mapsto G_i^\psi\in C^1(\overline{\ell Q};\mathbb{R}^2) \tag{3.1.24}$$

is continuous moreover, we have the estimate

$$|\nabla G_i^\psi(y)-\nabla G_i^0(y)|\leqslant R\|\psi\|_{C^1(\overline{U})}. \tag{3.1.25}$$

Lemma 3.5's uniform local level set coordinates permit us to obtain sharp density estimates on level sets.

Corollary 3.6 (Refined estimates on level sets). *Let δ_-, $\widetilde{\varepsilon}_0$, and $\widetilde{r}_0$ be the parameters from Lemma 3.5. There exists a constant $C\in\mathbb{R}^+$ such that for all $|t|\leqslant\widetilde{\varepsilon}_0$, $0<r\leqslant\delta_-$, and $\psi\in C^1_{(m)}(\overline{U})$ with $\|\psi\|_{C^1(\overline{U})}\leqslant(M+1)\widetilde{r}_0$ obeying the following properties, in which we use $\mathcal{H}^1$ to denote the one-dimensional Hausdorff measure.*

(1) We have the estimates

$$\sup_{x\in\overline{U}}\mathcal{H}^1(U\cap\{\Psi+\psi=t\}\cap B(x,r))\leqslant Cr \text{ and } \mathcal{H}^1(U\cap\{\Psi+\psi=t\})\leqslant C. \tag{3.1.26}$$

(2) We have the estimate

$$\sup_{x\in\overline{U}}\int_{U\cap\{\Psi+\psi=t\}\cap B(x,r)}|\log|x-y||\,\mathrm{d}\mathcal{H}^1(y)\leqslant Cr(1+\log(1/r)). \tag{3.1.27}$$

(3) We have the estimates

$$\sup_{x\in\overline{U}}\int_{(U\setminus B(x,r))\cap\{\Psi+\psi=t\}}\frac{1}{|x-y|}\,\mathrm{d}\mathcal{H}^1(y)\leqslant C(1+\log(1/r)). \tag{3.1.28}$$

(4) We have the estimates

$$|\mathcal{H}^1(U\cap\{\Psi+\psi=t\})-\mathcal{H}^1(U\cap\{\Psi=0\})|\leqslant C(t+\|\psi\|_{C^1(\overline{U})}). \tag{3.1.29}$$

Proof. We begin by proving the first item. Let $x\in\overline{U}$. From the first item of Lemma 3.5 we have

$$\mathcal{H}^1(U\cap\{\Psi+\psi=t\}\cap B(x,r))\leqslant\sum_{i=1}^{N}\mathcal{H}^1(B(x_i,\delta_-)\cap\{\Psi+\psi=t\}\cap B(x,r)) \tag{3.1.30}$$

so it suffices to estimate the i^{th}-term in the sum above. For this we use the local parametrization of the t-level set determined by $G_i^{\psi}(\cdot,t)$ where the latter is from the collection of coordinate functions produced by Lemma 3.5. We get

$$\mathcal{H}^1(B(x_i,\delta_-)\cap\{\Psi+\psi=t\}\cap B(x,r))\leqslant\int_{-\ell}^{\ell}\mathbb{1}_{G_i^{\psi}(\cdot,t)\in B(x,r)}(y_1)|\partial_1G_i^{\psi}(y_1,t)|\,\mathrm{d}y_1. \tag{3.1.31}$$

From estimate (3.1.23) we get $|\partial_1G_i^{\psi}(y_1,t)|\leqslant R$. On the other hand, by inspection of identity (A.1.10) from the proof of Lemma 3.5, we deduce

$$\forall\,y_1,\widetilde{y}_1\in[-\ell,\ell],\ |G_i^{\psi}(y_1,t)-G_i^{\psi}(\widetilde{y}_1,t)|\geqslant|y_1-\widetilde{y}_1| \tag{3.1.32}$$

and hence, by bounding the one-dimensional Hausdorff measure above by the length,

$$\mathcal{H}^1(\{y_1\in[-\ell,\ell]\ :\ G_i^{\psi}(y_1,t)\in B(x,r)\})\leqslant 2r \tag{3.1.33}$$

Synthesizing the estimates (3.1.30), (3.1.31), and (3.1.33) yields

$$\mathcal{H}^1(U\cap\{\Psi+\psi=t\}\cap B(x,r))\leqslant(2RM)r \tag{3.1.34}$$

which is the bound claimed on the left hand side of (3.1.26). The right hand bound follows from the left by setting $r=\delta_-$, and summing over $x\in\{x_1,\dots,x_N\}$.

We turn our attention to the second item, so again let $x\in\overline{U}$. We break up the region of integration according to dyadic annuli:

$$\int_{U\cap\{\Psi+\psi=t\}\cap B(x,r)}|\log|x-y||\,\mathrm{d}\mathcal{H}^1(y)=\sum_{n=0}^{\infty}\int_{U\cap\{\Psi+\psi=t\}\cap A(x,r/2^n)}|\log|x-y||\,\mathrm{d}\mathcal{H}^1(y) \tag{3.1.35}$$

with $A(x,r/2^n)=B(x,r/2^n)\setminus B(x,r/2^{n+1})$. On the n^{th} annulus we have the (crude) inequality $|\log|x-y||\leqslant 2(n+1)\log(1/r)$ and so (3.1.35) lends to us the bound:

$$\int_{U\cap\{\Psi+\psi=t\}\cap B(x,r)}|\log|x-y||\,\mathrm{d}\mathcal{H}^1(y)\leqslant 2\log(1/r)\sum_{n=0}^{\infty}(n+1)\mathcal{H}^1(U\cap\{\Psi+\psi=t\}\cap B(x,r/2^n)). \tag{3.1.36}$$

The right hand side sum's n^{th} Hausdorff measure can then be estimated by $Cr/2^n$, thanks to the first item. Thus we get the desired estimate (3.1.27) with a constant $2C\sum_{n=0}^{\infty}(n+1)2^{-n}<\infty$.

We prove the third item via a similar strategy as above. For $x\in\overline{U}$ we dyadically decompose in a similar fashion, thus obtaining:

$$\int_{(U\setminus B(x,r))\cap\{\Psi+\psi=t\}}\frac{1}{|x-y|}\,\mathrm{d}\mathcal{H}^1(y)\leqslant\frac{1}{\delta_-}\mathcal{H}^1(\{\Psi+\psi=t\})+\sum_{n=0}^{N}\frac{2^{n+1}}{\delta_-}\mathcal{H}^1(\{\Psi+\psi=t\}\cap B(x,\delta_-/2^n)). \tag{3.1.37}$$

where $N\in\mathbb{N}^+$ is the smallest positive integer for which $\delta_-/2^N<r$. We then use the first item again to bound the n^{th} term's Hausdorff measure by $C\delta_-/2^n$. This gives us

$$\int_{(U\setminus B(x,r))\cap\{\Psi+\psi=t\}}\frac{1}{|x-y|}\,\mathrm{d}\mathcal{H}^1(y)\leqslant\frac{C}{\delta_-}+2C(N+1). \tag{3.1.38}$$

We conclude by noting that $N \leqslant 1 + \log(\delta_-/r)$.

Finally, we consider the fourth item. We begin by using the triangle inequality to write

$$|\mathcal{H}^1(\{\Psi + \psi = t\}) - \mathcal{H}^1(\{\Psi = 0\})| \leqslant \mathbf{I}_1 + \mathbf{I}_2 \tag{3.1.39}$$

with

$$\mathbf{I}_1 = |\mathcal{H}^1(\{\Psi + \psi = t\}) - \mathcal{H}^1(\{\Psi = t\})| \text{ and } \mathbf{I}_2 = |\mathcal{H}^1(\{\Psi = t\}) - \mathcal{H}^1(\{\Psi = 0\})|. \tag{3.1.40}$$

Estimating $\mathbf{I}_2$ is simpler. Consider the case that $t > 0$ (we omit the proof of the case $t < 0$, as it is similar) We deduce from Definition 1.2 that $\nabla\Psi/|\nabla\Psi| \in C^{0,1}(\overline{U})$ and so we can use the divergence theorem to derive:

$$\mathbf{I}_2 = \Big| \int_{U\cap\{0<\Psi<t\}} \nabla \cdot (\nabla\Psi/|\nabla\Psi|)\Big| \leqslant C\mathcal{H}^2(U \cap \{0 < \Psi < t\}). \tag{3.1.41}$$

Then, the coarea formula (see, e.g., Theorem 3.2.11 in Federer [30] or Theorem 3.13 in Evans and Gariepy [29]) in conjunction with the first item in turn gives us

$$\mathcal{H}^2(U \cap \{0 < \Psi < t\}) = \int_0^t \int_{\{\Psi=\tau\}} |\nabla\Psi(y)|^{-1} \, \mathrm{d}\mathcal{H}^1(y) \, \mathrm{d}\tau \leqslant Ct. \tag{3.1.42}$$

Synthesizing (3.1.41) and (3.1.42) shows $\mathbf{I}_2 \leqslant Ct$.

Estimating $\mathbf{I}_1$ is more involved. Let $\{\chi_i\}_{i=1}^N \subset C_{\mathrm{c}}^\infty(U)$ be any fixed sequence of functions that satisfy

$$\mathrm{supp}\chi_i \subset B(x_i, 2\delta_-) \text{ for every } i \in \{1, \dots, N\} \text{ and } \sum_{i=1}^N \chi_i = 1 \text{ on the set } \bigcup_{i=1}^N B(x_i, \delta_-). \tag{3.1.43}$$

We use the uniform local level set coordinates constructed by Lemma 3.5 to locally parametrize the t-level sets; in doing so we get

$$\mathcal{H}^1(\{\Psi + \psi = t\}) - \mathcal{H}^1(\{\Psi = t\}) = \sum_{i=1}^N \int_0^\ell (\chi(G_i^\psi(y_1, t))|\partial_1 G_i^\psi(y_1, t)| - \chi_i(G_i^0(y_1, t))|\partial_1 G_i^0(y_1, t)|) \, \mathrm{d}y_1. \tag{3.1.44}$$

Then, using the special continuity estimate (3.1.25) from Lemma 3.5, we arrive at the bound

$$\mathbf{I}_1 \leqslant C\|\psi\|_{C^1(\overline{U})}. \tag{3.1.45}$$

Synthesizing the estimates on $\mathbf{I}_1$ and $\mathbf{I}_2$ completes the proof of estimate (3.1.29). □

The level set coordinate functions constructed by Lemma 3.5 combine with the density estimates of the first three items of Corollary 3.6 to allow us to make the subsequent important boundedness and regularity estimates.

Proposition 3.7 (Uniform integral estimates for the Newtonian potential over thin domains)**.** *Let $\widetilde{\varepsilon}_0$ and $\widetilde{r}_0$ be the positive values granted by Lemma 3.5. There exists $C \in \mathbb{R}^+$ such that for all $\psi \in C^1_{(m)}(\overline{U})$ with $\|\psi\|_{C^1(\overline{U})} \leqslant (M+1)\widetilde{r}_0$ and all $0 < \varepsilon \leqslant \widetilde{\varepsilon}_0$ we have the estimates*

$$\sup_{x\in\overline{U}} \frac{1}{\varepsilon} \int_{\{|\Psi+\psi|\leqslant\varepsilon\}} (|\log|x-y|| + 1) \, \mathrm{d}y \leqslant C \tag{3.1.46}$$

and

$$\sup_{\substack{x,z\in\overline{U}\\ x\neq z}} \frac{1}{\varepsilon} \int_{\{|\Psi+\psi|\leqslant\varepsilon\}} \Big(\frac{|\log|x-y| - \log|z-y||}{|x-z|(1+|\log|x-z||)} + 1 \Big) \, \mathrm{d}y \leqslant C. \tag{3.1.47}$$

Proof. We claim first, with $\mathcal{H}^2$ denoting the two-dimensional Hausdorff measure, that the area estimate

$$\mathcal{H}^2(\{|\Psi + \psi| \leqslant \varepsilon\}) \leqslant C\varepsilon \tag{3.1.48}$$

holds. By the coarea formula (e.g. Theorem 3.2.11 in [30] or Theorem 3.13 in [29]) we have

$$\mathcal{H}^2(\{|\Psi+\psi|\leqslant\varepsilon\})\leqslant 2\varepsilon\cdot\sup_{|t|\leqslant\varepsilon}\int_{\{\Psi+\psi=t\}}|\nabla(\Psi+\psi)(y)|^{-1}\,\mathrm{d}\mathcal{H}^1(y) \tag{3.1.49}$$

The definition of $\widetilde{r_0}$ ensures us that $\min_{\overline{U}}|\nabla(\Psi+\psi)|\geqslant\min_{\overline{U}}|\nabla\Psi|/2$ and so it follows from (3.1.49) that

$$\mathcal{H}^2(\{|\Psi+\psi|\leqslant\varepsilon\})\leqslant C\varepsilon\cdot\sup_{|t|\leqslant\varepsilon}\mathcal{H}^1(\{\Psi+\psi=t\}). \tag{3.1.50}$$

The bound (3.1.48) now follows by combining (3.1.50) with the first item of Corollary 3.6.

We continue with the proof of estimate (3.1.46). The coarea formula again lends us the initial estimate for any $x\in\overline{U}$

$$\frac{1}{\varepsilon}\int_{\{|\Psi+\psi|\leqslant\varepsilon\}}|\log|x-y||\,\mathrm{d}y\leqslant 2\sup_{|t|\leqslant\varepsilon}\int_{\{\Psi+\psi=t\}}\frac{|\log|x-y||}{|\nabla(\Psi+\psi)(y)|}\,\mathrm{d}\mathcal{H}^1(y). \tag{3.1.51}$$

We can, for $|t|\leqslant\varepsilon$ fixed, split $\{\Psi+\psi=t\}$ relative to the ball $B(x,\delta_-)$ and use the first item of Corollary 3.6 to bound

$$\int_{\{\Psi+\psi=t\}}|\log|x-y||\,\mathrm{d}\mathcal{H}^1(y)\leqslant C|\log\delta_-|+\int_{\{\Psi+\psi=t\}\cap B(x,\delta_-)}|\log|x-y||\,\mathrm{d}\mathcal{H}^1(y). \tag{3.1.52}$$

The final term above is at most $C\delta_-|\log\delta_-|$ (by the second item of Corollary 3.6) and so we conclude that estimate (3.1.46) holds.

We turn our attention to the proof of (3.1.47). Let $x,z\in\overline{U}$ with $x\neq z$. If $|x-z|\geqslant\delta_-/2$, then we do not exploit the difference between the logarithms. The estimate

$$\frac{1}{\varepsilon}\int_{\{|\Psi+\psi|\leqslant\varepsilon\}}\frac{|\log|x-y|-\log|z-y||}{|x-z|(1+|\log|x-z||)}\mathrm{d}y\leqslant\frac{4C}{\delta_-\log(2/\delta_-)} \tag{3.1.53}$$

simply follows from (3.1.46). Thus we need only consider the case that $0<|x-z|<\delta_-/2$.

By the coarea formula again, we have

$$\frac{1}{\varepsilon}\int_{\{|\Psi+\psi|\leqslant\varepsilon\}}\frac{|\log|x-y|-\log|z-y||}{|x-z|(1+|\log|x-z||)}\mathrm{d}y\leqslant\frac{2\sup_{|t|\leqslant\varepsilon}}{\min_{\overline{U}}|\nabla\Psi|}\int_{\{\Psi+\psi=t\}}\frac{|\log|x-y|-\log|z-y||}{|x-z|(1+|\log|x-z||)}\,\mathrm{d}\mathcal{H}^1(y). \tag{3.1.54}$$

Fix $|t|\leqslant\varepsilon$. We make the region splitting

$$\{\Psi+\psi=t\}=[\{\Psi+\psi=t\}\cap B((x+z)/2,2|x-z|)]\cup[\{\Psi+\psi=t\}\setminus B((x+z)/2,2|x-z|)]. \tag{3.1.55}$$

In the first case that $y\in B((x+z)/2,2|x-z|)$, we again do not exploit the difference and instead use the second item of Corollary 3.6, giving us

$$\int_{\{\Psi+\psi=t\}\cap B((x+z)/2,2|x-z|)}|\log|x-y|-\log|z-y||\,\mathrm{d}\mathcal{H}^1(y)\leqslant 4C|x-z|(1+|\log|2|x-z|||). \tag{3.1.56}$$

On the other hand, in the second case of (3.1.55) we have the fundamental theorem of calculus estimate:

$$|\log|x-y|-\log|z-y||\leqslant\int_0^1\frac{|x-z|}{|\tau x+(1-\tau)z-y|}\,\mathrm{d}\tau\leqslant\frac{4|x-z|}{3|y-(x+z)/2|} \tag{3.1.57}$$

for $y\in\{\Psi+\psi=t\}\setminus B((x+z)/2,2|x-z|)$. We combine (3.1.57) with the third item of Corollary 3.6 to deduce

$$\int_{\{\Psi+\psi=t\}\setminus B((x+z)/2,2|x-z|)}|\log|x-y|-\log|z-y||\,\mathrm{d}\mathcal{H}^1(y)\leqslant\frac{4C}{3}|x-z|(1+|\log|2|x-z|||). \tag{3.1.58}$$

Synthesizing the above casework gives the desired bound (3.1.47). □

The level set coordinates of Lemma 3.5 also find an application in the identification of certain limits.

Proposition 3.8 (Limit identification). *Assume that* $g \in C^0(\overline{U})$, $\{\varepsilon_n\}_{n\in\mathbb{N}} \subset (0, \widetilde{\varepsilon}_0]$, $\psi, \{\psi_n\}_{n\in\mathbb{N}} \subset \overline{B(0,(M+1)\widetilde{r}_0)} \subset C^1_{(m)}(\overline{U})$, *and* $\phi, \{\phi_n\}_{n\in\mathbb{N}} \subset C^0_{(m)}(\overline{U})$ *satisfy*

$$\varepsilon_n \to 0,\ \|\psi_n - \psi\|_{C^1(\overline{U})} \to 0,\ \text{and } \|\phi_n - \phi\|_{C^0(\overline{U})} \to 0 \text{ as } n \to \infty. \tag{3.1.59}$$

Then

$$-\int_U g(x)[(\Gamma^\Psi_{\varepsilon_n})'(\psi_n)](x)\phi_n(x)\,\mathrm{d}x \to \int_{U\cap\{\Psi+\psi=0\}} \frac{g(y)\phi(y)}{|\nabla(\Psi+\psi)(y)|}\,\mathrm{d}\mathcal{H}^1(y) \text{ as } n\to\infty \tag{3.1.60}$$

and

$$\int_U g(x)[\Gamma^\Psi_{\varepsilon_n}(\psi_n)](x)\phi_n(x)\,\mathrm{d}x \to \int_{U\cap\{\Psi+\psi<0\}} g(x)\phi(x)\,\mathrm{d}x \text{ and } n\to\infty. \tag{3.1.61}$$

Proof. We only prove (3.1.60) in the special case that $\psi = 0$. The remaining claims in the general case follow by similar arguments. We shall use the uniform local level set coordinates constructed by Lemma 3.5 and the partition of unity (3.1.43).

Lemmas 3.2 and 3.5 imply that $(\Gamma^\Psi_{\varepsilon_n})'(\psi_n)$ is supported in $\bigcup_{i=1}^N B(x_i,\delta_-)$ for every $n\in\mathbb{N}$. In turn we can decompose into a sum of N terms and in the i^{th}-term make a change of variables with $G_i^{\psi_n}$:

$$\int_U g(x)[(\Gamma^\Psi_{\varepsilon_n})'(\psi_n)](x)\phi_n(x)\,\mathrm{d}x = \sum_{i=1}^N I_n^i,$$
$$\text{where } I^i_n = \frac{1}{\varepsilon_n}\int_{-\ell}^{\ell}\int_{-\varepsilon_n}^{\varepsilon_n} \chi_i(G_i^{\psi_n}(s))g(G_i^{\psi_n}(s))\gamma'(-s_2/\varepsilon_n)\phi_n(G_i^{\psi_n}(s))\det\nabla G_i^{\psi_n}(s)\,\mathrm{d}s_2\,\mathrm{d}s_1. \tag{3.1.62}$$

Now, it is suggested to us by the structure above that we ought to make a further change of variables by replacing s_2/ε_n by s_2. Doing so gives:

$$I^i_n = \int_{-\ell}^{\ell}\int_{-1}^{1} \gamma'(-s_2)\chi_i(G_i^{\psi_n}(s_1,\varepsilon_n s_2))g(G_i^{\psi_n}(s_1,\varepsilon_n s_2))\phi_n(G_i^{\psi_n}(s_1,\varepsilon_n s_2))\det\nabla G_i^{\psi_n}(s_1,\varepsilon_n s_2)\,\mathrm{d}s_2\,\mathrm{d}s_1. \tag{3.1.63}$$

Next, we use the dominated convergence theorem in conjunction with the hypotheses (3.1.59) and the third item of Lemma 3.5 to deduce that the sequence $\{I^i_n\}_{n\in\mathbb{N}}\subset\mathbb{R}$ converges to the limit $I^i\in\mathbb{R}$ given by

$$\lim_{n\to\infty} I^i_n = I^i = \int_{-\ell}^{\ell} \chi_i(G_i^0(s_1,0))g(G_i^0(s_1,0))\phi(G_i^0(s_1,0))\det\nabla G_i^0(s_1,0)\,\mathrm{d}s_1. \tag{3.1.64}$$

We next use the right hand equality of (3.1.22) to introduce a line element to the above integral and rewrite it in the coordinate-free fashion:

$$I^i = \int_\Sigma \chi_i(y)\frac{g(y)\phi(y)}{|\nabla\Psi(y)|}\,\mathrm{d}\mathcal{H}^1(y). \tag{3.1.65}$$

The proof is then completed upon summing over $i\in\{1,\dots,N\}$. □

This subsection's final result provides essential logarithmic-Lipschitz estimates on the operators $\boldsymbol{K}_{\varepsilon,\rho,♀}$ that we recall are defined in equation (3.1.7).

Proposition 3.9 (Preliminary mapping estimates, II). *With* $\widetilde{\varepsilon}_0$ *and* $\widetilde{r}_0$ *as in Proposition 3.7 and* $♀_0$ *as in Lemma 3.2, there exists* $C\in\mathbb{R}^+$ *with the property that for all* $0<\varepsilon\leqslant\widetilde{\varepsilon}_0$, $0\leqslant\rho\leqslant\widetilde{r}_0$, $0\leqslant ♀\leqslant ♀_0$, *and* $\psi,\widetilde{\psi}\in\overline{B(0,\widetilde{r}_0)}\subset C^1_{(m)}(\overline{U})$ *we have the continuity estimate*

$$\|\boldsymbol{K}_{\varepsilon,\rho,♀}(\psi)-\boldsymbol{K}_{\varepsilon,\rho,♀}(\widetilde{\psi})\|_{C^1(\overline{U})} \leqslant C\|\psi-\widetilde{\psi}\|_{C^0(\overline{U})}(1+|\log\|\psi-\widetilde{\psi}\|_{C^0(\overline{U})}|). \tag{3.1.66}$$

Proof. The first part of the proof is to establish two auxiliary claims. The first claim is: there exists $C\in\mathbb{R}^+$ such that for all $0<\varepsilon\leqslant\widetilde{\varepsilon}_0$, $0\leqslant\rho\leqslant\widetilde{r}_0$, $0\leqslant ♀\leqslant ♀_0$, and $\psi\in\overline{B(0,\widetilde{r}_0)}\subset C^1_{(m)}(\overline{U})$ we have the estimate

$$\|\boldsymbol{K}_{\varepsilon,\rho,♀}(\psi)\|_{\mathrm{LL}^1(\overline{U})}\leqslant C. \tag{3.1.67}$$

To prove the claim (3.1.67), we begin by noting that the pointwise estimates

$$|[\boldsymbol{K}_{\varepsilon,\rho,\female}(\psi)](x)| \leqslant C\int_U (|\log|x-y||+1)\,\mathrm{d}y,\ |\nabla[\boldsymbol{K}_{\varepsilon,\rho,\female}(\psi)](x)| \leqslant C\int_U \Big(\frac{1}{|x-y|}+1\Big)\,\mathrm{d}y \tag{3.1.68}$$

hold for $x\in U$. If $\widetilde{x}\in U\setminus\{x\}$ we also have

$$|\nabla[\boldsymbol{K}_{\varepsilon,\rho,\female}(\psi)](x) - \nabla[\boldsymbol{K}_{\varepsilon,\rho,\female}(\psi)](\widetilde{x})| \leqslant C\Big(\int_U\Big|\frac{x-y}{|x-y|^2}-\frac{\widetilde{x}-y}{|\widetilde{x}-y|^2}\Big|\,\mathrm{d}y + |x-\widetilde{x}|\Big). \tag{3.1.69}$$

Estimates (3.1.68) evidently establish the variant of (3.1.67) obtained by replacing $\|\cdot\|_{\mathrm{LL}^1(\overline{U})}$ by $\|\cdot\|_{C^1(\overline{U})}$. To obtain the remaining $\mathrm{LL}(\overline{U})$-norm on the derivative as a consequence of (3.1.69), further analysis is required. We shall split the integral on the right of (3.1.69) over the following two regions:

$$E_{x,\widetilde{x}} = \{y\in U\ :\ |y-(x+\widetilde{x})/2|\geqslant 2|x-\widetilde{x}|\},\ F_{x,\widetilde{x}} = \{y\in U\ :\ |y-(x+\widetilde{x})/2|< 2|x-\widetilde{x}|\}. \tag{3.1.70}$$

If $y\in E_{x,\widetilde{x}}$ then $2\min_{0\leqslant\tau\leqslant 1}|y-\tau x-(1-\tau)\widetilde{x}|\geqslant |y-(x+\widetilde{x})/2|+|x-\widetilde{x}|$ and so the fundamental theorem of calculus leads to the estimate

$$\int_{E_{x,\widetilde{x}}}\Big|\frac{x-y}{|x-y|^2}-\frac{\widetilde{x}-y}{|\widetilde{x}-y|^2}\Big|\,\mathrm{d}y \leqslant C|x-\widetilde{x}|\int_{E_{x,\widetilde{x}}}\frac{1}{|x-\widetilde{x}|^2+|y-(x+\widetilde{x})/2|^2}\,\mathrm{d}y \leqslant C|x-\widetilde{x}|(1+|\log|x-\widetilde{x}||). \tag{3.1.71}$$

where C is a constant depending only on U. On the other hand, for the region $F_{x,\widetilde{x}}$ there is no need to exploit the difference, instead we simply bound

$$\int_{F_{x,\widetilde{x}}}\Big|\frac{x-y}{|x-y|^2}-\frac{\widetilde{x}-y}{|\widetilde{x}-y|^2}\Big|\,\mathrm{d}y \leqslant 2\int_{B(0,4|x-\widetilde{x}|)}\frac{1}{|y|}\,\mathrm{d}y \leqslant C|x-\widetilde{x}|. \tag{3.1.72}$$

Synthesizing the above gives the first auxiliary claim (3.1.67).

Let us move on to the second auxiliary claim: there exists $C\in\mathbb{R}^+$ such that for all $0<\varepsilon\leqslant\widetilde{\varepsilon}_0$, $0\leqslant\rho\leqslant\widetilde{r}_0$, $0\leqslant\female\leqslant\female_0$, and $\psi,\widetilde{\psi}\in\overline{B(0,\widetilde{r}_0)}\subset C^1_{(m)}(\overline{U})$ we have the estimate

$$\|\boldsymbol{K}_{\varepsilon,\rho,\female}(\psi)-\boldsymbol{K}_{\varepsilon,\rho,\female}(\widetilde{\psi})\|_{\mathrm{LL}(\overline{U})} \leqslant C\|\psi-\widetilde{\psi}\|_{C^0(\overline{U})}. \tag{3.1.73}$$

Thanks to the fundamental theorem of calculus, we have the following pointwise expression for $x\in U$:

$$\begin{aligned}&[\boldsymbol{K}_{\varepsilon,\rho,\female}(\psi)-\boldsymbol{K}_{\varepsilon,\rho,\female}(\widetilde{\psi})](x)\\ &\quad=\sum_{k=-M}^{M}\frac{\sigma_k}{\varepsilon}\int_0^1\int_U \mathbf{N}_\female(x,y)\gamma'\Big(-\frac{\Psi(y)+k\rho+\tau\psi(y)+(1-\tau)\widetilde{\psi}(y)}{\varepsilon}\Big)(\psi-\widetilde{\psi})(y)\,\mathrm{d}y\,\mathrm{d}\tau.\end{aligned} \tag{3.1.74}$$

Therefore, if we define the regions indexed by $k\in\{-M,\dots,M\}$

$$R_k(\tau) = \{y\in U\ :\ |\Psi(y)+\tau\psi(y)+(1-\tau)\widetilde{\psi}(y)+k\rho|\leqslant\varepsilon\}, \tag{3.1.75}$$

we see that (3.1.74) lends us the estimates

$$|[\boldsymbol{K}_{\varepsilon,\rho,\female}(\psi)-\boldsymbol{K}_{\varepsilon,\rho,\female}(\widetilde{\psi})](x)| \leqslant \frac{C}{\varepsilon}\|\psi-\widetilde{\psi}\|_{C^0(\overline{U})}\sum_{k=-M}^{M}\int_0^1\int_U \mathbb{1}_{R_k(\tau)}(s)(|\log|x-y||+1)\,\mathrm{d}y \tag{3.1.76}$$

and

$$\begin{aligned}&\frac{|[\boldsymbol{K}_{\varepsilon,\rho,\female}(\psi)-\boldsymbol{K}_{\varepsilon,\rho,\female}(\widetilde{\psi})](x)-[\boldsymbol{K}_{\varepsilon,\rho,\female}(\psi)-\boldsymbol{K}_{\varepsilon,\rho,\female}(\widetilde{\psi})](\widetilde{x})|}{|x-\widetilde{x}|(1+|\log|x-\widetilde{x}||)}\\ &\qquad\leqslant\frac{C}{\varepsilon}\|\psi-\widetilde{\psi}\|_{C^0(\overline{U})}\sum_{k=-M}^{M}\int_0^1\int_U \mathbb{1}_{R_k(\tau)}(s)\frac{(|\log|x-y|-\log|\widetilde{x}-y||+|x-\widetilde{x}|)}{|x-\widetilde{x}|(1+|\log|x-\widetilde{x}||)}\,\mathrm{d}y\,\mathrm{d}\tau\end{aligned} \tag{3.1.77}$$

for $x,\widetilde{x}\in U$ with $x\neq\widetilde{x}$. Proposition 3.7 can now be applied to estimate the right hand sides of (3.1.76) and (3.1.77); the result is that the second auxiliary claim (3.1.73) holds.

The final thrust of the proof is to combine the above auxiliary claims to deduce (3.1.66). First we note that thanks to the interpolation estimates of Appendix A.2, we have

$$\begin{aligned}&\|\boldsymbol{K}_{\varepsilon,\rho,♀}(\psi)-\boldsymbol{K}_{\varepsilon,\rho,♀}(\widetilde{\psi})\|_{C^1(\overline{U})}\\&\qquad\leqslant C\|\boldsymbol{K}_{\varepsilon,\rho,♀}(\psi)-\boldsymbol{K}_{\varepsilon,\rho,♀}(\widetilde{\psi})\|_{\mathrm{LL}(\overline{U})}\Big(1+\Big|\log\Big|\frac{c\|\boldsymbol{K}_{\varepsilon,\rho,♀}(\psi)-\boldsymbol{K}_{\varepsilon,\rho,♀}(\widetilde{\psi})\|_{\mathrm{LL}(\overline{U})}}{\|\boldsymbol{K}_{\varepsilon,\rho,♀}(\psi)-\boldsymbol{K}_{\varepsilon,\rho,♀}(\widetilde{\psi})\|_{\mathrm{LL}^1(\overline{U})}}\Big|\Big|\Big).\end{aligned}\tag{3.1.78}$$

Now for the $\|\cdot\|_{\mathrm{LL}(\overline{U})}$ and $\|\cdot\|_{\mathrm{LL}^1(\overline{U})}$ factors on the right hand side we employ the bounds from the second and first auxiliary claims, respectively. □

3.2. **Key consequences of nondegeneracy.** In this subsection the nondegeneracy condition (1.3.4) from Definition 1.2 will now be exploited, unveiling the heart of our construction. Our first result combines nondegeneracy with certain compactness properties of the operators $\boldsymbol{\kappa}$ from (3.1.9) to establish uniformly coercive operator bounds in the supremum norm. We recall that the parameters $\widetilde{\varepsilon}_0$ and $\widetilde{r}_0$ are given by Lemma 3.5 and $♀_0$ is from Lemma 3.2.

Proposition 3.10 (Quantitative closed range estimates)**.** *There exists $C\in\mathbb{R}^+$, $0<\varepsilon_1\leqslant\widetilde{\varepsilon}_0$, $0<\rho_1,r_1<\widetilde{r}_0$, and $0<♀_1\leqslant♀_0$ with the property that for all $0<\varepsilon\leqslant\varepsilon_1$, $0\leqslant\rho\leqslant\rho_1$, $0\leqslant♀\leqslant♀_1$, $\psi^0,\psi^1\in\overline{B(0,r_1)}\subset C^1_{(m)}(\overline{U})$, and $\phi\in C^0_{(m)}(\overline{U})$ we have the estimate*

$$\|\phi\|_{C^0(\overline{U})}\leqslant C\Big\|\phi+\Big[\int_0^1\boldsymbol{\kappa}_{\varepsilon,\rho,♀}((1-\tau)\psi^0+\tau\psi^1)\,\mathrm{d}\tau\Big]\phi\Big\|_{C^0(\overline{U})}\tag{3.2.1}$$

Proof. We shall argue by way of contradiction, so let us suppose that the claim of the proposition is false. Then we would be able to find sequences

$$\begin{aligned}&\{\varepsilon_n\}_{n\in\mathbb{N}}\subset(0,\widetilde{\varepsilon}_0],\ \{\rho_n\}_{n\in\mathbb{N}}\subset[0,\widetilde{r}_0),\ \{♀_n\}_{n\in\mathbb{N}}\subset[0,♀_0],\\&\qquad\{\psi^0_n\}_{n\in\mathbb{N}},\{\psi^1_n\}_{n\in\mathbb{N}}\subset\overline{B(0,\widetilde{r}_0)}\subset C^1_{(m)}(\overline{U}),\ \{\phi_n\}_{n\in\mathbb{N}}\subset C^0_{(m)}(\overline{U})\end{aligned}\tag{3.2.2}$$

that satisfy $\|\phi_n\|_{C^0(\overline{U})}=1$ for all $n\in\mathbb{N}$ but

$$\lim_{n\to\infty}[|\varepsilon_n|+|\rho_n|+|♀_n|+\|\psi^0_n\|_{C^1(\overline{U})}+\|\psi^1_n\|_{C^1(\overline{U})}+\|\phi_n-\varphi_n\|_{C^0(\overline{U})}]=0\tag{3.2.3}$$

where for $n\in\mathbb{N}$ we have defined

$$\varphi_n=-\Big[\int_0^1\boldsymbol{\kappa}_{\varepsilon_n,\rho_n,♀_n}((1-\tau)\psi^0_n+\tau\psi^1_n)\,\mathrm{d}\tau\Big]\phi_n\in C^0(\overline{U}).\tag{3.2.4}$$

We claim first that the sequence $\{\varphi_n\}_{n\in\mathbb{N}}\subset C^0_{(m)}(\overline{U})$ has compact closure. Thanks to the Arzelà-Ascoli theorem, it is sufficient to establish that the sequence $\{\varphi_n\}_{n\in\mathbb{N}}\subset\mathrm{LL}(\overline{U})$ is bounded. Due to $\|\phi_n\|_{C^0(\overline{U})}=1$, support considerations, and Tonelli's theorem we can directly make the estimates

$$|\varphi_n(x)|\leqslant\frac{C}{\varepsilon_n}\sum_{k=-M}^{M}\int_0^1\int_U\mathbb{1}_{E^k_n(\tau)}(y)(|\log|x-y||+1)\,\mathrm{d}y\,\mathrm{d}\tau\tag{3.2.5}$$

for any $x\in\overline{U}$. For $\tau\in[0,1]$ and $k\in\{-M,\dots,M\}$ we have defined the sets

$$E^k_n(\tau)=\{x\in U\ :\ |\Psi(x)+(1-\tau)\psi^0_n(x)+\tau\psi^1_n(x)+k\rho_n|\leqslant\varepsilon_n\}.\tag{3.2.6}$$

Similarly, if $z\in\overline{U}\setminus\{x\}$ we have the bounds

$$\frac{|\varphi_n(x)-\varphi_n(z)|}{|x-z|(1+|\log|x-z||)}\leqslant\frac{C}{\varepsilon_n}\sum_{k=-M}^{M}\int_0^1\int_U\mathbb{1}_{E^k_n(\tau)}(y)\Big(\frac{|\log|x-y|-\log|z-y||}{|x-z|(1+|\log|x-z||)}+1\Big)\,\mathrm{d}y\,\mathrm{d}\tau.\tag{3.2.7}$$

By using (3.2.5) and (3.2.7) in conjunction with Proposition 3.7 and equation (3.2.3) we conclude that $\{\varphi_n\}_{n\in\mathbb{N}}$ is indeed bounded in the space $\mathrm{LL}(\overline{U})$.

Thanks to the aforementioned compactness, we are assured the existence of $\phi \in C^0_{(m)}(\overline{U})$ that we may suppose, after extraction of a subsequence and relabeling, satisfies $\varphi_n \to \phi$ as $n \to \infty$ in the $C^0(\overline{U})$ norm. From the final limit of (3.2.3) we deduce further that $\phi_n \to \phi$ as $n \to \infty$ in the same norm. Since $\|\phi_n\|_{C^0(\overline{U})} = 1$ for all $n \in \mathbb{N}$ we must have $\|\phi\|_{C^0(\overline{U})} = 1$ as well.

We next claim that, in the sense of distributions on U, we have the convergence of $\varphi_n \to \phi_\star$ as $n \to \infty$ where $\phi_\star \in C^0_{(m)}(\overline{U})$ is determined via

$$\phi_\star(x) = -\int_\Sigma \mathbf{N}(x-y)\frac{\phi(y)}{\nu(y)\cdot\nabla\Psi(y)}\,\mathrm{d}\mathcal{H}^1(y),\ x \in U, \tag{3.2.8}$$

where we recall that $\mathbf{N} = (2\pi)^{-1}\log|\cdot|$ is the $\mathbb{R}^2$-Newtonian potential. So let us fix an arbitrary test function $g \in C^\infty_{\mathrm{c}}(U)$ and consider the limit of the $L^2(U)$-pairings $\{\langle\varphi_n, g\rangle_{L^2(U)}\}_{n\in\mathbb{N}} \subset \mathbb{R}$. Thanks to Fubini's theorem and symmetry of the operators from equation (3.0.5), we first have the identity

$$\langle\varphi_n, g\rangle_{L^2(U)} = \sum_{k=-M}^{M}\sigma_k\int_0^1 [\langle(\Gamma^\Psi_{\varepsilon_n})'((1-\tau)\psi^0_n + \tau\psi^1_n + k\rho_n)\phi_n, \mathbf{N}_{\varrho_n}[g]\rangle_{L^2(\mathbb{R}^2)}\,\mathrm{d}\tau \tag{3.2.9}$$

for any $n \in \mathbb{N}$. The integrand above is a continuous function in $\tau \in [0,1]$. We deduce uniform boundedness in $n \in \mathbb{N}$ and convergence pointwise thanks to Propositions 3.7 and 3.8. Thus, the dominated convergence theorem applies and we identify (tacitly using (3.0.1)) the limit of (3.2.9) as $n \to \infty$ to be

$$\langle\phi, g\rangle_{L^2(U)} = \lim_{n\to\infty}\langle\varphi_n, g\rangle_{L^2(U)} = -\int_\Sigma \frac{\phi(y)}{\nu(y)\cdot\nabla\Psi(y)}(\mathbf{N}*g)(y)\,\mathrm{d}\mathcal{H}^1(y) = \langle\phi_\star, g\rangle_{L^2(U)}. \tag{3.2.10}$$

As (3.2.10) holds for all $g \in C^\infty_{\mathrm{c}}(U)$ we deduce that $\phi = \phi_\star$ where the latter is defined in (3.2.8). This is a contradiction of the nondegeneracy condition from the fourth item of Definition 1.2. Therefore our contradiction hypothesis, that the statement of the proposition does not hold, must be false. □

Our first consequence of Proposition 3.10 allows us to deduce invertibility of the derivative of the map $\boldsymbol{F}$ (defined in (3.1.7)) on an open subset of stream function perturbations. While the estimates (3.2.11) and (3.2.12) that follow are manifestly *not* uniform in ε, the size of the neighborhood on which we have invertibility is uniform.

Corollary 3.11 (Invertibility and estimates on the derivative)**.** *Let ε_1, ρ_1, r_1, and ϱ_1 be the positive values granted by Proposition 3.10. There exists $C \in \mathbb{R}^+$ such that for all $0 < \varepsilon \leqslant \varepsilon_1$, $0 \leqslant \rho \leqslant \rho_1$, $0 \leqslant \varrho \leqslant \varrho_1$, and $\psi, \widetilde{\psi} \in \overline{B(0,r_1)} \subset C^1_{(m)}(\overline{U})$ the derivative $D\boldsymbol{F}_{\varepsilon,\rho,\varrho}(\psi) : C^1_{(m)}(\overline{U}) \to C^1_{(m)}(\overline{U})$ is invertible and we have the estimates*

$$\|[D\boldsymbol{F}_{\varepsilon,\rho,\varrho}(\psi)]^{-1}\phi\|_{C^1(\overline{U})} + \|[D\boldsymbol{F}_{\varepsilon,\rho,\varrho}(\psi)]\phi\|_{C^1(\overline{U})} \leqslant (C/\varepsilon)\|\phi\|_{C^1(\overline{U})} \tag{3.2.11}$$

and

$$\|[D\boldsymbol{F}_{\varepsilon,\rho,\varrho}(\psi)]^{-1}\phi - [D\boldsymbol{F}_{\varepsilon,\rho,\varrho}(\widetilde{\psi})]^{-1}\phi\|_{C^1(\overline{U})} \leqslant (C/\varepsilon^3)\|\psi - \widetilde{\psi}\|_{C^1(\overline{U})}\|\phi\|_{C^0(\overline{U})} \tag{3.2.12}$$

for all $\phi \in C^1_{(m)}(\overline{U})$.

Proof. Identity (3.1.8) and estimate (3.1.15) show that the map $D\boldsymbol{F}_{\varepsilon,\rho,\varrho}(\psi)$ as a linear operator is a compact perturbation of the identity and thus has vanishing Fredholm index. Invertibility of $D\boldsymbol{F}_{\varepsilon,\rho,\varrho}(\psi)$ follows as soon as we know that $\ker D\boldsymbol{F}_{\varepsilon,\rho,\varrho}(\psi) = \{0\}$. If $\phi \in C^1_{(m)}(\overline{U})$ belongs to this kernel, then we can use estimate (3.2.1) from Proposition 3.10 (with $\psi^0 = \psi^1 = \psi$) to deduce that

$$\|\phi\|_{C^0(\overline{U})} \leqslant C\|\phi + \boldsymbol{\kappa}_{\varepsilon,\rho,\varrho}(\psi)\phi\|_{C^0(\overline{U})} = C\|D\boldsymbol{F}_{\varepsilon,\rho,\varrho}(\psi)\phi\|_{C^0(\overline{U})} = 0. \tag{3.2.13}$$

The kernel is indeed trivial.

We now prove estimate (3.2.11). That the second summand on the left hand side is bounded by $(C/\varepsilon)\|\phi\|_{C^1(\overline{U})}$ is a consequence of estimate (3.1.14) from Lemma 3.4. We now focus on the first summand. Let $\varphi = [D\boldsymbol{F}_{\varepsilon,\rho,\varrho}(\psi)]^{-1}\phi$. Thanks to Proposition 3.10 (again with $\psi^0 = \psi^1 = \psi$) we have initially that

$\|\varphi\|_{C^0(\overline{U})} \leqslant C\|\phi\|_{C^0(\overline{U})}$. On the other hand, we have $\varphi = \phi - \boldsymbol{\kappa}_{\varepsilon,\rho,\qoppa}(\psi)\varphi$ and so estimate (3.1.14) from Lemma 3.4 provides the bound

$$\|\varphi\|_{C^1(\overline{U})} \leqslant C\|\phi\|_{C^1(\overline{U})} + (C/\varepsilon)\|\varphi\|_{C^0(\overline{U})} \leqslant (C/\varepsilon)\|\phi\|_{C^1(\overline{U})}. \tag{3.2.14}$$

Estimate (3.2.11) is now established.

We conclude by proving (3.2.12). Let $\varphi = [D\boldsymbol{F}_{\varepsilon,\rho,\qoppa}(\psi)]^{-1}\phi$ and $\widetilde{\varphi} = [D\boldsymbol{F}_{\varepsilon,\rho,\qoppa}(\widetilde{\psi})]^{-1}\phi$. We compute first that

$$D\boldsymbol{F}_{\varepsilon,\rho,\qoppa}(\psi)(\varphi - \widetilde{\varphi}) = -(\boldsymbol{\kappa}_{\varepsilon,\rho,\qoppa}(\psi) - \boldsymbol{\kappa}_{\varepsilon,\rho,\qoppa}(\widetilde{\psi}))\widetilde{\varphi}. \tag{3.2.15}$$

Now take the norm in $C^1(\overline{U})$ on both sides of the above equation. On the left we use (3.2.11) while on the right we use estimate (3.1.16) from Lemma 3.4. This leaves us with

$$\|\varphi - \widetilde{\varphi}\|_{C^1(\overline{U})} \leqslant (C/\varepsilon^3)\|\psi - \widetilde{\psi}\|_{C^1(\overline{U})}\|\widetilde{\varphi}\|_{C^0(\overline{U})}. \tag{3.2.16}$$

Estimate (3.2.12) follows by using Proposition 3.10 to bound $\|\widetilde{\varphi}\|_{C^0(\overline{U})} \leqslant C\|\phi\|_{C^0(\overline{U})}$. □

Our second consequence of Proposition 3.10 gives us a reverse uniform continuity estimate on $\boldsymbol{F}$.

Corollary 3.12 (Reverse logarithmic-Lipschitz estimate)**.** *Let ε_1, ρ_1, $\qoppa_1$, and r_1 be the positive values granted by Proposition 3.10. There exists $C \in \mathbb{R}^+$ such that for all $0 < \varepsilon \leqslant \varepsilon_1$, $0 \leqslant \rho \leqslant \rho_1$, $0 \leqslant \qoppa \leqslant \qoppa_1$, and $\psi, \widetilde{\psi} \in \overline{B(0,r_1)} \subset C^1_{(m)}(\overline{U})$ we have the estimate*

$$\begin{aligned}\|\psi - \widetilde{\psi}\|_{C^1(\overline{U})} \leqslant{}& C\|\boldsymbol{F}_{\varepsilon,\rho,\qoppa}(\psi) - \boldsymbol{F}_{\varepsilon,\rho,\qoppa}(\widetilde{\psi})\|_{C^1(\overline{U})} \\ &+ C\|\boldsymbol{F}_{\varepsilon,\rho,\qoppa}(\psi) - \boldsymbol{F}_{\varepsilon,\rho,\qoppa}(\widetilde{\psi})\|_{C^0(\overline{U})}(1 + |\log\|\boldsymbol{F}_{\varepsilon,\rho,\qoppa}(\psi) - \boldsymbol{F}_{\varepsilon,\rho,\qoppa}(\widetilde{\psi})\|_{C^0(\overline{U})}|). \end{aligned}\tag{3.2.17}$$

Proof. We begin by using Lemma 3.3 and the fundamental theorem of calculus to write

$$\boldsymbol{F}_{\varepsilon,\rho,\qoppa}(\psi) - \boldsymbol{F}_{\varepsilon,\rho,\qoppa}(\widetilde{\psi}) = (\psi - \widetilde{\psi}) + \Big[\int_0^1 \boldsymbol{\kappa}_{\varepsilon,\rho,\qoppa}((1-\tau)\widetilde{\psi} + \tau\psi)\,\mathrm{d}\tau\Big](\psi - \widetilde{\psi}). \tag{3.2.18}$$

We then take the norm in $C^0(\overline{U})$ of the above expression and use Proposition 3.10 to deduce that

$$\|\psi - \widetilde{\psi}\|_{C^0(\overline{U})} \leqslant C\|\boldsymbol{F}_{\varepsilon,\rho,\qoppa}(\psi) - \boldsymbol{F}_{\varepsilon,\rho,\qoppa}(\widetilde{\psi})\|_{C^0(\overline{U})}. \tag{3.2.19}$$

On the other hand from (3.1.7) we also have the estimate

$$\|\psi - \widetilde{\psi}\|_{C^1(\overline{U})} \leqslant \|\boldsymbol{F}_{\varepsilon,\rho,\qoppa}(\psi) - \boldsymbol{F}_{\varepsilon,\rho,\qoppa}(\widetilde{\psi})\|_{C^1(\overline{U})} + \|\boldsymbol{K}_{\varepsilon,\rho,\qoppa}(\psi) - \boldsymbol{K}_{\varepsilon,\rho,\qoppa}(\widetilde{\psi})\|_{C^1(\overline{U})}. \tag{3.2.20}$$

Combining the above with Proposition 3.9 and the bound (3.2.19) gives the claim (3.2.17). □

3.3. **Quantitative local invertibility.** Somewhat remarkably, the conclusions of Corollaries 3.11 and 3.12 are all we need to deduce local invertibility near zero for the smooth maps $\boldsymbol{F}_{\varepsilon,\rho,\qoppa}$. The properties established by these results ensure local surjectivity by guaranteeing the convergence of Newton's method, despite the proximity to singularities. We have the following abstract result that is inspired by Hamilton's version of the Nash-Moser inverse function theorem [35].

Theorem 3.13 (Abstract quantitative local surjectivity)**.** *Let X and Y be Banach spaces, $r^+, r^-, \chi \in \mathbb{R}^+$ with $r^- < r^+$, and $F : B_X(0, r^+) \to Y$ be continuously differentiable and satisfy:*

(1) $F(0) = 0$ and if $x \in B_X(0, r^+)$ is such that $\|F(x)\|_Y \leqslant \chi$ then $\|x\|_X \leqslant r^-$.
(2) There exists $L_0 \in \mathbb{R}^+$ such that for all $x \in B_X(0, r^+)$ we have the estimates

$$\|F(x)\|_X + \|DF(x)\|_{\mathcal{L}(X;Y)} \leqslant L_0. \tag{3.3.1}$$

(3) There exists $L_1 \in \mathbb{R}^+$ and a map $R : B_X(0, r^+) \to \mathcal{L}(Y;X)$ such that for all $x, \widetilde{x} \in B_X(0, r^+)$ with $x \neq \widetilde{x}$ and all $y \in Y$ it holds that

$$\|R(x)\|_{\mathcal{L}(Y;X)} + \|R(x) - R(\widetilde{x})\|_{\mathcal{L}(Y;X)}/\|x - \widetilde{x}\|_X \leqslant L_1 \text{ and } DF(x)R(x)y = y. \tag{3.3.2}$$

Then, for each $y \in Y$ with $\|y\|_Y \leqslant \chi$ there exists $x \in X$ with $\|x\|_X \leqslant r^-$ with $F(x) = y$.

Proof. We begin by defining an auxiliary map. We set $r^0 = (r^- + r^+)/2$ and

$$T = \min\{(r^+ - r^-)/2L_1(\chi + L_0), 1/2L_1(\chi + 2L_0)\} \in \mathbb{R}^+. \tag{3.3.3}$$

Then, for any $T_0 \in [0, \infty)$ let us define

$$\Phi_{T_0} : \overline{B_X(0, r^-)} \times \overline{B_Y(0, \chi)} \times C^0([T_0, T_0 + T]; \overline{B_X(0, r^0)}) \to C^0([T_0, T_0 + T]; \overline{B_X(0, r^0)}) \tag{3.3.4}$$

to be the operator

$$\Phi_{T_0}(x, y, \gamma)(t) = x + \int_{T_0}^{t} R(\gamma(s))(y - F(\gamma(s)))\, \mathrm{d}s \tag{3.3.5}$$

for $x \in \overline{B_X(0, r^-)}$, $y \in \overline{B_Y(0, \chi)}$, $\gamma \in C^0([T_0, T_0 + T]; \overline{B_X(0, r^0)})$, and $t \in [T_0, T_0 + T]$. Let us verify that Φ_{T_0} is well defined in the sense that it maps into the codomain stated in (3.3.4). By using the hypotheses of the proposition and the definition of T in (3.3.3) we estimate for x, y, γ, and t belonging to the aforementioned sets

$$\|\Phi_{T_0}(x, y, \gamma)(t)\|_X \leqslant r^- + TL_1(\chi + L_0) \leqslant r^0. \tag{3.3.6}$$

So indeed Φ_{T_0} is well-defined.

We now claim that Φ_{T_0} is a contraction in its final argument. We estimate for any $x \in \overline{B_X(0, r^-)}$, $y \in \overline{B_Y(0, \chi)}$, $\gamma, \widetilde{\gamma} \in C^0([T_0, T_0 + T], \overline{B_X(0, r^0)})$, and $t \in [T_0, T]$ that

$$\|\Phi_{T_0}(x, y, \gamma)(t) - \Phi_{T_0}(x, y, \widetilde{\gamma})(t)\|_X \leqslant TL_1(\chi + 2L_0)\|\gamma - \widetilde{\gamma}\|_{C^0([T_0, T_0+T]; X)}. \tag{3.3.7}$$

Thanks to (3.3.3), we are assured that $TL_1(\chi + 2L_0) \leqslant 1/2$ and thus the above estimate shows the desired contraction estimates.

We are in a position to invoke the contraction mapping principle and deduce the existence of a function

$$\varphi_{T_0} : \overline{B_X(0, r^-)} \times \overline{B_Y(0, \chi)} \to C^0([T_0, T_0 + T]; \overline{B_X(0, r^0)}) \tag{3.3.8}$$

with the property that $\Phi_{T_0}(x, y, \varphi_{T_0}(x, y)) = \varphi_{T_0}(x, y)$ for all x and y belonging to the domain of (3.3.8). Immediately from (3.3.5) we deduce that $\varphi_{T_0}(x, y) \in C^1([T_0, T_0 + T]; \overline{B_X(0, r^0)})$ and is a solution to the ODE

$$\varphi_{T_0}(x, y)(T_0) = x \text{ and for all } t \in [T_0, T_0 + T]\ [\varphi_{T_0}(x, y)]'(t) = R(\varphi_{T_0}(x, y)(t))(y - F(\varphi_{T_0}(x, y)(t))). \tag{3.3.9}$$

Therefore, for any $z \in Y^\star$ (the dual space of Y) we can use (3.3.9) and the right hand identity of (3.3.2) to compute that

$$[\langle y - F(\varphi_{T_0}(x, y)(\cdot)), z\rangle_{Y,Y^*}]'(t) = -\langle y - F(\varphi_{T_0}(x, y)(t)), z\rangle_{Y,Y^*} \tag{3.3.10}$$

for $t \in [T_0, T_0 + T]$. Hence, for such t we have the identity

$$\langle y - F(\varphi_{T_0}(x, y)(t)), z\rangle_{Y,Y^*} = e^{-(t-T_0)}\langle y - F(x), z\rangle_{Y,Y^*}. \tag{3.3.11}$$

Since $z \in Y^*$ was arbitrary, we deduce that for all $t \in [T_0, T_0 + T]$

$$F(\varphi_{T_0}(x, y)(t)) = e^{-(t-T_0)}F(x) + (1 - e^{-(t-T_0)})y. \tag{3.3.12}$$

Let us move on to the next stage of our construction. We fix $y \in \overline{B_Y(0, \chi)}$ and define a sequence $\{x_n\}_{n\in\mathbb{N}} \subset \overline{B_X(0, r^-)}$ satisfying $F(x_n) = (1 - e^{-nT})y$, for all $n \in \mathbb{N}$, inductively as follows. We set $x_0 = 0$; since $F(0) = 0$ we have $F(x_0) = (1 - e^{-0})y$. Now if $n \in \mathbb{N}$ and $x_n \in \overline{B_X(0, r^-)}$ has been defined and satisfies the correct image identity, we take

$$x_{n+1} = \varphi_{nT}(x_n, y)((n + 1)T) \tag{3.3.13}$$

where T is from (3.3.3) and φ is the fixed point from (3.3.8). Initially we only know that $x_{n+1} \in \overline{B_X(0, r^0)}$. However, thanks to (3.3.12) and the induction hypothesis, we get

$$F(x_{n+1}) = e^{-T}F(x_n) + (1 - e^{-T})y = (1 - e^{-(n+1)T})y. \tag{3.3.14}$$

Then from the first hypothesis we find that $\|x_{n+1}\|_X \leqslant r^-$ since (3.3.14) implies that $\|F(x_{n+1})\|_Y \leqslant \chi$. This completes the inductive construction.

Let us now show that the sequence $\{x_n\}_{n\in\mathbb{N}} \subset \overline{B_X(0,r^-)}$ is Cauchy. To see this, we first use identity (3.3.13) and the fact that φ is a fixed point of Φ from (3.3.5) to obtain the difference identity

$$x_{n+1} - x_n = \int_{nT}^{(n+1)T} R(\varphi_{nT}(x_n,y)(s))(y - F(\varphi_{nT}(x_n,y)(s)))\,\mathrm{d}s. \tag{3.3.15}$$

Now from (3.3.12) it follows that for $s \in [nT,(n+1)T]$

$$y - F(\varphi_{nT}(x_n,y)(s)) = e^{-(s-nT)}(y - F(x_n)) = e^{-s}y. \tag{3.3.16}$$

Therefore (3.3.15) admits the upper bound

$$\|x_{n+1}-x_n\|_X \leqslant L_1\chi(1-e^{-T})e^{-nT} \text{ and hence } \sum_{n\in\mathbb{N}}\|x_{n+1}-x_n\|_X < \infty. \tag{3.3.17}$$

We conclude that the sequence in question is indeed Cauchy and so there exists $x \in \overline{B_X(0,r^-)}$ such that $x_n \to x$ as $n \to \infty$. From the identity $F(x_n) = (1-e^{-nT})y$, that holds for every $n \in \mathbb{N}$, we deduce that $F(x) = y$. This completes the proof. □

Corollary 3.14 (Local invertibility)**.** *Let ε_1, ρ_1, ♀_1, and r_1 be the positive values granted by Proposition 3.10. There exists $C, r_2 \in \mathbb{R}^+$ with $r_2 \leqslant r_1$ such that for all $0 < \varepsilon \leqslant \varepsilon_1$, $0 \leqslant \rho \leqslant \rho_1$, $0 \leqslant \text{♀} \leqslant \text{♀}_1$ the following hold.*

(1) For each $f \in \overline{B(0,r_2)} \subset C^1_{(m)}(\overline{U})$, there exists a unique $\psi \in \overline{B(0,r_1)} \subset C^1_{(m)}(\overline{U})$ such that $\boldsymbol{F}_{\varepsilon,\rho,\text{♀}}(\psi) = f$. This defines a left inverse $(\boldsymbol{F}_{\varepsilon,\rho,\text{♀}})^{-1} : \overline{B(0,r_2)} \to \overline{B(0,r_1)}$ mapping between subsets of $C^1_{(m)}(\overline{U})$.

(2) The left inverses from the previous item obey the logarithmic-Lipschitz estimate

$$\|(\boldsymbol{F}_{\varepsilon,\rho,\text{♀}})^{-1}(f) - (\boldsymbol{F}_{\varepsilon,\rho,\text{♀}})^{-1}(\widetilde{f})\|_{C^1(\overline{U})} \leqslant C\|f-\widetilde{f}\|_{C^1(\overline{U})} + C\|f-\widetilde{f}\|_{C^0(\overline{U})}(1+|\log\|f-\widetilde{f}\|_{C^0(\overline{U})}|) \tag{3.3.18}$$

for all $f, \widetilde{f} \in \overline{B(0,r_2)} \subset C^1_{(m)}(\overline{U})$.

Proof. We begin by proving the first item. The strategy is to show that the hypotheses of Theorem 3.13 are satisfied for $F = \boldsymbol{F}_{\varepsilon,\rho,\text{♀}}$, $X = Y = C^1_{(m)}(\overline{U})$, $r^+ = r_1$, and χ, r^- to be determined. Thanks to Proposition 3.12 in the special case that $\widetilde{\psi} = 0$, we know that

$$\|\psi\|_{C^1(\overline{U})} \leqslant C\|\boldsymbol{F}_{\varepsilon,\rho,\text{♀}}(\psi)\|_{C^1(\overline{U})}(1+|\log\|\boldsymbol{F}_{\varepsilon,\rho,\text{♀}}(\psi)\|_{C^1(\overline{U})}|) \leqslant C\|\boldsymbol{F}_{\varepsilon,\rho,\text{♀}}(\psi)\|^{1/2}_{C^1(\overline{U})} \tag{3.3.19}$$

for all $\|\psi\|_{C^1(\overline{U})} \leqslant r_1$. Therefore, we may take $r^- = r_1/2$ and $\chi = (r_1/2C)^2$ and conclude (by also implicitly using Corollary 3.11) that the hypotheses of Theorem 3.13 are met. We set $r_2 = \chi$.

We are thus granted for each $f \in \overline{B(0,r_2)} \subset C^1_{(m)}(\overline{U})$ at least one $\psi \in \overline{B(0,r_1)} \subset C^1_{(m)}(\overline{U})$ such that $\boldsymbol{F}_{\varepsilon,\rho,\text{♀}}(\psi) = f$. That there is at most one such ψ follows from Corollary 3.12's establishment of local injectivity. This proves the first item. The second item is an immediate consequence of the aforementioned corollary. □

We now recall the definition of the functions $f_{\varepsilon,\rho,\text{♀}}$ introduced in equation (3.1.10) of Lemma 3.3. Our next result, in a certain sense, proves Theorem I, II, and III simultaneously.

Corollary 3.15 (Existence and estimates for the perturbed equation)**.** *Let ε_1, ρ_1, ♀_1, and r_1 be the positive values granted by Proposition 3.10, and let $0 < r_2 \leqslant r_1$ be from Corollary 3.14. There exists $\varepsilon_\star, \rho_\star, \text{♀}_\star, C \in \mathbb{R}^+$ with $\varepsilon_\star \leqslant \varepsilon_1$, $\rho_\star \leqslant \rho_1$, $\text{♀}_\star \leqslant \text{♀}_1$ such that for all $0 < \varepsilon \leqslant \varepsilon_\star$, $0 \leqslant \rho \leqslant \rho_\star$, $0 \leqslant \text{♀} \leqslant \text{♀}_\star$ the following hold.*

(1) There exists $\psi_{\varepsilon,\rho,\text{♀}} \in \overline{B(0,r_1)} \subset C^1_{(m)}(\overline{U})$ satisfying $\boldsymbol{F}_{\varepsilon,\rho,\text{♀}}(\psi_{\varepsilon,\rho,\text{♀}}) = f_{\varepsilon,\rho,\text{♀}}$ along with the estimates

$$\|\psi_{\varepsilon,\rho,\text{♀}}\|_{\mathrm{LL}^1(\overline{U})} \leqslant C,\ \|\psi_{\varepsilon,\rho,\text{♀}}\|_{C^1(\overline{U})} \leqslant C(\varepsilon\log(1/\varepsilon) + \rho\log(1/\rho) + \text{♀}\log(1/\text{♀})). \tag{3.3.20}$$

(2) There exists $\Psi_{\varepsilon,\rho,\text{♀}} \in C^\infty_{(m)}(\overline{B(0,\text{♀}^{-1/2})})$ and $\omega_{\varepsilon,\rho,\text{♀}} \in C^\infty_{(m),\mathrm{c}}(B(0,\text{♀}^{-1/2}/2))$ satisfying

$$\Psi_{\varepsilon,\rho,\text{♀}} = \mathbf{N}_{\text{♀}}[\omega_{\varepsilon,\rho,\text{♀}}] - \frac{\Omega}{2}|\cdot|^2 - c \text{ and } \nabla^\perp\Psi_{\varepsilon,\rho,\text{♀}}\cdot\nabla\omega_{\varepsilon,\rho,\text{♀}} = 0 \text{ on } \overline{B(0,\text{♀}^{-1/2})}; \tag{3.3.21}$$

moreover, recalling Γ^Ψ_ε *from Lemma 3.2,*

$$\Psi_{\varepsilon,\rho,\Omega} = \Psi + \psi_{\varepsilon,\rho,\Omega} \text{ and } \omega_{\varepsilon,\rho,\Omega} = \sum_{k=-M}^{M} \sigma_k \Gamma^\Psi_\varepsilon(\psi_{\varepsilon,\rho,\Omega} + k\rho) \tag{3.3.22}$$

on the set $\overline{U}$. *In the case that* $\Omega > 0$ *we also have that* $\Psi_{\varepsilon,\rho,\Omega} + \Omega|\cdot|^2/2$ *is constant on* $\partial B(0, \Omega^{-1/2})$.

(3) We have the estimates

$$\|\omega_{\varepsilon,\rho,\Omega} - \mathbb{1}_{\widetilde{U}}\|_{L^1(B(0,\Omega^{-1/2}))} + \big|\|\nabla\omega_{\varepsilon,\rho,\Omega}\|_{\mathrm{TV}(B(0,\Omega^{-1/2}))} - \|\nabla \mathbb{1}_{\widetilde{U}}\|_{\mathrm{TV}(B(0,\Omega^{-1/2}))}\big| \\ + \|\nabla(\Psi_{\varepsilon,\rho,\Omega} - \Psi)\|_{L^\infty(B(0,\Omega^{-1/2}))} \leqslant C(\varepsilon\log(1/\varepsilon) + \rho\log(1/\rho) + \Omega\log(1/\Omega)), \tag{3.3.23}$$

with TV *referring to the total variation norm – see Section 1.4.*

Proof. For the first part of the proof, we shall estimate the norm of the source terms $f_{\varepsilon,\rho,\Omega}$, defined in (3.1.10), in the space $C^1_{(m)}(\overline{U})$. We begin by considering the decomposition

$$f_{\varepsilon,\rho,\Omega} = f^0_{\varepsilon,\rho,\Omega} + f^1_{\varepsilon,\rho} + f^2_\varepsilon \tag{3.3.24}$$

with

$$f^0_{\varepsilon,\rho,\Omega} = \sum_{k=-M}^{M} \sigma_k(\mathbf{N}_\Omega - \mathbf{N}_0)[\Gamma^\Psi_\varepsilon(k\rho)], \tag{3.3.25}$$

$$f^1_{\varepsilon,\rho} = \sum_{k=-M}^{M} \sigma_k \mathbf{N} * (\Gamma^\Psi_\varepsilon(k\rho) - \Gamma^\Psi_\varepsilon(0)), \tag{3.3.26}$$

and

$$f^2_\varepsilon = \mathbf{N} * (\Gamma^\Psi_\varepsilon(0) - \mathbb{1}_{\widetilde{U}}). \tag{3.3.27}$$

The claim is that there exists C such that for all $0 < \varepsilon \leqslant \varepsilon_1$, $0 \leqslant \Omega \leqslant \Omega_1$, and $0 \leqslant \rho \leqslant \rho_1$ we have the estimates

$$\|f^0_{\varepsilon,\rho,\Omega}\|_{C^0(\overline{U})} + (1/\log(1/\Omega))\|f^0_{\varepsilon,\rho,\Omega}\|_{C^1(\overline{U})} \leqslant C\Omega,\ \|f^1_{\varepsilon,\rho}\|_{C^0(\overline{U})} + (1/\log(1/\rho))\|f^1_{\varepsilon,\rho}\|_{C^1(\overline{U})} \leqslant C\rho, \\ \text{and } \|f^2_\varepsilon\|_{C^0(\overline{U})} + (1/\log(1/\varepsilon))\|f^2_\varepsilon\|_{C^1(\overline{U})} \leqslant C\varepsilon. \tag{3.3.28}$$

By arguing exactly as in the first part of the proof of Proposition 3.9, we find that

$$\|f^0_{\varepsilon,\rho,\Omega}\|_{\mathrm{LL}^1(\overline{U})} + \|f^1_{\varepsilon,\rho}\|_{\mathrm{LL}^1(\overline{U})} + \|f^2_\varepsilon\|_{\mathrm{LL}^1(\overline{U})} \leqslant C. \tag{3.3.29}$$

The interpolation results of Appendix A.2 combine with (3.3.29) to reduce the task of establishing estimate (3.3.28) to showing the following bounds:

$$\|f^0_{\varepsilon,\rho,\Omega}\|_{\mathrm{LL}(\overline{U})} \leqslant C\Omega,\ \|f^1_{\varepsilon,\rho}\|_{\mathrm{LL}(\overline{U})} \leqslant C\rho,\ \text{and } \|f^2_\varepsilon\|_{\mathrm{LL}(\overline{U})} \leqslant C\varepsilon. \tag{3.3.30}$$

The first bound in (3.3.30) for the source term $f^0_{\varepsilon,\rho,\Omega}$ holds thanks to the simple uniform kernel bounds

$$|\mathbf{N}_\Omega(x,y) - \mathbf{N}_0(x,y)| = |\log(1 - 2\Omega x\cdot y + \Omega^2|x|^2|y|^2)|/4\pi \leqslant C\Omega \tag{3.3.31}$$

and

$$|(\mathbf{N}_\Omega - \mathbf{N}_0)(x,y) - (\mathbf{N}_\Omega - \mathbf{N}_0)(\widetilde{x},y)|/|x - \widetilde{x}| \leqslant C\Omega, \tag{3.3.32}$$

that are valid for all $x, \widetilde{x}, y \in \overline{U \cup \widetilde{U}} \subset B(0, \Omega_0^{-1/2}/2)$ with $x \neq \widetilde{x}$.

The establishment of the second bound in (3.3.30) is our next task. For $x \in \overline{U}$ the coarea formula (Theorem 3.2.11 in [30] or Theorem 3.13 in [29]) and the fundamental theorem of calculus give us the identity

$$f^1_{\varepsilon,\rho}(x) = -\frac{\rho}{2\pi}\sum_{k=-M}^{M} k\sigma_k \int_0^1 \int_{-\varepsilon-|k\rho|}^{\varepsilon+|k\rho|} \frac{1}{\varepsilon}\gamma'\Big(-\frac{t+\tau k\rho}{\varepsilon}\Big)\int_{\{\Psi=t\}} \frac{\log|x-y|}{|\nabla\Psi(y)|}\,\mathrm{d}\mathcal{H}^1(y)\,\mathrm{d}t\,\mathrm{d}\tau. \tag{3.3.33}$$

By taking the norm in $\mathrm{LL}(\overline{U})$, arguing as in the proof of Proposition 3.7 to bound the logarithmic terms over the level sets, and performing a simple change of variables, we arrive at the desired estimate:

$$\|f^1_{\varepsilon,\rho}\|_{\mathrm{LL}(\overline{U})} \leqslant C\rho \sum_{k=-M}^{M} \int_0^1 \int_{-\varepsilon-|k\rho|}^{\varepsilon+|k\rho|} \frac{1}{\varepsilon}\Big|\gamma'\Big(-\frac{t+\tau k\rho}{\varepsilon}\Big)\Big| \,\mathrm{d}t\,\mathrm{d}\tau \leqslant C\rho. \tag{3.3.34}$$

Let us now discuss the proof of the third bound of (3.3.30). We again use the coarea formula to derive the pointwise equality

$$f^2_{\varepsilon}(x) = \frac{1}{2\pi}\int_{-\varepsilon}^{\varepsilon} (\gamma(-t/\varepsilon) - \mathrm{sgn}(-t)) \int_{\{\Psi=t\}} \frac{\log|x-y|}{|\nabla\Psi(y)|}\,\mathrm{d}\mathcal{H}^1(y)\,\mathrm{d}t \tag{3.3.35}$$

for $x \in \overline{U}$. Again we take the norm in $\mathrm{LL}(\overline{U})$ of (3.3.35), estimate the logarithmic contributions with the proof of Proposition 3.7, and make a change of variables to get the sought after bound:

$$\|f^2_{\varepsilon}\|_{\mathrm{LL}(\overline{U})} \leqslant C\int_{-\varepsilon}^{\varepsilon} |\gamma(-t/\varepsilon) - \mathrm{sgn}(-t)|\,\mathrm{d}t \leqslant C\varepsilon. \tag{3.3.36}$$

Estimate (3.3.28) paired with the decomposition (3.3.24) implies the existence of $0<\varepsilon_\star\leqslant\varepsilon_1$, $0<\rho_\star\leqslant\rho_1$, and $0<\qoppa_\star\leqslant\qoppa_1$ such that $\|f_{\varepsilon,\rho,\qoppa}\|_{C^1(\overline{U})}\leqslant r_2$ whenever $0<\varepsilon\leqslant\varepsilon_\star$, $0\leqslant\rho\leqslant\rho_\star$, and $0\leqslant\qoppa\leqslant\qoppa_\star$. The first item of Corollary 3.14 then allows us to define $\psi_{\varepsilon,\rho,\qoppa} = (\boldsymbol{F}_{\varepsilon,\rho,\qoppa})^{-1}(f_{\varepsilon,\rho,\qoppa}) \in \overline{B(0,r_1)} \subset C^1_{(m)}(\overline{U})$. In turn, the estimates (3.3.28) in conjunction with the second item of the aforementioned corollary imply the second estimate in (3.3.20). The first estimate of (3.3.20) follows from (3.3.29) and (3.1.67). The first item is now shown.

We turn our attention to the second item. Define $\Psi_{\varepsilon,\rho,\qoppa}\in C^1_{(m)}(\overline{U})$ via $\Psi_{\varepsilon,\rho,\qoppa} = \Psi + \psi_{\varepsilon,\rho,\qoppa}$ with the latter summand being the function constructed by the previous item. Then Lemma 3.2 allows us to define $\omega_{\varepsilon,\rho,\qoppa}\in C^1_{(m),\mathrm{c}}(B(0,\qoppa^{-1/2}/2))$ via

$$\omega_{\varepsilon,\rho,\qoppa} = \sum_{k=-M}^{M}\sigma_k\Gamma^{\Psi}_{\varepsilon}(\psi_{\varepsilon,\rho,\qoppa}+k\rho). \tag{3.3.37}$$

In fact, we have the inclusions

$$\mathrm{supp}(\omega_{\varepsilon,\rho,\qoppa})\subset\overline{U}\cup U^{\mathrm{in}} \text{ and } \mathrm{supp}(\nabla\omega_{\varepsilon,\rho,\qoppa})\subset U. \tag{3.3.38}$$

Next, we claim that the formula $\Psi_{\varepsilon,\rho,\qoppa} = \mathbf{N}_\qoppa[\omega_{\varepsilon,\rho,\qoppa}] + \Omega|\cdot|^2/2 - c$ constitutes a smooth extension of $\Psi_{\varepsilon,\rho,\qoppa}$ to all of $\overline{B(0,\qoppa^{-1/2})}$. Since $\omega_{\varepsilon,\rho,\qoppa}\in C^1_{(m),\mathrm{c}}(B(0,\qoppa^{-1/2}/2))$ and $\mathbf{N}_\qoppa$, being the Poisson kernel, is an operator of order -2, the function

$$\widetilde{\Psi}_{\varepsilon,\rho,\qoppa} = \mathbf{N}_\qoppa[\omega_{\varepsilon,\rho,\qoppa}] - \Omega|\cdot|^2/2 - c \tag{3.3.39}$$

is indeed in the regularity class $C^2_{(m)}(\overline{B(0,\qoppa^{-1/2})})$. That we have the equality $\Psi_{\varepsilon,\rho,\qoppa} = \widetilde{\Psi}_{\varepsilon,\rho,\qoppa}$ on the set $\overline{U}$ is then a consequence of $\psi_{\varepsilon,\rho,\qoppa} = (\boldsymbol{F}_{\varepsilon,\rho,\qoppa})^{-1}(f_{\varepsilon,\rho,\qoppa})$ and the equivalence in the third item of Lemma 3.3.

The second item is complete as soon as we verify that: 1) $\omega_{\varepsilon,\rho,\qoppa},\Psi_{\varepsilon,\rho,\qoppa}\in C^\infty(\overline{B(0,\qoppa^{-1/2})})$ and 2) the second equation of (3.3.21) is satisfied (the first equation is satisfied thanks to (3.3.39)). Item 1) follows from a simple induction argument, properties of the Poisson kernel, identities (3.3.37) and (3.3.39), and Lemma 3.2. By the second inclusion of (3.3.38), it suffices to verify that $\nabla^\perp\Psi_{\varepsilon,\rho,\qoppa}\cdot\nabla\omega_{\varepsilon,\rho,\qoppa}=0$ on U. But $\omega_{\varepsilon,\rho,\qoppa} = \mathsf{v}_{\varepsilon,\rho}(\Psi_{\varepsilon,\rho,\qoppa})$ where

$$\mathsf{v}_{\varepsilon,\rho}(t) = \sum_{k=-M}^{M}\sigma_k\gamma\Big(-\frac{t+k\rho}{\varepsilon}\Big) \text{ for } t\in\mathbb{R} \tag{3.3.40}$$

is a smooth function and so the equation is satisfied as a result of the chain rule and $\nabla^\perp\Psi_{\varepsilon,\rho,\qoppa}\cdot\nabla\Psi_{\varepsilon,\rho,\qoppa}=0$.

At last, we consider the third item. The claimed relative stream function estimate of (3.3.23) in the region $\overline{U}$, i.e. the bound $\|\nabla(\Psi_{\varepsilon,\rho,\qoppa}-\Psi)\|_{L^\infty(\overline{U})}\leqslant C(\varepsilon\log(1/\varepsilon)+\rho\log(1/\rho)+\qoppa\log(1/\qoppa))$, follows directly from (3.3.20), so it suffices to study the behavior in the region $\overline{B(0,\text{ʕ}^{-1/2})}\setminus\overline{U}$. By using familiar coarea formula based arguments and that

$$\sup\{|D_1\mathbf{N}_{\text{ʕ}}(x,y)|\ :\ x\in\overline{B(0,\text{ʕ}^{-1/2})}\setminus\overline{U},\ y\in\operatorname{supp}(\omega_{\varepsilon,\rho}-\mathbb{1}_{\widetilde{U}})\}\leqslant C,\tag{3.3.41}$$

we get for any $x\in\overline{B(0,\text{ʕ}^{-1/2})}\setminus\overline{U}$ the pointwise bound

$$|\nabla(\Psi_{\varepsilon,\rho,\text{ʕ}}-\Psi)(x)|\leqslant C\int_U|\omega_{\varepsilon,\rho,\text{ʕ}}(y)-\mathbb{1}_{\widetilde{U}}(y)|\,\mathrm{d}y+C\text{ʕ}\leqslant C\sum_{k=-M}^{M}\int_U\Big|\gamma\Big(-\frac{\Psi_{\varepsilon,\rho,\text{ʕ}}+k\rho}{\varepsilon}\Big)-\gamma\Big(-\frac{\Psi+k\rho}{\varepsilon}\Big)\Big|$$
$$+C\int_U\Big|\sum_{k=-M}^{M}\sigma_k\gamma\Big(-\frac{\Psi+k\rho}{\varepsilon}\Big)-\mathbb{1}_{\widetilde{U}}\Big|+C\text{ʕ}\leqslant C(\varepsilon\log 1/\varepsilon+\rho\log(1/\rho)+\text{ʕ}\log(1/\text{ʕ})).\tag{3.3.42}$$

The desired inequality is now apparent.

The last thing to prove are the $L^1(B(0,\text{ʕ}^{-1/2}))$ and $\mathrm{TV}(B(0,\text{ʕ}^{-1/2}))$ estimates on the vorticity. The $L^1(B(0,\text{ʕ}^{-1/2}))$ bounds are established by a computation similar to (3.3.42). On the other hand, for the $\mathrm{TV}(B(0,\text{ʕ}^{-1/2}))$-estimates, we use the coarea formula to identify

$$\|\nabla\mathbb{1}_{\widetilde{U}}\|_{\mathrm{TV}(B(0,\text{ʕ}^{-1/2}))}=\mathcal{H}^1(\{\Psi=0\})\text{ and }\|\nabla\omega_{\varepsilon,\rho,\text{ʕ}}\|_{\mathrm{TV}(B(0,\text{ʕ}^{-1/2}))}=\int_{-\varepsilon-M|\rho|}^{\varepsilon+M|\rho|}\mathsf{v}'_{\varepsilon,\rho}(t)\mathcal{H}^1(\{\Psi_{\varepsilon,\rho,\text{ʕ}}=t\})\,\mathrm{d}t.\tag{3.3.43}$$

The above can be combined with the fourth item of Corollary 3.6 to estimate

$$\big|\|\nabla\omega_{\varepsilon,\rho,\text{ʕ}}\|_{\mathrm{TV}(B(0,\text{ʕ}^{-1/2}))}-\|\nabla\mathbb{1}_{\widetilde{U}}\|_{\mathrm{TV}(B(0,\text{ʕ}^{-1/2}))}\big|\leqslant C\sum_{k=-M}^{M}\sup_{|t+k\rho|\leqslant\varepsilon}|\mathcal{H}^1(\{\Psi_{\varepsilon,\rho,\text{ʕ}}=t\})-\mathcal{H}^1(\{\Psi=0\})|$$
$$\leqslant C(\varepsilon\log(1/\varepsilon)+\rho\log(1/\rho)+\text{ʕ}\log(1/\text{ʕ})).\tag{3.3.44}$$

Estimate (3.3.23) is now shown. □

3.4. **Synthesis.** Our goal now is to combine the analyses of Sections 2 and 3 to complete the proofs of the main theorems stated in Section 1.3.

Proof of Theorem I. The desired result is a special consequence of Corollary 3.15. We take $\rho=0$, $\text{ʕ}=0$, and $0<\varepsilon\leqslant\varepsilon_\star$ then define $\Psi_\varepsilon=\Psi_{\varepsilon,0,0}$ and $\omega_\varepsilon=\omega_{\varepsilon,0,0}$ where the latter are the smooth functions guaranteed to exist by the second item of the corollary. The neighborhood U is taken as in Lemma 3.1. The validity of (3.3.21) immediately implies the first item of the theorem. Similarly, the second item of the theorem is a direct consequence of the estimate (3.3.23). The theorem's third item follows from (3.3.22) and Lemma 3.2 while the fourth item is a restatement of $\Psi_\varepsilon\in C^\infty_{(m)}(\mathbb{R}^2)$ and $\omega_\varepsilon\in C^\infty_{(m),\mathrm{c}}(\mathbb{R}^2)$. Lastly, the limits (1.3.5) hold by interpolation, equation (1.3.7), and the uniform bounds $\sup_{0<\varepsilon\leqslant\varepsilon_\star}\|\nabla\Psi_\varepsilon\|_{\mathrm{LL}(\mathbb{R}^2)}<\infty$. □

Proof of Theorem II. The idea is to pass to the limit $\varepsilon\to0$ in Corollary 3.15. Let $\varepsilon_\star$, $\rho_\star$, and $\text{ʕ}_\star$ be the small parameters granted by this result. Fix $\rho\in[0,\rho_\star]$ and $\text{ʕ}\in[0,\text{ʕ}_\star]$ and define the sequences $\{\Psi^n_{\rho,\text{ʕ}}\}\subset C^\infty_{(m)}(\overline{B(0,\text{ʕ}^{-1/2})})$ and $\{\omega^n_{\rho,\text{ʕ}}\}\subset C^\infty_{(m),\mathrm{c}}(B(0,\text{ʕ}^{-1/2}/2))$ via

$$\Psi^n_{\rho,\text{ʕ}}=\Psi_{\varepsilon_n,\rho,\text{ʕ}}\text{ and }\omega^n_{\rho,\text{ʕ}}=\omega_{\varepsilon_n,\rho,\text{ʕ}}\text{ where }\varepsilon_n=\varepsilon_\star/2^n\tag{3.4.1}$$

and $\Psi_{\cdot,\cdot,\cdot}$ and $\omega_{\cdot,\cdot,\cdot}$ are from the second item of Corollary 3.15.

The uniform estimates of equation (3.3.20) paired with the far-field strategy used in and around equation (3.3.42) in fact allows one to deduce the uniform bounds

$$\sup_{n\in\mathbb{N}}\|\Psi^n_{\rho,\text{ʕ}}\|_{\mathrm{LL}^1(\overline{U})}<\infty\text{ and }\sup_{n\in\mathbb{N}}\|\nabla\Psi^n_{\rho,\text{ʕ}}\|_{\mathrm{LL}(\overline{B(0,\text{ʕ}^{-1/2})})}<\infty.\tag{3.4.2}$$

On the other hand, from (3.3.23) and (3.3.38) we have

$$\sup_{n\in\mathbb{N}}\|\omega^n_{\rho,\text{ʕ}}\|_{(L^\infty\cap\mathrm{BV})(B(0,\text{ʕ}^{-1/2}/2))}<\infty\text{ and }\operatorname{supp}(\omega^n_{\rho,\text{ʕ}})\subset\overline{U}\cup U^{\mathrm{in}},\tag{3.4.3}$$

where we recall that U^{in} is from the fourth item of Lemma 3.1.

By the Arzelà-Ascoli theorem, compactness properties of functions of bounded variation (e.g. Theorem 3.23 in Ambrosio, Fusco, and Pallara [2]), and extraction and relabeling of a diagonal subsequence, we are bestowed with

$$\Psi_{\rho,\varrho} \in C^1_{(m)}(\overline{B(0,\varrho^{-1/2})}) \text{ satisfying } \nabla\Psi_{\rho,\varrho} \in \mathrm{LL}(\overline{B(0,\varrho^{-1/2})}) \text{ and } \omega_{\rho,\varrho} \in (L^\infty \cap \mathrm{BV})_{(m),\mathrm{c}}(\overline{B(0,\varrho^{-1/2})}) \tag{3.4.4}$$

and may suppose the satisfaction of the following strong limits (for all finite $1 \leqslant R \leqslant \varrho^{-1/2}$):

$$\|\omega^n_{\rho,\varrho} - \omega_{\rho,\varrho}\|_{L^1(B(0,\varrho^{-1/2}/2))} \to 0 \text{ and } \|\Psi^n_{\rho,\varrho} - \Psi_{\rho,\varrho}\|_{C^1(\overline{B(0,R)})} \to 0 \text{ as } n \to \infty, \tag{3.4.5}$$

as well as the following weak-$*$ convergences:

$$\nabla\omega^n_{\rho,\varrho} \overset{*}{\rightharpoonup} \nabla\omega_{\rho,\varrho} \text{ (in the space of Radon measures) and } \omega^n_{\rho,\varrho} \overset{*}{\rightharpoonup} \omega_{\rho,\varrho} \text{ (in } L^\infty(B(0,\varrho^{-1/2}))) \text{ as } n \to \infty. \tag{3.4.6}$$

One can directly pass to the limit $n \to \infty$ in the first identity of (3.3.21) to deduce that

$$\Psi_{\rho,\varrho} = \mathbf{N}_\varrho[\omega_{\rho,\varrho}] - \Omega|\cdot|^2/2 - c \text{ in } \overline{B(0,\varrho^{-1/2})} \tag{3.4.7}$$

and hence, upon differentiation of (3.4.7), for almost every $x \in \overline{B(0,\varrho^{-1/2})}$

$$\nabla\Psi_{\rho,\varrho}(x) = -\Omega x - \frac{1}{2\pi}\int_{B(0,\varrho^{-1/2})} \mathbf{K}_{\varrho^{-1/2}}(x,y)^\perp \omega_{\rho,\varrho}(y)\,\mathrm{d}y \tag{3.4.8}$$

where $\mathbf{K}$ is the Biot-Savart kernel defined in (1.1.11). On the other hand, for all $f \in C^\infty_{\mathrm{c}}(\overline{B(0,\varrho^{-1/2})})$ we may use the strong convergences in equation (3.4.5) to justify passing to the limit $n \to \infty$ in the weak form of the second identity in equation (3.3.21). Executing this procedure gives

$$\lim_{n\to\infty}\int_{B(0,\varrho^{-1/2})} \omega^n_{\rho,\varrho}\nabla^\perp\Psi^n_{\rho,\varrho}\cdot\nabla f = \int_{B(0,\varrho^{-1/2})} \omega_{\rho,\varrho}\nabla^\perp\Psi_{\rho,\varrho}\cdot\nabla f = 0. \tag{3.4.9}$$

By synthesizing (3.4.7), (3.4.8), and (3.4.9) we see that the triple $(\Omega, \Psi_{\rho,\varrho}, \omega_{\rho,\varrho})$ form a steady rotating weak solution to the Euler system, in the sense of Definition 1.1, in the domain $\overline{B(0,\varrho^{-1/2})}$. Therefore the first and fifth items of the theorem follows by taking $\varrho = 0$, $\omega_\rho = \omega_{\rho,0}$, and $\Psi_\rho = \Psi_{\rho,0}$.

We may also use (3.1.61) from Proposition 3.8 in conjunction with the convergences (3.4.5) and the vorticity identity (3.3.22) to pass to the limit in the space of tempered distributions and deduce

$$\omega_{\rho,\varrho} = \sum_{k=-M}^{M} \sigma_k \mathbb{1}_{(0,\infty)}(-\Psi_{\rho,\varrho} - k\rho) \text{ a.e. in } U,\ \omega_{\rho,\varrho} = 1 \text{ a.e. in } \overline{U^{\mathrm{in}}},\ \omega_{\rho,\varrho} = 0 \text{ a.e. in } \overline{U^{\mathrm{out}}}. \tag{3.4.10}$$

We now turn our attention to the proof of the second item. Define the sequence of open sets $\{W^k_\rho\}^M_{k=-M}$ via

$$W^k_\rho = \{x \in U \ :\ \Psi_\rho(x) + k\rho < 0\} \cup \overline{U^{\mathrm{in}}} \text{ for all } k \in \{-M,\dots,M\}. \tag{3.4.11}$$

The monotone inclusion relations (1.3.10) as well as the vorticity identities (1.3.11) and (1.3.14) are now immediate from (3.4.10) and (3.4.11). As Ψ_ρ is class $\bigcap_{0<\alpha<1} C^{1,\alpha}$ with $\min_{\overline{U}} |\nabla\Psi_\rho| > 0$ (see the first item of Lemma 3.2 and (3.4.5)) we deduce from the identity

$$\partial W^k_\rho = \{x \in U \ :\ \Psi_\rho(x) + k\rho = 0\} \tag{3.4.12}$$

and the implicit function theorem that ∂W^k_ρ is class $\bigcap_{0<\alpha<1} C^{1,\alpha}$ as well.

The distance estimate (1.3.12) follows from the following observation. Fix $k \in \{-M,\dots,M-1\}$ and let $x \in \partial W^k_\rho$, $y \in \partial W^{k+1}_\rho$ satisfy

$$|x-y| = \mathrm{dist}(\partial W^k_\rho, \partial W^{k+1}_\rho). \tag{3.4.13}$$

By the fundamental theorem of calculus, the lower bound

$$\rho = \Psi_\rho(x) - \Psi_\rho(y) \leqslant \|\nabla\Psi_\rho\|_{L^\infty(\mathbb{R}^2)}|x-y| \leqslant C\cdot\mathrm{dist}(\partial W^k_\rho, \partial W^{k+1}_\rho) \tag{3.4.14}$$

holds.

With the first, second, fourth, and fifth items of the theorem established, we complete the proof by justifying the third item and the limit (1.3.9). The first and third estimates of (1.3.13) are immediate from the strong convergences (3.4.5) and estimate (3.3.23). The claimed total variation estimate

$$\left|\|\nabla\omega_\rho\|_{\mathrm{TV}(\mathbb{R}^2)} - \|\nabla\mathbb{1}_{\widetilde{U}}\|_{\mathrm{TV}(\mathbb{R}^2)}\right| \leqslant C\rho\log(1/\rho) \tag{3.4.15}$$

follows by an argument similar to that used in equations (3.3.43) and (3.3.44) in the proof of Corollary 3.15. The limit (1.3.9) is again a consequence of (3.4.15), interpolation, and a uniform bound on $\nabla\Psi_\rho$ in $\mathrm{LL}(\mathbb{R}^2)$. We omit the repetitive details. □

Proof of Theorem III. Most of the work was already carried out in the first step of the proof of Theorem II. Recall that we took the limit $\varepsilon \to 0$ in Corollary 3.15 and obtained the functions $\Psi_{\rho,♀}$, $\omega_{\rho,♀}$ of equation (3.4.4), which are defined for $0 \leqslant \rho \leqslant \rho_\star$ and $0 \leqslant ♀ \leqslant ♀_\star$. We take $R_\star = ♀_\star^{-1/2}$ and for $R_\star \leqslant R < \infty$ define

$$\omega_R = \omega_{0,R^{-2}} \in (L^\infty \cap \mathrm{BV})_{(m),\mathrm{c}}(B(0,R/2)) \text{ and } \Psi_R = \Psi_{0,R^{-2}} \in \mathrm{LL}^1_{(m)}(\overline{B(0,R)}). \tag{3.4.16}$$

The satisfaction of the first and fourth items of the theorem are an immediate consequences of the proof of Theorem II, specifically the paragraph following (3.4.9), since it was shown that $(\Omega, \Psi_{\rho,♀}, \omega_{\rho,♀})$ is a steady rotating weak solution in the domain $\overline{B(0,♀^{-1/2})}$. We require here only the special case of $\rho = 0$ and $♀ = R^{-2}$.

The proof of the second item is similar to that for the corresponding item of Theorem II. We define

$$W_R = \{x \in U \ : \ \Psi_R(x) < 0\} \cup \overline{U^{\mathrm{in}}}. \tag{3.4.17}$$

The proof of Lemma 3.2 guarantees the existence of $C_1 \leqslant R_\star$ such that $W_R \subset B(0,C_1)$. As in the proof of Theorem II we observe also that

$$\partial W_R = \{x \in U \ : \ \Psi_R(x) = 0\} \text{ and } \min_{\overline{U}} |\nabla\Psi_R| > 0 \tag{3.4.18}$$

from which it follows that ∂W_R is class $\bigcap_{0<\alpha<1} C^{1,\alpha}$.

The proofs of the estimate (1.3.17) and the limit (1.3.15) are extremely similar to that of (1.3.13) and (1.3.9) in the proof of Theorem II. We again omit the repetitive details. □

Proof of Theorem IV. The first item is Theorem 2.4. The second item is Theorem 2.9. □

Appendix A. Tools from analysis

A.1. **Uniform local level set coordinates.** In this appendix we give the proof of Lemma 3.5 and construct the uniform local level set coordinates. Throughout the proof we shall employ the following bracket notation. For any $d \in \mathbb{N}^+$ the function $\langle\cdot\rangle : \mathbb{C}^d \to \mathbb{R}^+$ has the action

$$\langle z_1, \dots, z_d\rangle = (1 + |z_1|^2 + \dots + |z_d|^2)^{1/2}. \tag{A.1.1}$$

Proof of Lemma 3.5. We define the following positive values

$$\delta_+ = \mathrm{dist}(\Sigma, \partial U)/4,\ \upmu = \min_{\Sigma}(\nu\cdot\nabla\Psi),\ \upchi = \exp(\max_{x\in\overline{U}\setminus\Sigma} |\log|\Psi(x)/\mathrm{dist}(x,\Sigma)||),$$
$$\uplambda = (\min\{1, \upmu, \delta_+\}/20\langle\|\Psi\|_{C^{1,1/2}(\overline{U})}\rangle),\ L = \uplambda^2,\ \ell = \uplambda^3, \text{ and } \widetilde{r}_0 = (1/2\upchi)\uplambda^5. \tag{A.1.2}$$

These determine the following complete metric spaces

$$\mathcal{X} = \{g \in C^0(\overline{\ell Q}) \ : \ \|g\|_{C^0(\overline{\ell Q})} \leqslant L\} \text{ and } \mathcal{B} = \{\psi \in C^1(\overline{U}) \ : \ \|\psi\|_{C^1(\overline{U})} \leqslant 2\widetilde{r}_0\}. \tag{A.1.3}$$

Now let $x \in \Sigma$. It must be the case that either $|\partial_1\Psi(x)| \geqslant |\nabla\Psi(x)|/2$ or $|\partial_2\Psi(x)| \geqslant |\nabla\Psi(x)|/2$. Let us suppose that the latter inequality is satisfied. The construction that follows can be modified in the obvious way for points x obeying the alternative inequality.

Let us define the mapping $\aleph_x : \mathcal{B} \times \mathcal{X} \to \mathcal{X}$ via

$$[\aleph_x(\psi,g)](y) = -(1/\partial_2\Psi(x))(\Psi(x+(y_1,g(y))) - \Psi(x) - \nabla\Psi(x)\cdot(y_1,g(y)) + \psi(x+(y_1,g(y))) + y_1\partial_1\Psi(x) - y_2) \tag{A.1.4}$$

for all $(\psi, g) \in \mathcal{B} \times \mathcal{X}$ and $y \in \overline{\ell Q}$. The values ℓ, L, and $\widetilde{r}_0$ from equation (A.1.2) are chosen much more conservatively than what is required to ensure that $\aleph_x$ is well defined in the sense that the compositions in its definition are permissible and it maps into the stated codomain in the sense that $\|\aleph_x(\psi, g)\|_{C^0(\overline{\ell Q})} \leqslant L$. Furthermore, this selection of parameters also permit the following contraction and Lipschitz estimates:

$$\|\aleph_x(\psi, g) - \aleph_x(\psi, \widetilde{g})\|_{C^0(\overline{\ell Q})} \leqslant (2/3)\|g - \widetilde{g}\|_{C^0(\overline{\ell Q})}, \ \|\aleph_x(\psi, g) - \aleph_x(\widetilde{\psi}, g)\|_{C^0(\overline{\ell Q})} \leqslant (2/\mu)\|\psi - \widetilde{\psi}\|_{C^0(\overline{U})}. \tag{A.1.5}$$

for all $\psi, \widetilde{\psi} \in \mathcal{B}$ and all $g, \widetilde{g} \in \mathcal{X}$. The contraction mapping principle with parameter thus applies (see, for instance, Theorem C.7 in Irwin [42]) and we are assured the existence of a Lipschitz continuous mapping $\mathbf{g}_x : \mathcal{B} \to \mathcal{X}$ that satisfies

$$\mathbf{g}_x(\psi) = \aleph_x(\psi, \mathbf{g}_x(\psi)), \text{ for all } \psi \in \mathcal{B}. \tag{A.1.6}$$

By rearrangement of identity (A.1.4), we compute that the above fixed point (A.1.6) satisfies the level set parametrization identity

$$(\Psi + \psi)(x + (y_1, [\mathbf{g}_x(\psi)](y))) = y_2, \tag{A.1.7}$$

whenever $\psi \in \mathcal{B}$ and $y \in \overline{\ell Q}$. By using (A.1.7) we can perform a standard argument via difference quotients to deduce that $\mathbf{g}_x(\psi) \in C^1(\overline{\ell Q})$ and its partial derivatives satisfy

$$\partial_1[\mathbf{g}_x(\psi)](y) = -\frac{\partial_1(\Psi + \psi)(x + (y_1, [\mathbf{g}_x(\psi)](y)))}{\partial_2(\Psi + \psi)(x + (y_1, [\mathbf{g}_x(\psi)](y)))} \text{ and } \partial_2[\mathbf{g}_x(\psi)](y) = \frac{1}{\partial_2(\Psi + \psi)(x + (y_1, [\mathbf{g}_x(\psi)](y)))} \tag{A.1.8}$$

for all $y \in \overline{\ell Q}$. Identities (A.1.8) readily imply the estimates

$$\|\partial_1[\mathbf{g}_x(\psi)]\|_{C^0(\overline{\ell Q})} \leqslant \frac{5}{2\mu}\|\nabla\Psi\|_{C^0(\overline{U})}, \ \|(\partial_2[\mathbf{g}_x(\psi)])^{-1}\|_{C^0(\overline{U})} \leqslant \frac{10}{9}\|\nabla\Psi\|_{C^0(\overline{U})}, \ \|\partial_2[\mathbf{g}_x(\psi)]\|_{C^0(\overline{U})} \leqslant \frac{9}{4\mu}; \tag{A.1.9}$$

That the map $\mathcal{B} \ni \psi \mapsto \mathbf{g}_x(\psi) \in C^1(\overline{\ell Q})$ is continuous can also be deduced from (A.1.8).

From the fixed point functions above we now construct a continuous family of diffeomorphisms. Define $\mathbf{G}_x : \mathcal{B} \to C^1(\overline{\ell Q}; \mathbb{R}^2)$ via

$$[\mathbf{G}_x(\psi)](y) = x + (y_1, [\mathbf{g}_x(\psi)](y)) \text{ for } y \in \overline{\ell Q}, \ \psi \in \mathcal{B}. \tag{A.1.10}$$

An elementary calculation using (A.1.8) and the graphical structure of $\mathbf{G}_x(\psi)$ shows that the map $\mathbf{G}_x(\psi) : \overline{\ell Q} \to \mathbb{R}^2$ is a diffeomorphism onto its image; moreover, $\det \nabla[\mathbf{G}_x(\psi)] = \partial_2[\mathbf{g}_x(\psi)] > 0$ and we have the derivative estimates for every $y \in \overline{\ell Q}$

$$|\nabla[\mathbf{G}_x(\psi)](y)| + |(\nabla[\mathbf{G}_x(\psi)])^{-1}(y)| \leqslant R \text{ for } R = (10/\min\{1, \mu\})\langle\|\nabla\Psi\|_{C^0(\overline{U})}\rangle, \tag{A.1.11}$$

where the -1 exponent above refers to matrix inversion. We also compute from the formulae (A.1.8) that

$$|\partial_1[\mathbf{G}_x(\psi)](y)| = (1 + |\partial_1[\mathbf{g}_x(\psi)](y)|^2)^{1/2} = |\nabla(\Psi + \psi)([\mathbf{G}_x(\psi)](y))| \det \nabla[\mathbf{G}_x(\psi)](y). \tag{A.1.12}$$

We now need to find a universal radius for a ball centered at x which remains a subset of the images of the family of diffeomorphisms constructed above. We claim that we can take

$$\delta_- = \ell \min\{1, \mu\}/40\langle\|\Psi\|_{C^{1,1/2}(\overline{U})}\rangle^2 \text{ and satisfy } B(x, 2\delta_-) \subset [\mathbf{G}_x(\psi)](\overline{\ell Q}) \tag{A.1.13}$$

for all $\psi \in \mathcal{B}$. The first thing we need to estimate is the points $[\mathbf{g}_x(\psi)](0) \in \mathbb{R}$. From the fixed point identity (A.1.6) and the definition (A.1.4) we find that

$$|[\mathbf{g}_x(\psi)](0)| \leqslant (2/\mu)(\|\nabla\Psi\|_{C^{1/2}(\overline{U})}|[\mathbf{g}_x(\psi)](0)|^{3/2} + 2\widetilde{r}_0). \tag{A.1.14}$$

On the other hand we know that $|[\mathbf{g}_x(\psi)](0)|^{1/2} \leqslant L^{1/2} = \lambda$. So inserting this bound into the right hand side of the above (A.1.14) and solving for $|[\mathbf{g}_x(\psi)](0)|$ gives:

$$|[\mathbf{g}_x(\psi)](0)| \leqslant \frac{(4/\mu)\widetilde{r}_0}{1 - (2/\mu)\|\nabla\Psi\|_{C^{1/2}(\overline{U})}L^{1/2}} \leqslant \frac{20}{9\mu\chi}\lambda^5. \tag{A.1.15}$$

The next ingredient is a deviation estimate. We use, for each fixed $y_1 \in [-\ell, \ell]$, the fundamental theorem of calculus applied to the map $[-\ell, \ell] \ni y_2 \mapsto [\mathbf{g}_x(\psi)](y_1, y_2) - [\mathbf{g}_x(\psi)](y_1, 0) \in \mathbb{R}$ in conjunction with the positive lower bound on $\partial_2[\mathbf{g}_x(\psi)]$ from (A.1.9); doing this gives the inclusion

$$[\mathbf{G}_x(\psi)](\overline{\ell Q}) - x \supseteq \{(y_1, h + [\mathbf{g}_x(\psi)](y_1, 0)) \;:\; |y_1| \leqslant \ell,\ |h| \leqslant 9\ell/10\|\nabla\Psi\|_{C^0(\overline{U})}\}. \tag{A.1.16}$$

The set on the right hand side of (A.1.16) is the region between the graphs of Lipschitz functions. Therefore we can deduce, for the values $h_\star = 9\ell/10\|\nabla\Psi\|_{C^0(\overline{U})}$ and $s_\star = 5\|\nabla\Psi\|_{C^0(\overline{U})}/2\mu$, the following polygonal region inclusion:

$$[\mathbf{G}_x(\psi)](\overline{\ell Q}) - x - (0, [\mathbf{g}_x(\psi)](0)) \supseteq \{y \in \mathbb{R}^2 \;:\; |y_1| \leqslant \ell, |y_2| \leqslant h_\star - s_\star|y_1|\}. \tag{A.1.17}$$

Elementary planar geometry now permits us to conclude that the right hand side polygonal region above contains the ball centered at the origin of radius

$$\delta = \min\{\ell, h_\star/\langle s_\star\rangle\} \geqslant 9\ell\min\{1, \mu\}/50\langle\|\Psi\|_{C^{1,1/2}(\overline{U})}\rangle^2. \tag{A.1.18}$$

By synthesizing equations (A.1.15), (A.1.17), and (A.1.18) with the estimate

$$\delta - |[\mathbf{g}_x(\psi)](0)| \geqslant (157/900)(\ell\min\{1, \mu\}/\langle\|\Psi\|_{C^{1,1/2}(\overline{U})}\rangle^2) \geqslant 2\delta_-, \tag{A.1.19}$$

we conclude that the inclusion asserted in (A.1.13) indeed holds.

We are now ready to complete the construction of the uniform local level set coordinates. The set $\widetilde{\Sigma} = \{x \in \overline{U} \;:\; \mathrm{dist}(x, \Sigma) \leqslant \delta_-\}$ is compact and is covered by the collection of open balls $\{B(x, 2\delta_-)\}_{x\in\Sigma}$ and so there exists $N \in \mathbb{N}$ and a finite collection of points $\{x_i\}_{i=1}^N \subset \Sigma$ with $\widetilde{\Sigma} \subset \bigcup_{i=1}^N B(x_i, 2\delta_-)$.

Define $\widetilde{\varepsilon}_0 = \min\{\varepsilon_0, \widetilde{r}_0\}$. If $0 < \varepsilon \leqslant \widetilde{\varepsilon}_0$, $\psi \in \mathcal{B}$, and $x \in \overline{U}$ is such that $|\Psi(x) + \psi(x)| \leqslant \varepsilon$, then

$$|\Psi(x)| \leqslant 3\widetilde{r}_0 = \frac{3\ell}{2\chi}\lambda^2 < \frac{\delta_-}{\chi} \;\Rightarrow\; \mathrm{dist}(x, \Sigma) < \delta_- \;\Rightarrow\; x \in \bigcup_{i=1}^N B(x_i, 2\delta_-). \tag{A.1.20}$$

The first item is now shown.

For the second item we simply define for $\psi \in \mathcal{B}$ and $i \in \{1, \dots, N\}$ the map $G_i^\psi \in C^1(\overline{\ell Q}; \mathbb{R}^2)$ via $G_i^\psi = \mathbf{G}_{x_i}(\psi)$ where the latter map is defined in (A.1.10). The claimed properties (3.1.21), (3.1.22), (3.1.23), (3.1.24), and (3.1.25) along with the third item follow from the previous construction and elementary calculations. □

A.2. **A logarithmic interpolation inequality.**

Lemma A.1. *Let $U \subset \mathbb{R}^d$ be a smooth bounded domain. There exists constants $C, c \in \mathbb{R}^+$ such that the following hold for all $f \in \mathrm{LL}^1(\overline{U})$:*

(1) We have $\|f\|_{\mathrm{LL}(\overline{U})} \leqslant c\|f\|_{\mathrm{LL}^1(\overline{U})}$.

(2) We have

$$\|f\|_{C^1(\overline{U})} \leqslant C\|f\|_{\mathrm{LL}(\overline{U})}(1 + \log(c\|f\|_{\mathrm{LL}^1(\overline{U})}/\|f\|_{\mathrm{LL}(\overline{U})})). \tag{A.2.1}$$

Proof. Since U is a domain with $C^{1,1}$ boundary, the extension operator $\mathfrak{E}$ constructed for Theorem 5 of Chapter 6 in Stein [59] can be shown to map boundedly

$$\mathfrak{E} : X(\overline{U}) \to X(\mathbb{R}^2) \cap \{f \;:\; \mathrm{supp}(f) \subset B(0, R)\} \text{ for } X = \{\mathrm{LL}, C^1, \mathrm{LL}^1\}, \tag{A.2.2}$$

for some fixed $R < \infty$ chosen sufficiently large depending on U. By using $\mathfrak{E}$ appropriately, we can reduce to proving the first and second items for the case $U = \mathbb{R}^2$; in this case the first item is trivial.

To prove the second item (when $U = \mathbb{R}^2$) and $f \in \mathrm{LL}^1(\overline{U}) \setminus \{0\}$, we let $0 < r < 1/e$ and begin with the following pointwise formula for $x \in \mathbb{R}^2$:

$$\nabla f(x) = -\frac{1}{\pi r^2}\int_{\partial B(x,r)} (f(y) - f(x))\frac{y - x}{|y - x|}\,\mathrm{d}y + \frac{1}{\pi r^2}\int_{B(x,r)} (\nabla f(x) - \nabla f(y))\,\mathrm{d}y. \tag{A.2.3}$$

In turn, we arrive at the pointwise estimate

$$|\nabla f(x)| \leqslant 2\log(1/r)\|f\|_{\mathrm{LL}(\mathbb{R}^2)} + r\log(1/r)\|f\|_{\mathrm{LL}^1(\mathbb{R}^2)}. \tag{A.2.4}$$

The estimate (A.2.1) follows from (A.2.4) by selecting

$$r = \min\{1/e, \|f\|_{\mathrm{LL}(\mathbb{R}^2)}/c\|f\|_{\mathrm{LL}^1(\mathbb{R}^2)}\}. \tag{A.2.5}$$

□

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References

[1] R. Abraham, J. E. Marsden, and T. Ratiu. *Manifolds, Tensor Analysis, and Applications*, volume 75 of *Applied Mathematical Sciences*. Springer-Verlag, New York, second edition, 1988.

[2] L. Ambrosio, N. Fusco, and D. Pallara. *Functions of Bounded Variation and Free Discontinuity Problems*. Oxford Mathematical Monographs. The Clarendon Press, Oxford University Press, New York, 2000.

[3] C. Bardos. Existence et unicité de la solution de l'équation d'Euler en dimension deux. *J. Math. Anal. Appl.*, 40:769–790, 1972.

[4] J. Bedrossian and V. Vicol. *The Mathematical Analysis of the Incompressible Euler and Navier-Stokes Equations—An Introduction*, volume 225 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI, 2022.

[5] A. L. Bertozzi and P. Constantin. Global regularity for vortex patches. *Comm. Math. Phys.*, 152(1):19–28, 1993.

[6] A. L. Bertozzi and A. J. Majda. *Vorticity and Incompressible Flow*, volume 27 of *Cambridge Texts in Applied Mathematics*. Cambridge University Press, Cambridge, 2002.

[7] B. Buffoni and J. Toland. *Analytic Theory of Global Bifurcation*. Princeton Series in Applied Mathematics. Princeton University Press, Princeton, NJ, 2003. An introduction.

[8] J. Burbea. Motions of vortex patches. *Lett. Math. Phys.*, 6(1):1–16, 1982.

[9] A. Castro, D. Córdoba, and J. Gómez-Serrano. Existence and regularity of rotating global solutions for the generalized surface quasi-geostrophic equations. *Duke Math. J.*, 165(5):935–984, 2016.

[10] A. Castro, D. Córdoba, and J. Gómez-Serrano. Uniformly rotating analytic global patch solutions for active scalars. *Ann. PDE*, 2(1):Art. 1, 34, 2016.

[11] A. Castro, D. Córdoba, and J. Gómez-Serrano. Uniformly rotating smooth solutions for the incompressible 2D Euler equations. *Arch. Ration. Mech. Anal.*, 231(2):719–785, 2019.

[12] J.-Y. Chemin. Persistance des structures géométriques liées aux poches de tourbillon. In *Séminaire sur les Équations aux Dérivées Partielles, 1990–1991*, pages Exp. No. XIII, 11. École Polytech., Palaiseau, 1991.

[13] J.-Y. Chemin. Persistance de structures géométriques dans les fluides incompressibles bidimensionnels. *Ann. Sci. École Norm. Sup. (4)*, 26(4):517–542, 1993.

[14] A. Choffrut and L. Székelyhidi, Jr. Weak solutions to the stationary incompressible Euler equations. *SIAM J. Math. Anal.*, 46(6):4060–4074, 2014.

[15] A. Choffrut and V. Šverák. Local structure of the set of steady-state solutions to the 2D incompressible Euler equations. *Geom. Funct. Anal.*, 22(1):136–201, 2012.

[16] P. Constantin, T. D. Drivas, and D. Ginsberg. Flexibility and rigidity in steady fluid motion. *Comm. Math. Phys.*, 385(1):521–563, 2021.

[17] M. Coti Zelati, T. M. Elgindi, and K. Widmayer. Stationary structures near the Kolmogorov and Poiseuille flows in the 2d Euler equations. *Arch. Ration. Mech. Anal.*, 247(1):Paper No. 12, 37, 2023.

[18] M. G. Crandall and P. H. Rabinowitz. Bifurcation from simple eigenvalues. *J. Functional Analysis*, 8:321–340, 1971.

[19] R. Danchin. Évolution temporelle d'une poche de tourbillon singulière. *Comm. Partial Differential Equations*, 22(5-6):685–721, 1997.

[20] R. Danchin. Évolution d'une singularité de type cusp dans une poche de tourbillon. *Rev. Mat. Iberoamericana*, 16(2):281–329, 2000.

[21] F. de la Hoz, Z. Hassainia, T. Hmidi, and J. Mateu. An analytical and numerical study of steady patches in the disc. *Anal. PDE*, 9(7):1609–1670, 2016.

[22] F. de la Hoz, T. Hmidi, J. Mateu, and J. Verdera. Doubly connected V-states for the planar Euler equations. *SIAM J. Math. Anal.*, 48(3):1892–1928, 2016.

[23] G. S. Deem and N. J. Zabusky. Vortex waves: Stationary "v-states", interactions, recurrence, and breaking. *Phys. Rev. Lett.*, 40(13):859–862, 1978.

[24] T. M. Elgindi and Y. Huang. Regular and singular steady states of the 2D incompressible Euler equations near the Bahouri-Chemin patch. *Arch. Ration. Mech. Anal.*, 249(1):Paper No. 2, 31, 2025.

[25] T. M. Elgindi and I.-J. Jeong. On singular vortex patches, II: long-time dynamics. *Trans. Amer. Math. Soc.*, 373(9):6757–6775, 2020.

[26] T. M. Elgindi and I.-J. Jeong. On singular vortex patches, I: Well-posedness issues. *Mem. Amer. Math. Soc.*, 283(1400):v+89, 2023.

[27] T. M. Elgindi and M. J. Jo. Cusp formation in vortex patches. 2025. Preprint, arXiv:2504.02705.
[28] T. M. Elgindi, R. W. Murray, and A. R. Said. Wellposedness and singularity formation beyond the yudovich class. *J. Eur. Math. Soc.*, 2025.
[29] L. C. Evans and R. F. Gariepy. *Measure Theory and Fine Properties of Functions*. Textbooks in Mathematics. CRC Press, Boca Raton, FL, revised edition, 2015.
[30] H. Federer. *Geometric Measure Theory*, volume Band 153 of *Die Grundlehren der mathematischen Wissenschaften*. Springer-Verlag New York, Inc., New York, 1969.
[31] G. B. Folland. *Introduction to Partial Differential Equations*. Princeton University Press, Princeton, NJ, second edition, 1995.
[32] G. B. Folland. *Real Analysis*. Pure and Applied Mathematics (New York). John Wiley & Sons, Inc., New York, second edition, 1999. Modern techniques and their applications, A Wiley-Interscience Publication.
[33] D. Gilbarg and N. S. Trudinger. *Elliptic Partial Differential Equations of Second Order*. Classics in Mathematics. Springer-Verlag, Berlin, 2001. Reprint of the 1998 edition.
[34] Y. Guo, C. Hallstrom, and D. Spirn. Dynamics near an unstable Kirchhoff ellipse. *Comm. Math. Phys.*, 245(2):297–354, 2004.
[35] R. S. Hamilton. The inverse function theorem of Nash and Moser. *Bull. Amer. Math. Soc. (N.S.)*, 7(1):65–222, 1982.
[36] Z. Hassainia, N. Masmoudi, and M. H. Wheeler. Global bifurcation of rotating vortex patches. *Comm. Pure Appl. Math.*, 73(9):1933–1980, 2020.
[37] T. Hmidi and J. Mateu. Bifurcation of rotating patches from Kirchhoff vortices. *Discrete Contin. Dyn. Syst.*, 36(10):5401–5422, 2016.
[38] T. Hmidi and J. Mateu. Degenerate bifurcation of the rotating patches. *Adv. Math.*, 302:799–850, 2016.
[39] T. Hmidi and J. Mateu. Existence of corotating and counter-rotating vortex pairs for active scalar equations. *Comm. Math. Phys.*, 350(2):699–747, 2017.
[40] T. Hmidi, J. Mateu, and J. Verdera. Boundary regularity of rotating vortex patches. *Arch. Ration. Mech. Anal.*, 209(1):171–208, 2013.
[41] E. Hölder. Über die unbeschränkte Fortsetzbarkeit einer stetigen ebenen Bewegung in einer unbegrenzten inkompressiblen Flüssigkeit. *Math. Z.*, 37(1):727–738, 1933.
[42] M. C. Irwin. *Smooth Dynamical Systems*, volume 17 of *Advanced Series in Nonlinear Dynamics*. World Scientific Publishing Co., Inc., River Edge, NJ, 2001.
[43] T. Kato and G. Ponce. Well-posedness of the Euler and Navier-Stokes equations in the Lebesgue spaces $L^p_s(\mathbf{R}^2)$. *Rev. Mat. Iberoamericana*, 2(1-2):73–88, 1986.
[44] G. Kirchhoff. *Vorlesungen Über Mathematische Physik. Bd. I: Mechanik*. B. G. Teubner, Leipzig, 1874.
[45] H. Lamb. *Hydrodynamics*. Cambridge Mathematical Library. Cambridge University Press, Cambridge, sixth edition, 1993. With a foreword by R. A. Caflisch [Russel E. Caflisch].
[46] M. Lanza de Cristoforis and L. Preciso. On the analyticity of the Cauchy integral in Schauder spaces. *J. Integral Equations Appl.*, 11(3):363–391, 1999.
[47] L. Lichtenstein. Über einige Existenzprobleme der Hydrodynamik homogener, unzusammendrückbarer, reibungsloser Flüssigkeiten und die Helmholtzschen Wirbelsätze. *Math. Z.*, 23(1):89–154, 1925.
[48] A. E. H. Love. On the Stability of certain Vortex Motions. *Proc. Lond. Math. Soc.*, 25:18–42, 1893/94.
[49] A. Majda. Vorticity and the mathematical theory of incompressible fluid flow. volume 39, pages S187–S220. 1986. Frontiers of the mathematical sciences: 1985 (New York, 1985).
[50] C. Marchioro and M. Pulvirenti. *Mathematical Theory of Incompressible Nonviscous Fluids*, volume 96 of *Applied Mathematical Sciences*. Springer-Verlag, New York, 1994.
[51] N. I. Muskhelishvili. *Singular Integral Equations. Boundary Problems of Function Theory and Their Application to Mathematical Physics*. P. Noordhoff N. V., Groningen, 1953. Translation by J. R. M. Radok.
[52] E. A. Overman, II. Steady-state solutions of the Euler equations in two dimensions. II. Local analysis of limiting V-states. *SIAM J. Appl. Math.*, 46(5):765–800, 1986.
[53] C. Pommerenke. *Boundary Behaviour of Conformal Maps*, volume 299 of *Grundlehren der mathematischen Wissenschaften*. Springer-Verlag, Berlin, 1992.
[54] R.-O. Radu. Existence and regularity for vortex patch solutions of the 2D Euler equations. *Indiana Univ. Math. J.*, 71(5):2195–2230, 2022.
[55] W. J. M. Rankine. *A Manual of Applied Mechanics*. R. Griffin, London, 1858.
[56] P. G. Saffman. *Vortex Dynamics*. Cambridge Monographs on Mechanics and Applied Mathematics. Cambridge University Press, New York, 1992.
[57] P. Serfati. Pertes de régularité pour le laplacien et l'équation d'euler sur $\mathbb{R}^n$. 1994. Preprint.
[58] P. Serfati. Une preuve directe d'existence globale des vortex patches 2d. *C. R. Acad. Sci. Paris Sér. I Math.*, 318(6):515–518, 1994.
[59] E. M. Stein. *Singular Integrals and Differentiability Properties of Functions*, volume No. 30 of *Princeton Mathematical Series*. Princeton University Press, Princeton, NJ, 1970.
[60] Y. Tang. Nonlinear stability of vortex patches. *Trans. Amer. Math. Soc.*, 304(2):617–638, 1987.
[61] Y. H. Wan. The stability of rotating vortex patches. *Comm. Math. Phys.*, 107(1):1–20, 1986.

[62] Y. H. Wan and M. Pulvirenti. Nonlinear stability of circular vortex patches. *Comm. Math. Phys.*, 99(3):435–450, 1985.
[63] W. Wolibner. Un theorème sur l'existence du mouvement plan d'un fluide parfait, homogène, incompressible, pendant un temps infiniment long. *Math. Z.*, 37(1):698–726, 1933.
[64] H. M. Wu, E. A. Overman, II, and N. J. Zabusky. Steady-state solutions of the Euler equations in two dimensions: rotating and translating V-states with limiting cases. I. Numerical algorithms and results. *J. Comput. Phys.*, 53(1):42–71, 1984.
[65] B. B. Xue, E. R. Johnson, and N. R. McDonald. New families of vortex patch equilibria for the two-dimensional Euler equations. *Phys. Fluids*, 29(12):123602, 2017.
[66] V. I. Yudovich. Non-stationary flows of an ideal incompressible fluid. *Ž. Vyčisl. Mat i Mat. Fiz.*, 3:1032–1066, 1963.
[67] V. I. Yudovich. Uniqueness theorem for the basic nonstationary problem in the dynamics of an ideal incompressible fluid. *Math. Res. Lett.*, 2(1):27–38, 1995.

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